已停干净了。现在确认没有任何 S19/S20/mpail2.train.train 训练进程残留，S19 那个 orphan Python 也已经 kill -9 掉了。

这次真正的 S20 已跑完：

logs/obs22_S20_bcActionPrior_h3_policy50_fifo50k_stdfloor003_raw8_gp50_lr1e6_r1c0p05_clip5_rewfreq1_8env_50iter_20260525_103001/models/model_50.pt

我做了三件事：

1. 加了离线 checkpoint evaluator：
   scripts/offline_checkpoint_evaluator.py
2. 加了 S20 action-prior 实验脚本：
   scripts/run_s20_bc_action_prior_h3_gp50_50.sh
3. 给 learner 加了 optional demo-action BC prior：
   mpail2/learner.py

关键结论：

S20 的 BC/action prior 确实把 policy 动作拉向 expert action manifold：

- S18 policy_teacher_raw_cos 约 0.075
- S20 model_25 约 0.485
- S20 model_50 约 0.571

但是任务还是 success=0，而且离线 evaluator 显示最核心问题还在：

- teacher_reward_dyn_mean 和 zero_reward_dyn_mean 差很小：S20 model_50 约 0.570 vs 0.554
- teacher_q_mean_mean 和 zero_q_mean_mean 差也很小：约 -2.943 vs -2.955
- dyn_demo_delta_cos 对 zero/policy/teacher 都差不多，约 0.093

这说明：即使 policy 更像 expert action，learned dynamics/reward/value 仍然没有学出“expert action 会把 latent state 推向 demo transition direction”的结构。下一步不该继续调 h/gp/r1/value，而应该查/改 dynamics：现在代码里 update_dynamics() 对 demo transition 用的是 zero placeholder，而且 JEP loss 只监督 agent replay，不用 demo action 监督 dynamics。这很可能就是 planner 看到 expert-like action 也不会预测 expert-like next state 的原因。

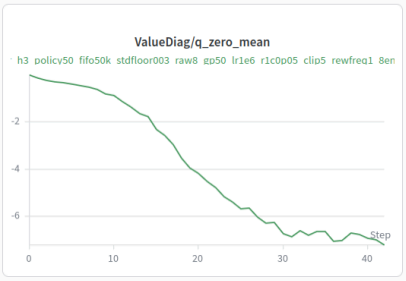
验证情况：

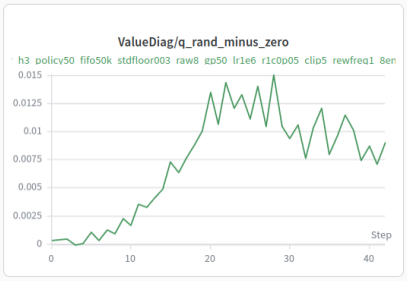
- py_compile 通过。
- bash -n S20 脚本通过。
  相关 pytest 里有 1 个已有 Hydra config override 失败，和这次 S20 路径无关。

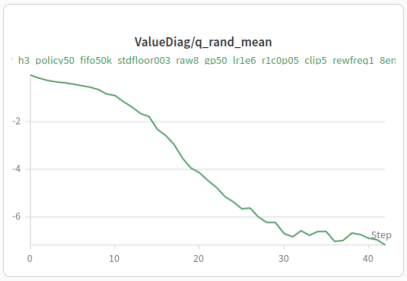
Search panels with regex

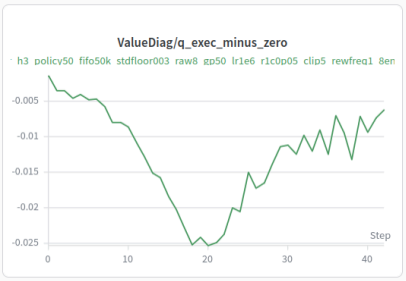
Settings | New report | Add panels

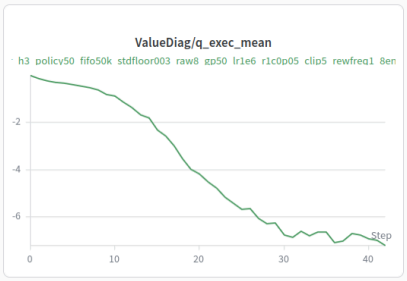
ValueDiag 7 1-6 of 7

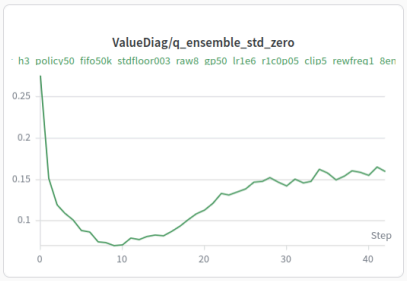










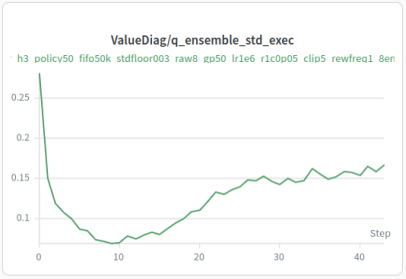


ActionDiag 2

Search panels with regex

Settings | New report | Add panels

ValueDiag 7 7-7 of 7



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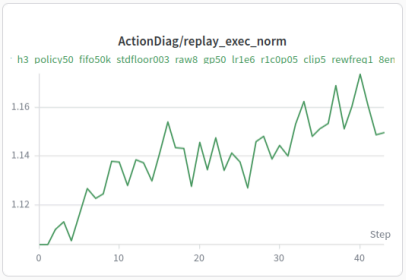

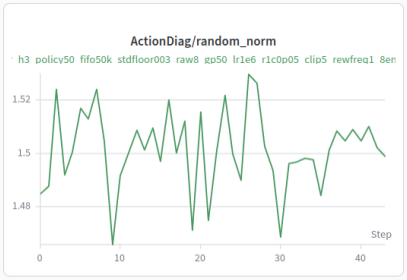

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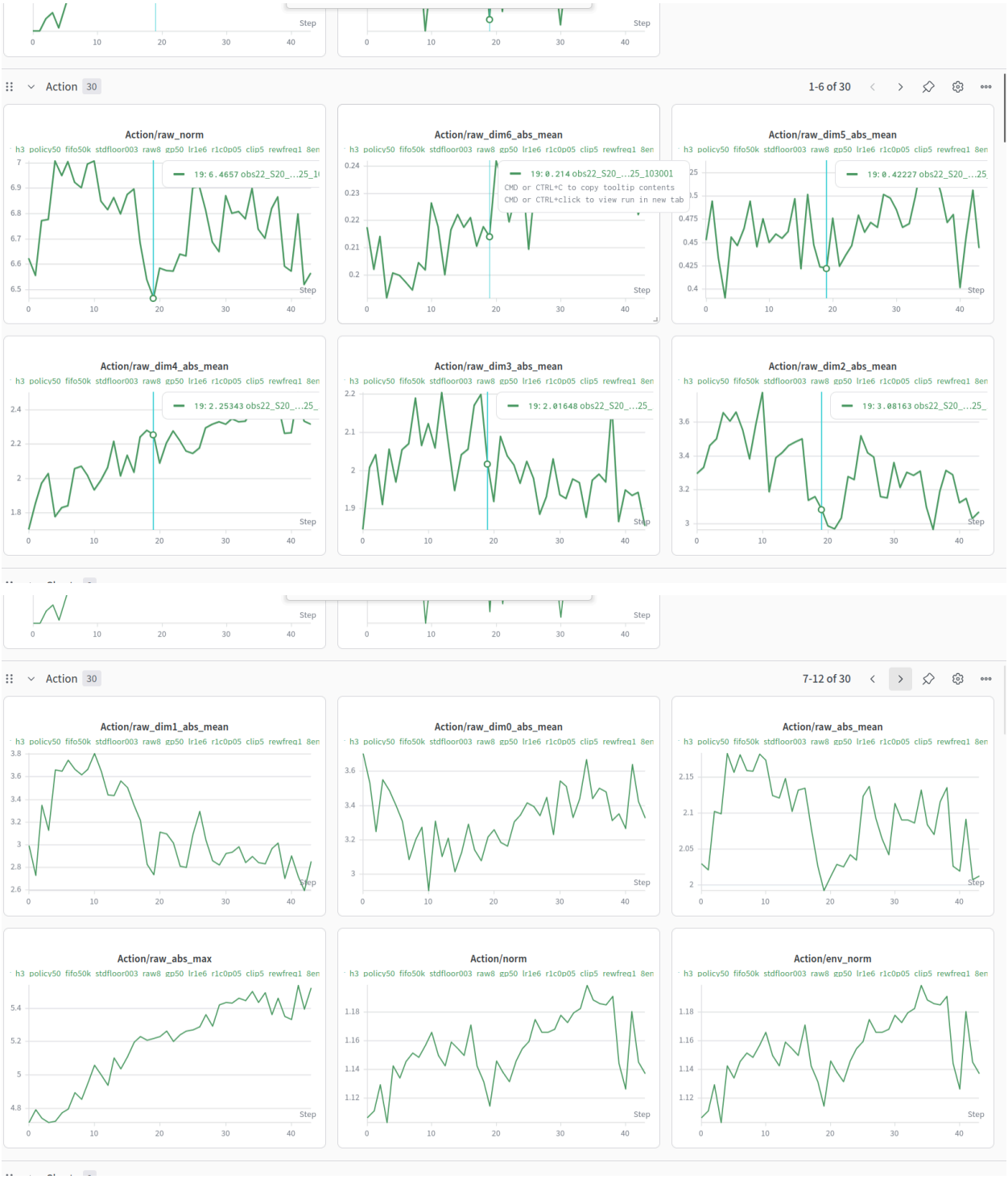
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1-6 of 30
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Action/raw_dim6_abs_mean
Action/raw_dim5_abs_mean
Action/raw_dim4_abs_mean
Action/raw_dim3_abs_mean
Action/raw_dim2_abs_mean
Action 30
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Action/raw_dim1_abs_mean
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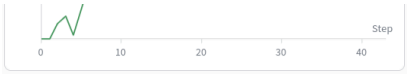
Action/norm
Action/env_norm
Step



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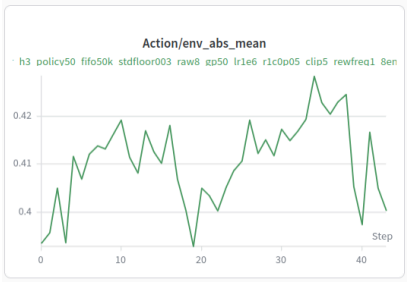

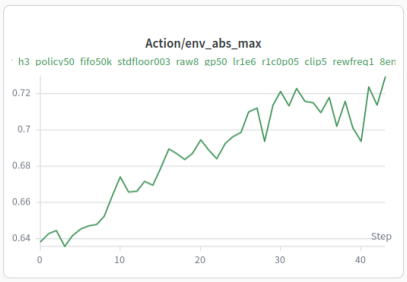

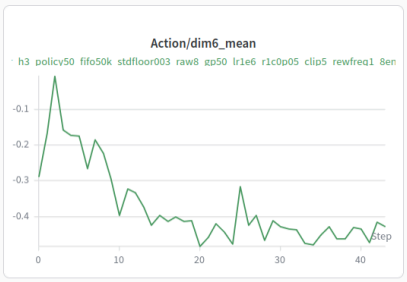

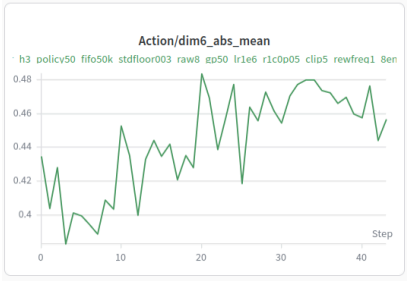

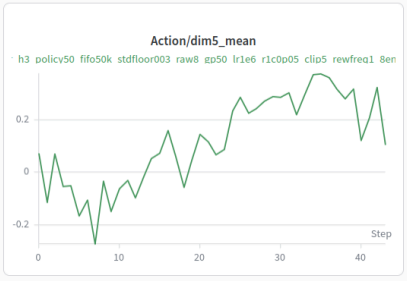

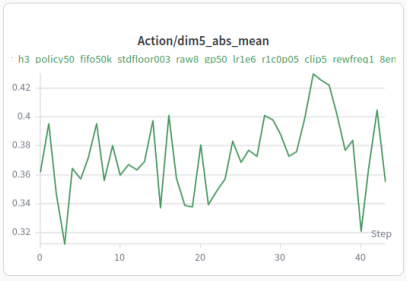

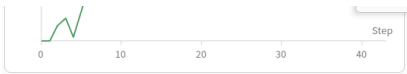

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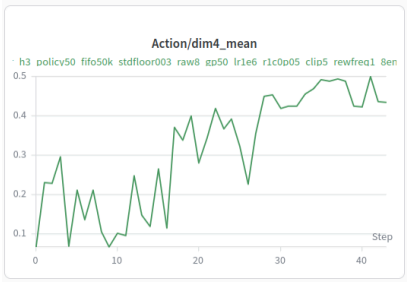

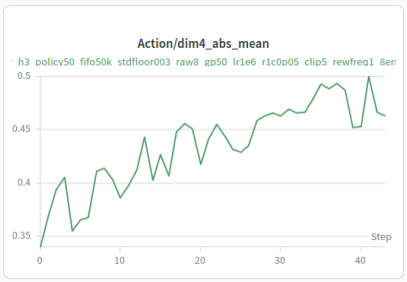

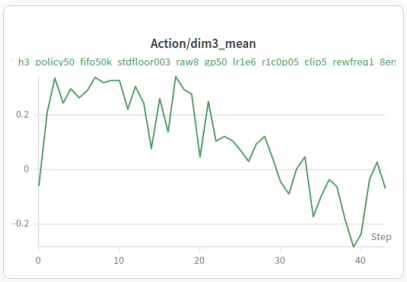

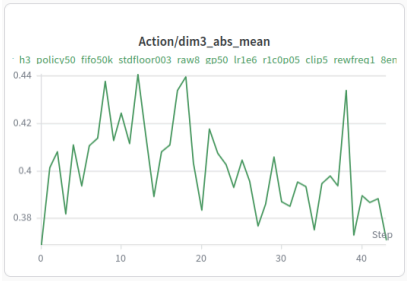

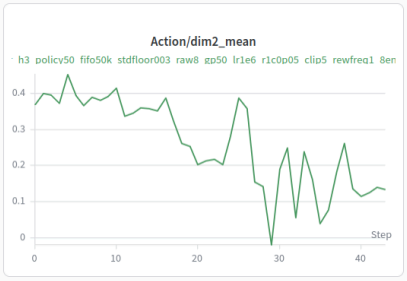

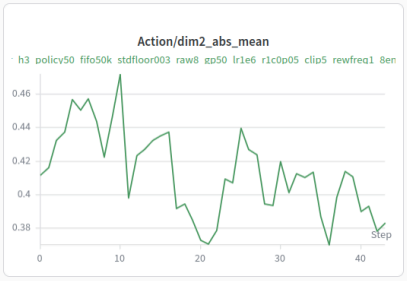

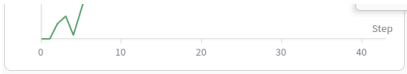

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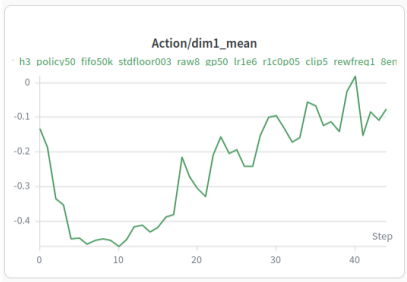

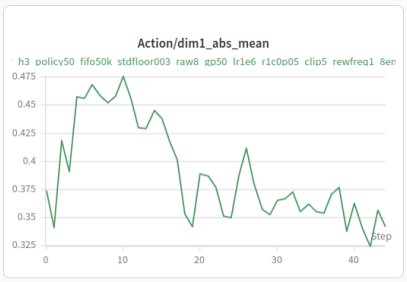

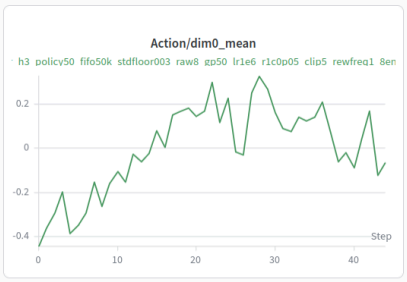

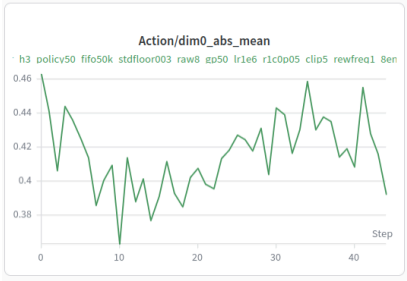

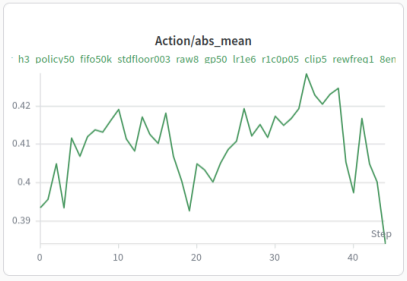

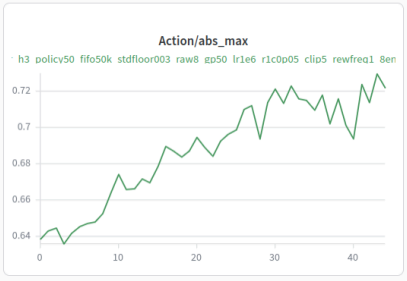

Data 2

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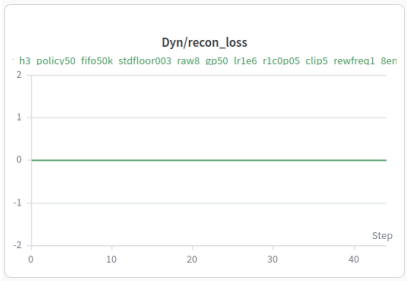
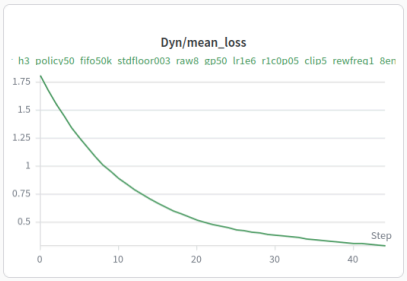
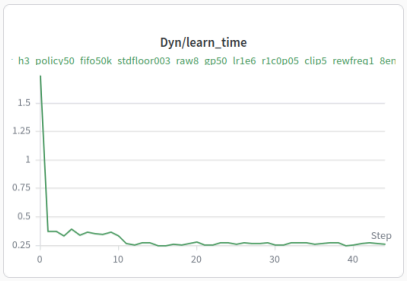
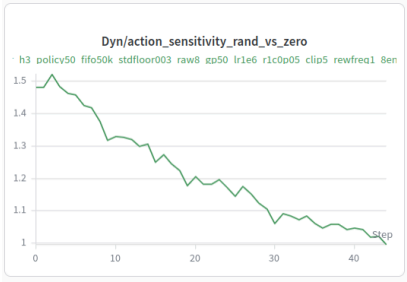
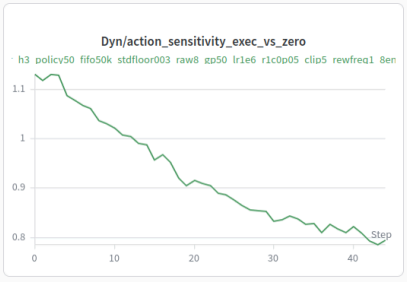
1-5 of 5





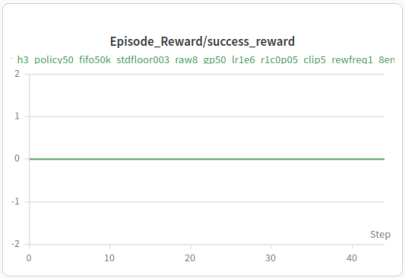


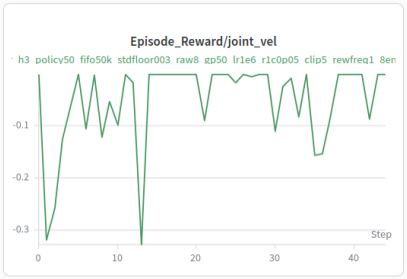


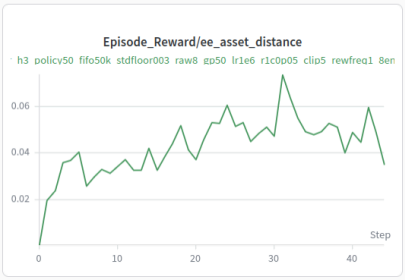


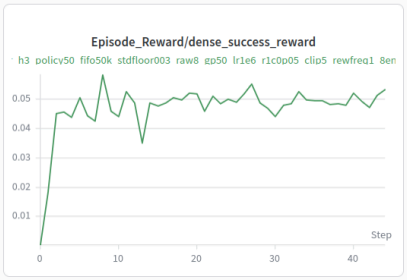
Env 11

Episode_Reward 8 1-6 of 8

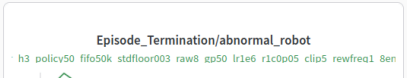




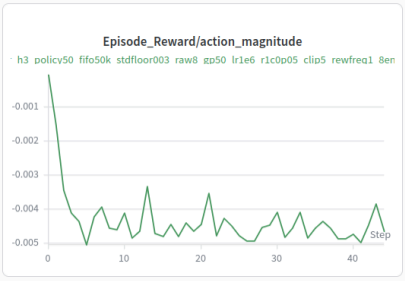


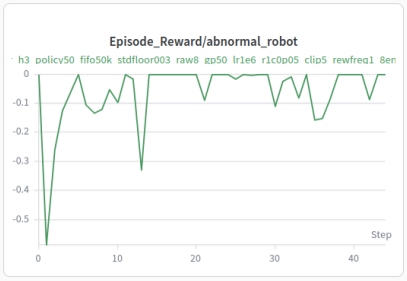


Episode_Termination 2



Episode_Reward 8 1-8 of 8





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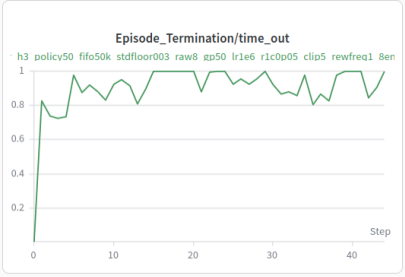
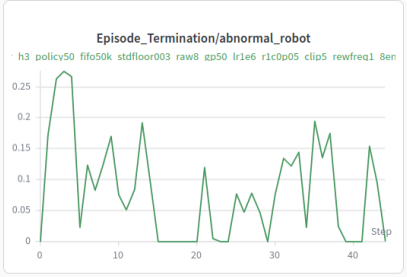
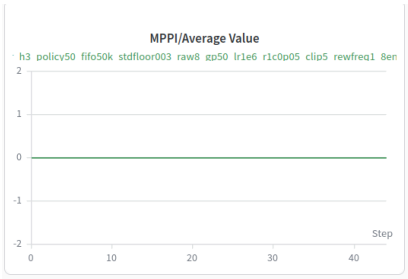
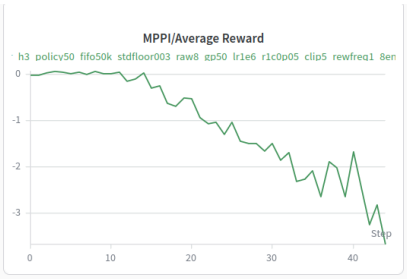




Metrics 5

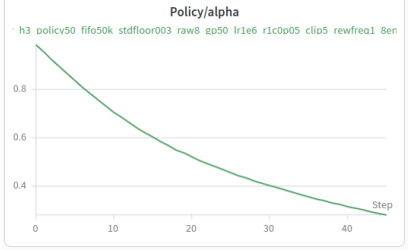
MPPI 20 19-20 of 20





Perf 3

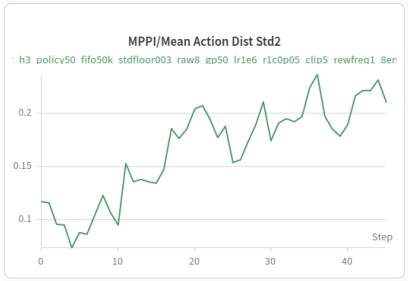
Policy 7 7-7 of 7

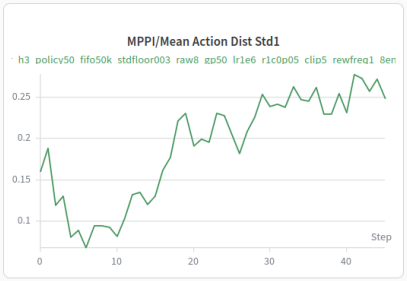


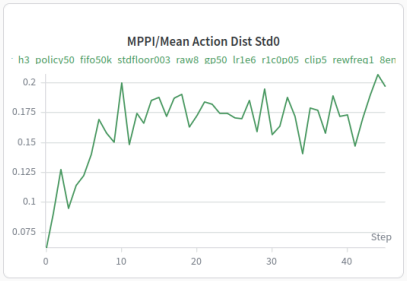
Reward 22 1-6 of 22

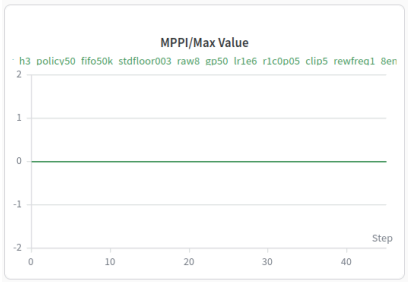
Metrics 5

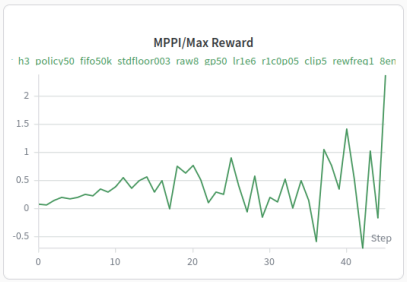
MPPI 20 13-18 of 20

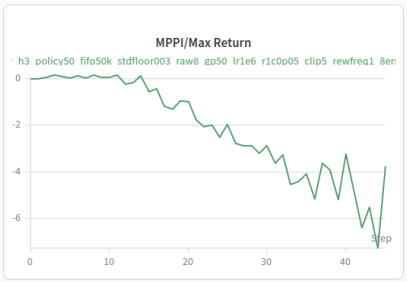








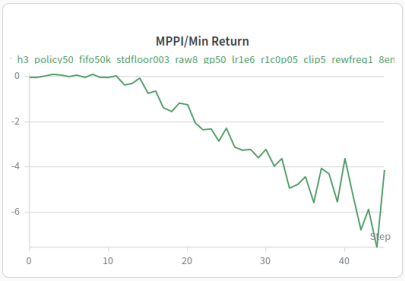


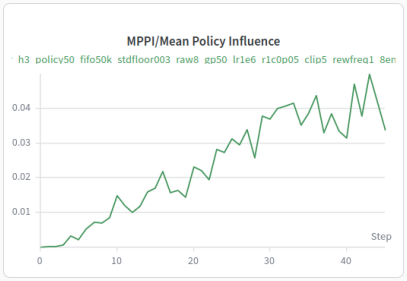


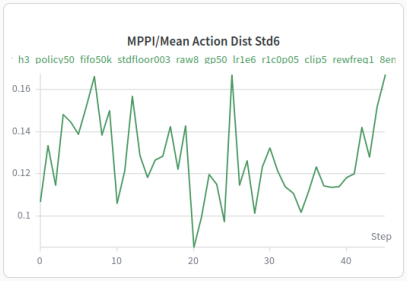
Perf 3

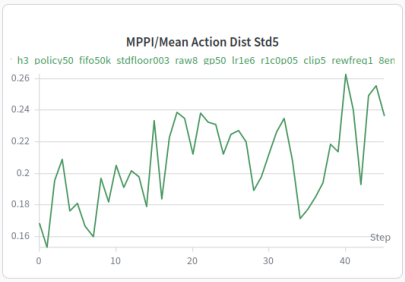
Metrics 5

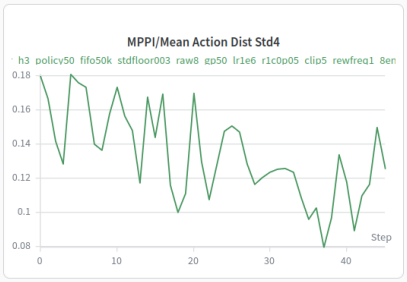
MPPI 20 7-12 of 20

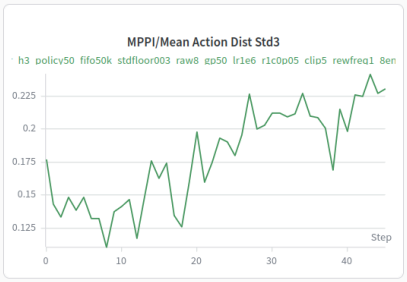






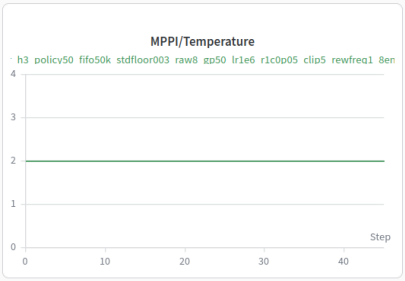


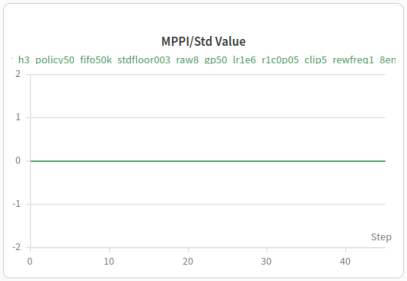


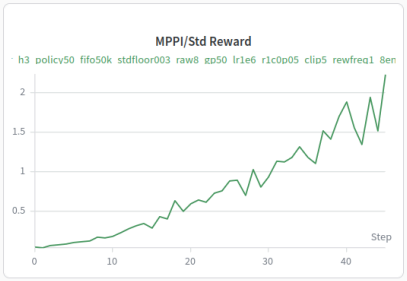


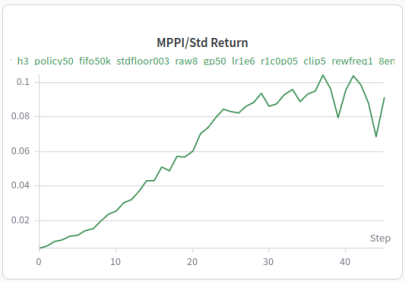
Metrics 5

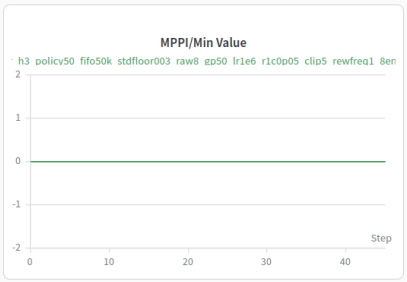
MPPI 20 1-6 of 20

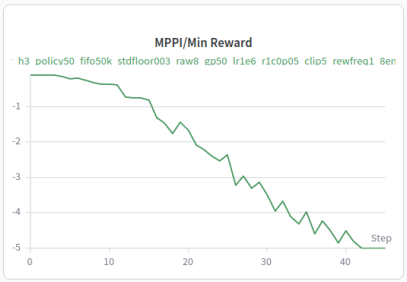




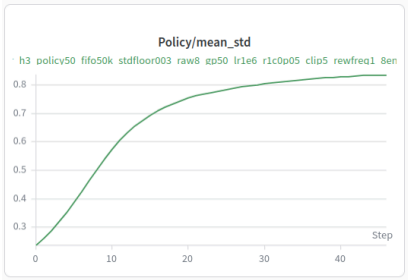


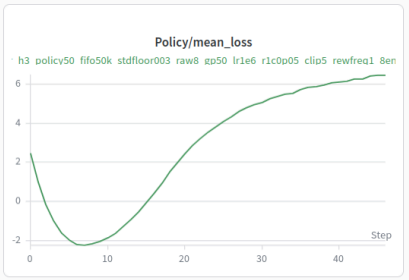


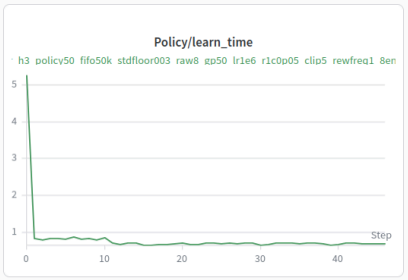


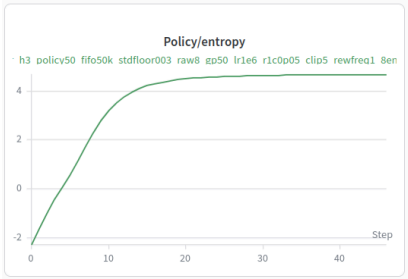


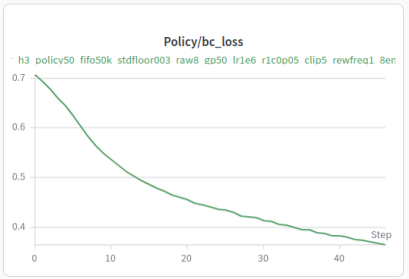
Policy 7 1-6 of 7

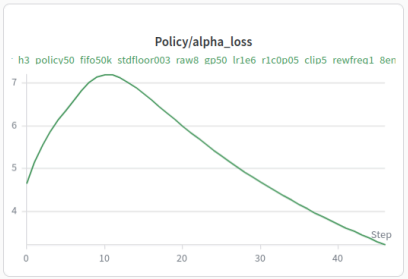






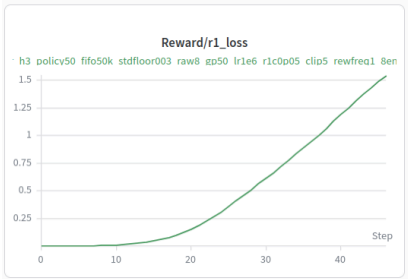


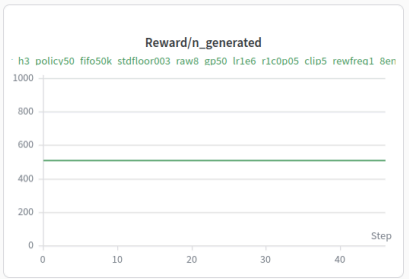


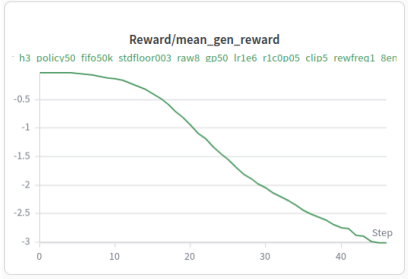


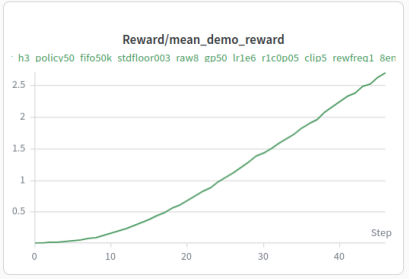
Reward 22 1-6 of 22

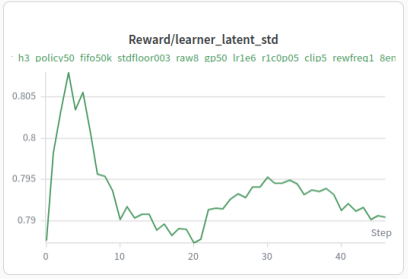
Reward 22 1-6 of 22







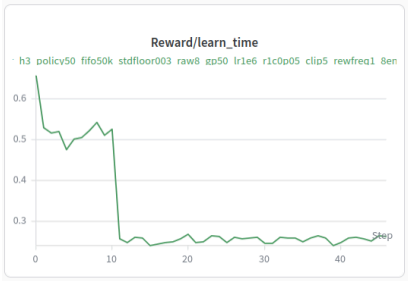


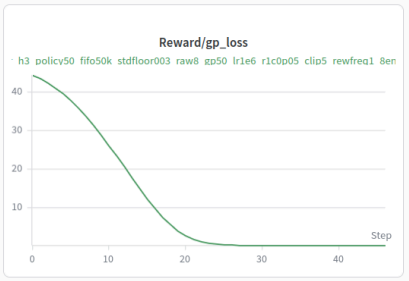


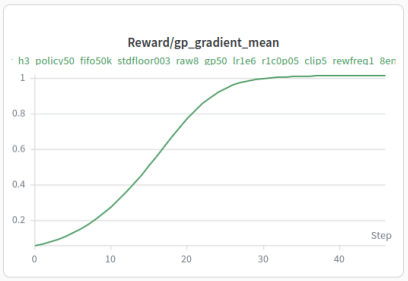
Storage 6

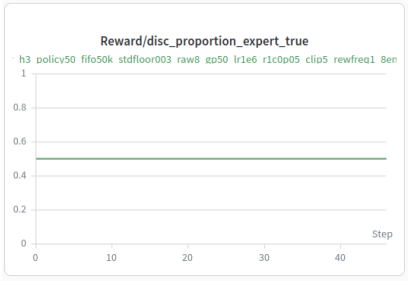
Training 2

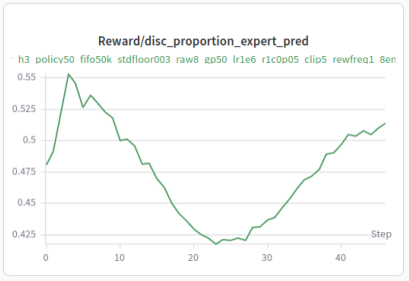
Reward 22 7-12 of 22

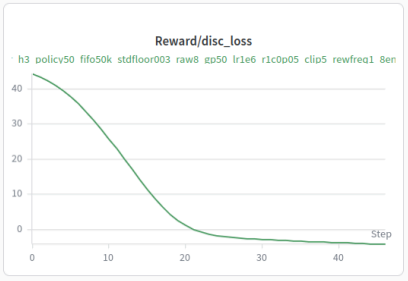








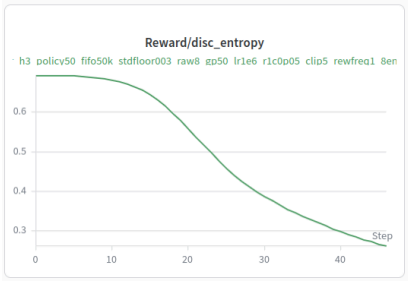


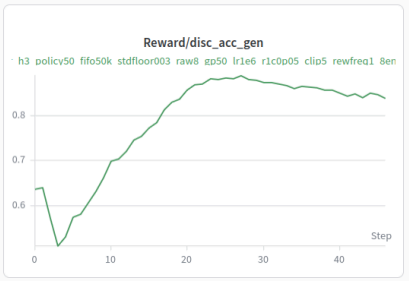


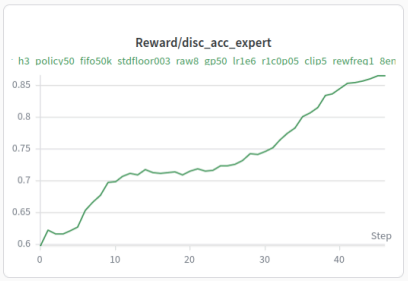
Storage 6

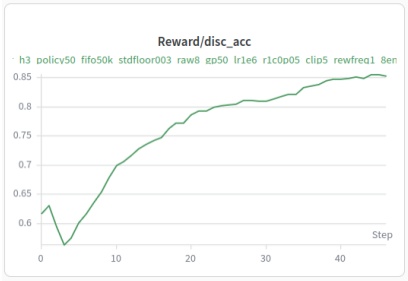
Training 2

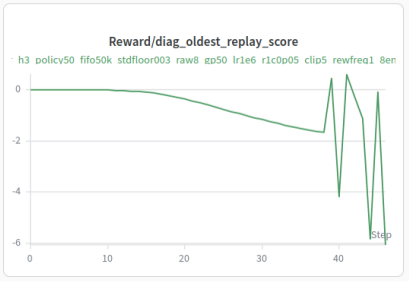
Reward 22 13-18 of 22

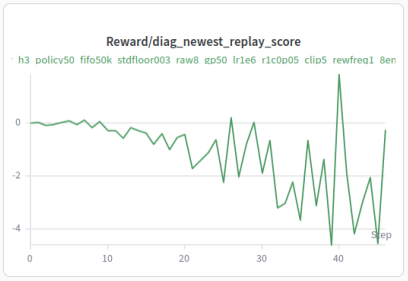












Storage 6

Training 2

深入的根据算法训练逻辑分析这次的现象，证明了什么，现在这个实验结果把整个训练算法的逻辑链条推到了哪一步？

Monday 10:58 AM

这次 S20 结果非常有诊断价值。它把问题从“value/MPPI 是不是主因”进一步推到了：

现在主断点在 dynamics/action-conditioning 和 reward 可规划性之间，而不是单纯 action 太小、value terminal、horizon、gp/r1。

先说结论。

S20 证明了三件事：

1. action manifold 问题存在，但不是唯一根因。
   加 BC action prior 后，policy 动作确实更像 expert，动作幅度也明显变大；所以之前 policy/MPPI 的动作分布确实偏离 expert action manifold。
2. 但即使动作更像 expert，任务仍然没有成功，说明 planner 仍然没有把“expert-like action”转化成真实 progress。
   这排除了“只是动作太小 / policy 不像 expert”作为主因。
3. 真正断点更靠近 learned dynamics：模型没有学到 action 会把 latent state 推向 demo transition direction。
   离线 evaluator 里最关键的是：`dyn_demo_delta_cos` 对 zero、policy、teacher action 都差不多，约 `0.09`。这说明 dynamics 预测的 next latent direction 基本不随 action 变成 demo-like direction。

按算法链条拆

MPAIL 这套闭环是：

`env rollout -> replay -> dynamics/reward/value/policy update -> MPPI sampling -> env action -> 新 replay`

S20 之前我们已经验证：

- S18/S19：关掉 MPPI terminal value，h=3，value 不直接参与 planning，任务仍不进展。
  所以 terminal value 是放大器，不是根因。
- S20：加入 BC action prior，让 policy 更靠近 expert action。
  结果 action 确实变大、policy 更像 expert，但 success 仍为 0，MPPI/Average Reward 后期继续下降。

所以链条不是断在：

- `terminal value`
- `horizon`
- `policy action magnitude`
- `policy 是否能被 BC 拉向 expert`

而是断在：

`action -> learned dynamics next latent -> reward/value scoring`

为什么 value 继续崩

你截图里的 value 很典型：

- `Value/td_target_mean` 从接近 0 掉到约 -10
- `Value/mean_q_value` 同步掉到约 -7
- `Value/mean_loss` 后期涨到 14+

这说明 critic 不是简单过估计或欠估计，而是在追一个持续变差的 target。

代码上 value target 是：

```
td_target = calibrated_reward(s, a, s') + gamma * lambda_return(predicted future)
```

而 future 又来自：

```
policy plan -> dynamics rollout -> reward/value
```

如果 dynamics rollout 对 action 的任务相关方向不可靠，那么 value 学到的就是：“这些 replay 状态未来都很差，而且动作之间差别很小”。这和你的图完全一致：

- `ValueDiag/q_exec_minus_zero` 基本接近 0 或负
- `q_rand_minus_zero` 也只有 0.01 量级
- value 能整体变负，但不能学出“哪个动作更好”

所以 value 崩不是第一因，而是 dynamics/reward 给它的可控结构太弱之后的结果。

Reward 图说明什么

Reward 本身在分类层面是工作的：

- `Reward/mean_demo_reward` 上升到 2.8+
- `Reward/mean_gen_reward` 下降到 -3 左右
- `Reward/disc_acc` 上升到 0.85 左右
- `Reward/disc_entropy` 下降
- `gp_gradient_mean` 收敛到 1

这说明 WGAN discriminator 可以区分 demo 和 generated replay。

但问题是：

能分类 demo/replay，不等于能给 MPPI 提供一条可执行梯度路径。

你看 `Reward/diag_newest_replay_score` 后期剧烈抖动，`oldest_replay_score` 也出现大幅负值。这说明 reward 对 replay 分布越来越像“支撑集分类器”：demo 高、replay 低，但在 replay 附近不同 action rollout 之间没有稳定、平滑、可规划的改进方向。

所以 reward 不是完全坏，而是“不够可规划”。

MPPI 图最关键

S20 里：

- `MPPI/Average Value = 0`，说明 terminal value 已经真的关掉
- `MPPI/Average Reward` 后期一路下降到约 -3
- `MPPI/Std Reward` 上升
- `MPPI/Mean Policy Influence` 上升到约 0.04
- `MPPI/Max Reward` 偶尔能 spike，但 `Max Return/Min Return` 整体恶化

这说明：

BC prior 让 policy proposal 终于有一点进入 elite，但 MPPI 仍然没有找到稳定高 reward path。它只是偶尔采到 reward spike，不是稳定找到真实可执行轨迹。

这和 offline evaluator 完全对上：

- policy/teacher action 的 reward_dyn 只比 zero 高一点点
- Q 几乎不区分 teacher/policy/zero
- dynamics predicted direction 和 demo direction cosine 都很低

S20 最重要的证明

S20 证明：

“planner 不动/失败”不是因为它完全没看到 expert-like actions。

现在即使把 *policy sampler* 拉向 *expert action*，*learned dynamics/reward/value* 仍然不能把这些 *action* 排序成有任务进展的轨迹。

也就是说，问题已经推进到：

```
demo/replay reward 分类可学
-> policy action prior 可学
-> 但 dynamics 对 action-conditioned demo-like transition 没学好
-> reward/value 在 model rollout 上无法区分 expert action vs zero/random
-> MPPI 只能在坏模型地形里找假高分或低质量动作
-> env success 仍为 0
```

下一步最严谨的实验方向

现在不该继续扫 `gp/r1/h/value_return_type`。下一步应该直接修/验证 dynamics 的监督来源：

1. 在 `update_dynamics()` 里加入 demo action dynamics loss。
   现在 demo transition 被拼进 encoder，但 demo action 是 zero placeholder，而且 JEP loss 只算 agent portion。也就是说 expert transition 没有用 expert action 监督 dynamics。
2. 用 `full_object_anywhere_bc_raw.pt` 的 `obs, action, next_obs` 加一项：

3. `dynamics(z_demo, a_demo) -> z_demo_next`

   作为 auxiliary demo dynamics loss。
4. 跑短实验，然后看 offline evaluator：
   - `teacher_dyn_demo_delta_cos` 是否明显高于 zero/random
   - `teacher_reward_dyn_mean - zero_reward_dyn_mean` 是否变大
   - `teacher_q_mean - zero_q_mean` 是否变大
   - MPPI/Average Reward 是否不再一路塌

如果这些指标改善，再看 env success。
如果这些指标不改善，就说明 latent encoder/reward geometry 本身不支持 action-conditioned planning，需要进一步约束 latent space。

Monday 10:58 AM
ok直接帮我跑下一步实验
Monday 11:01 AM
Worked for 31m 11s

S21 已经跑完，我没有再跑 S19。

我做了这一步实验：在 S20 的 expert-action policy prior 基础上，又加了 `demo_action_dyn_coeff=1.0`，让 dynamics 显式学习：

```
z_demo + a_demo -> z_demo_next
```

改动在 mpail2/learner.py 和配置里，启动脚本是 scripts/run_s21_demo_action_dynamics_h3_gp50_50.sh。

S21 运行目录：

```
logs/obs22_S21_demoActionDyn_h3_policy50_fifo50k_stdfloor003_raw8_gp50_lr1e6_r1
c0p05_clip5_rewfreq1_8env_50iter_20260525_111039
```

checkpoint 已保存：

```
model_0.pt
model_25.pt
model_50.pt
```

离线评估也跑完了：

```
logs/offline_checkpoint_eval_s21_model25_teacher_actions.csv
logs/offline_checkpoint_eval_s21_model50_teacher_actions.csv
```

关键结果：

```
model25:
policy_teacher_raw_cos        0.484
teacher_dyn_demo_delta_cos    0.1197
zero_dyn_demo_delta_cos       0.1187
teacher_reward_dyn - zero     +0.0075
```

```
teacher_q - zero              +0.0052

model50:
policy_teacher_raw_cos        0.568
teacher_dyn_demo_delta_cos    0.1388
zero_dyn_demo_delta_cos       0.1364
teacher_reward_dyn - zero     +0.0150
teacher_q - zero              -0.0100
```

结论很清楚：S21 证明了 action prior 和 demo-action dynamics loss 都确实在优化，`dyn_loss` 从约 `3.8` 降到 `0.63`，policy 也更接近 teacher action。但这仍然没有让 planner 在 latent dynamics 里把 `teacher action` 和 `zero/random/policy action` 明显区分开。

所以现在训练逻辑链条被推到这里：

不是 value terminal bootstrap
不是 horizon h=3/h=7
不是单纯 policy action manifold
也不是 reward 完全不会分 demo/gen

当前断点更像是：
learned dynamics 的 latent transition 虽然对动作幅度有响应，
但没有学出“专家动作会朝 demo next latent 方向走”的可规划方向。
因此 reward/Q 在 planner 假想转移上几乎无法偏好 teacher action。

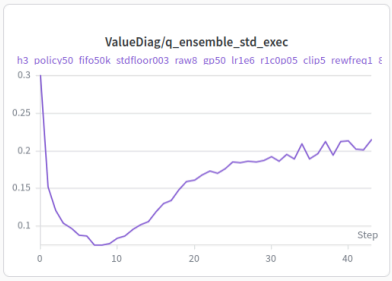
换句话说：S21 把问题从“policy 没给出专家动作”进一步排除掉了。现在更应该查 `encoder + dynamics` 的训练目标是否在制造一个“动作可变但方向不可用”的 latent space，下一步实验应直接冻结/预训练 dynamics 或改成监督真实 demo transition 的更强约束，而不是继续调 value。后台我也确认了，没有 S19/S20/S21 训练进程残留。

Search panels with regex

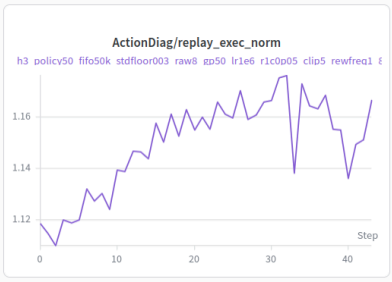
Settings | New report | Add panels

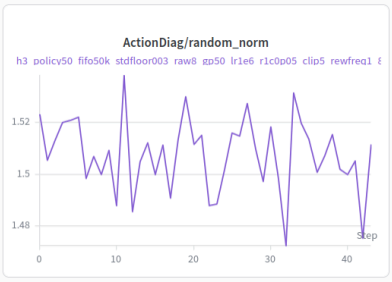
ValueDiag 7

7-7 of 7



ActionDiag 2





Action 30

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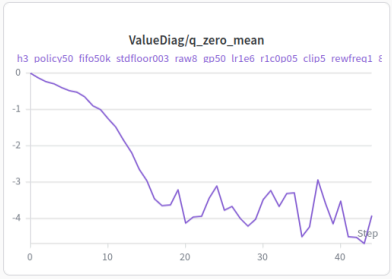
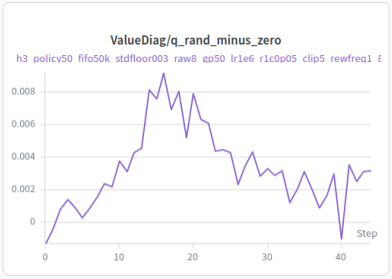
Search panels with regex

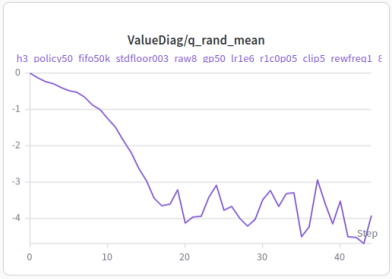
Settings | New report | Add panels

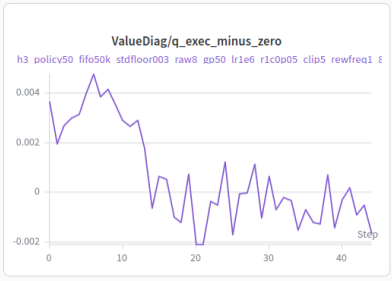
ValueDiag 7

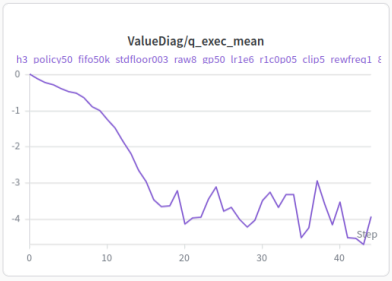
1-6 of 7

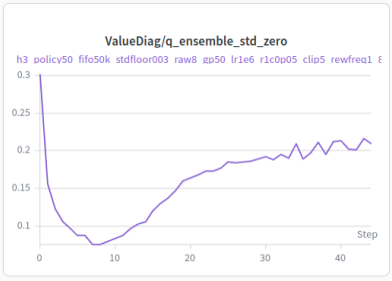






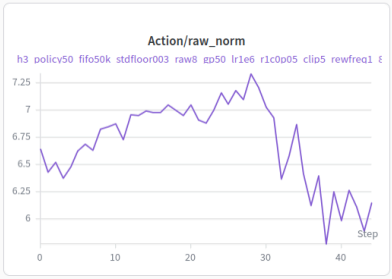


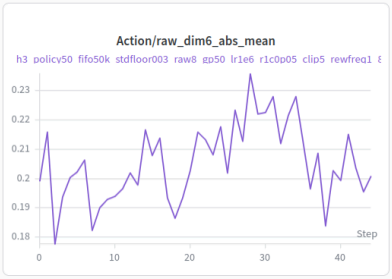


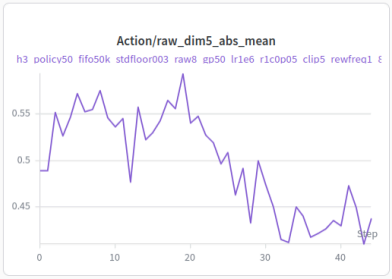


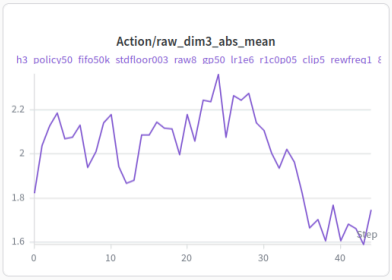
ActionDiag 2

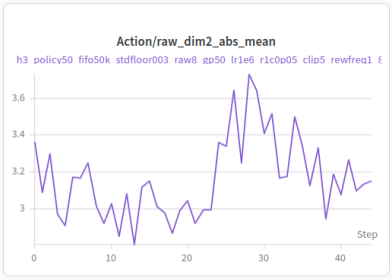
Action 30 1-6 of 30







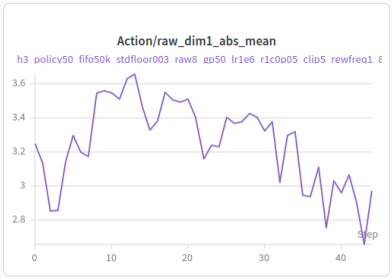


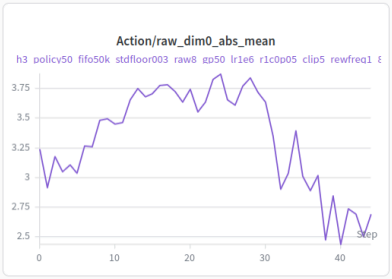


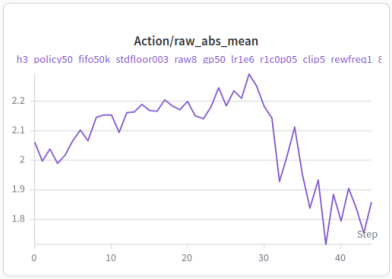
Charts 2

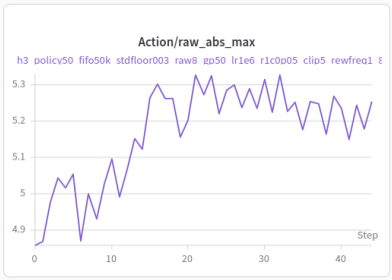
Data 2

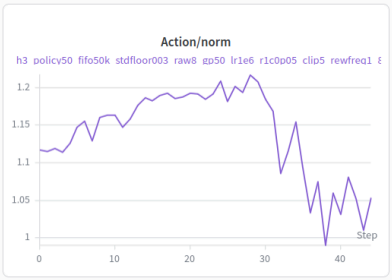
Action 30 7-12 of 30

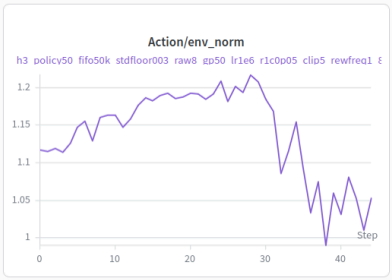












Charts 2

Data 2

Action 30

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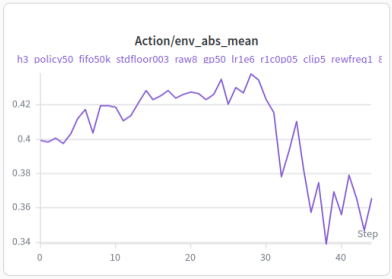

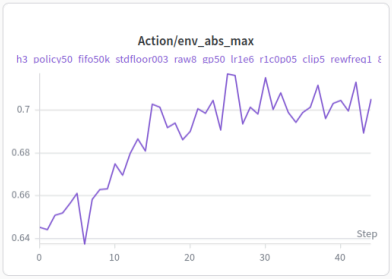

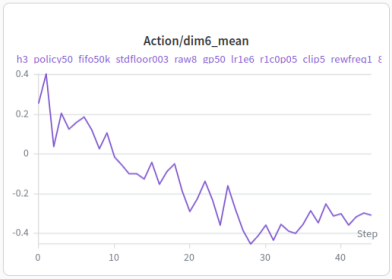

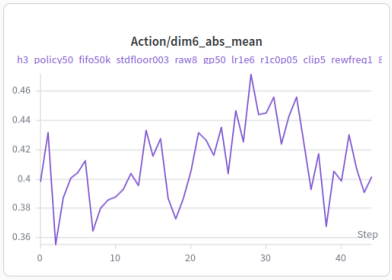

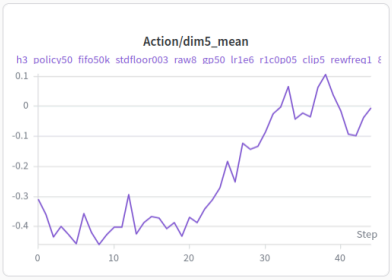

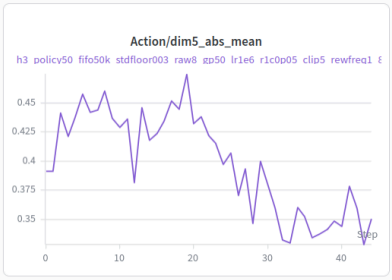

Charts 2

Data 2

Action 30

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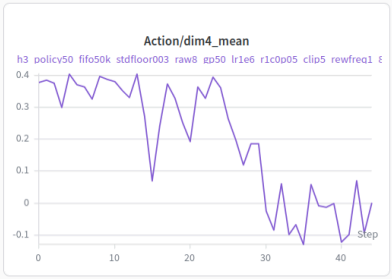
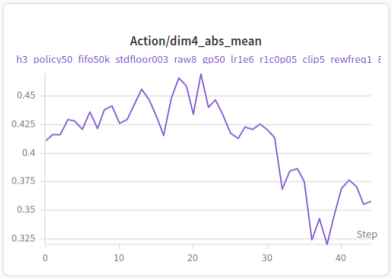
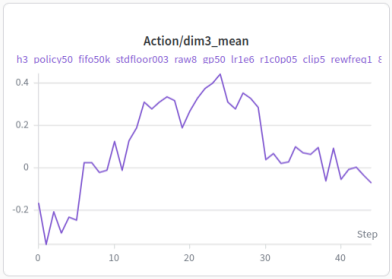
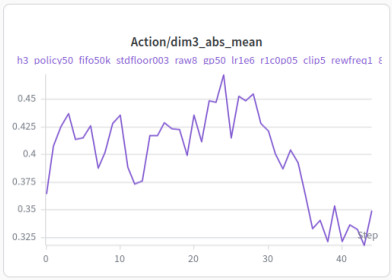
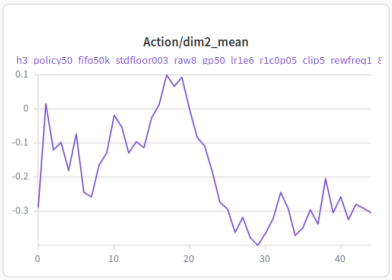
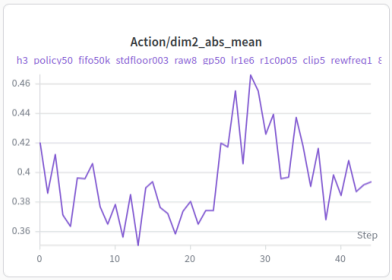








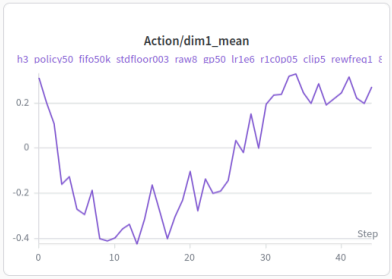


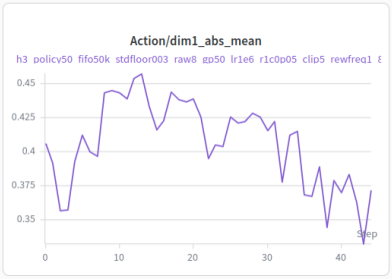


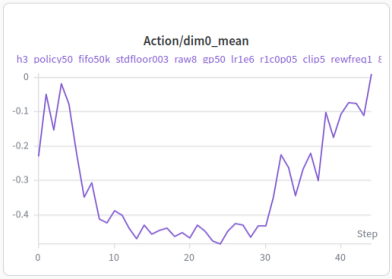
Charts 2

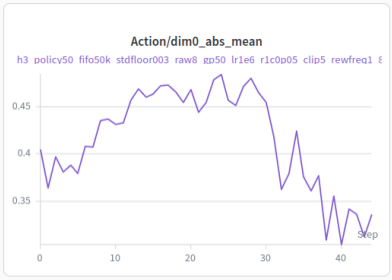
Data 2

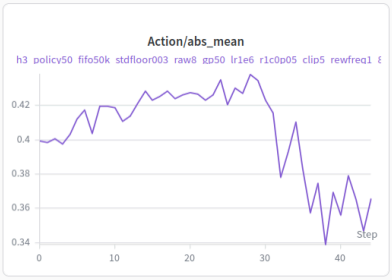
Action 30 25-30 of 30

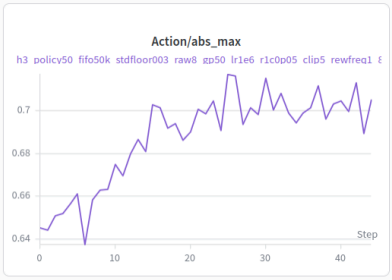












Charts 2

Data 2

Charts 2

Data 2

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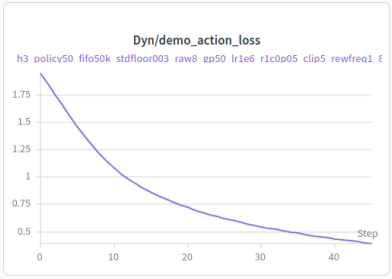

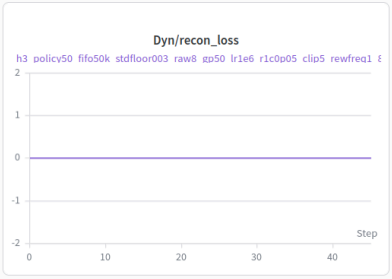

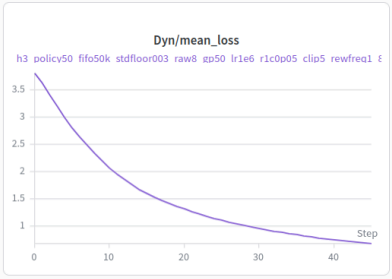

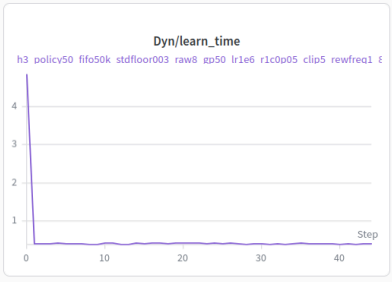

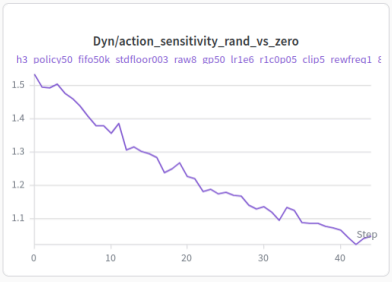

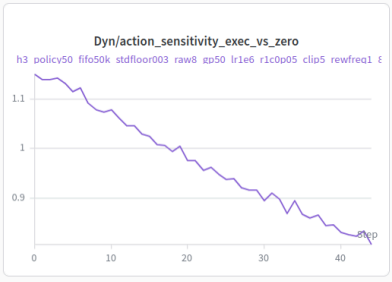

Episode_Reward 8

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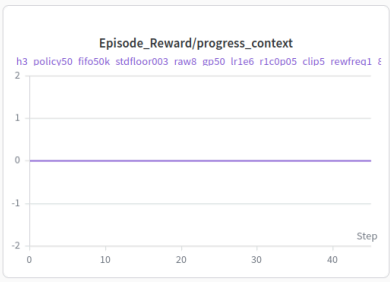

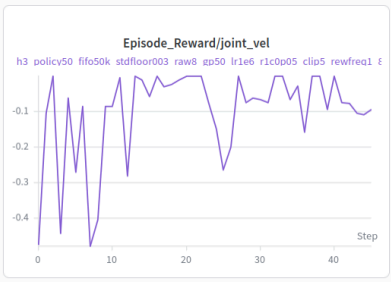

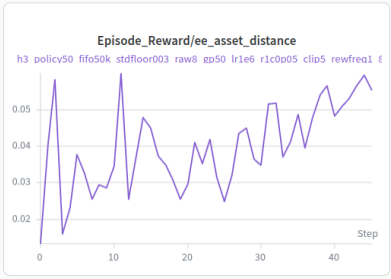

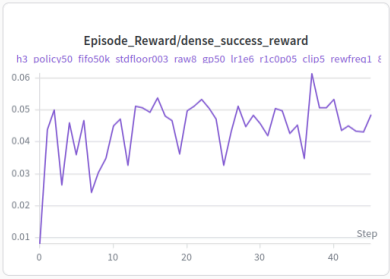

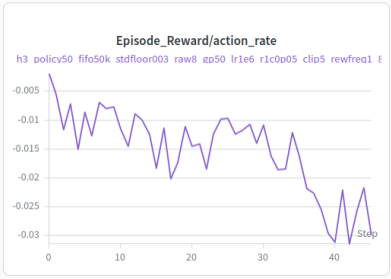

Episode_Termination 2

Episode_Reward 8

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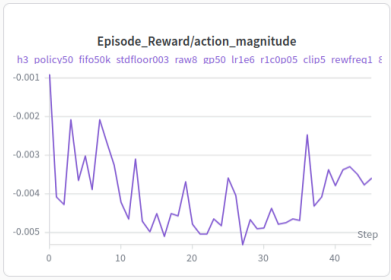
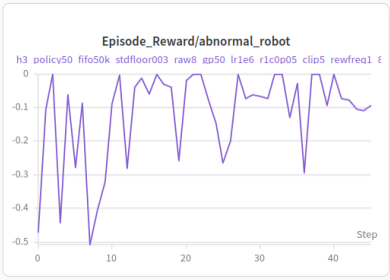
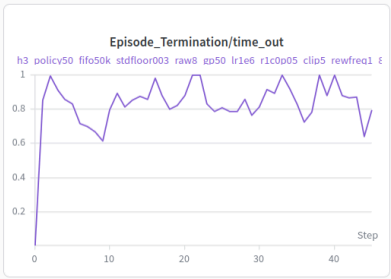
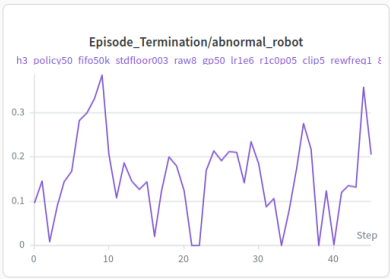




Episode_Termination 2





Metrics 5

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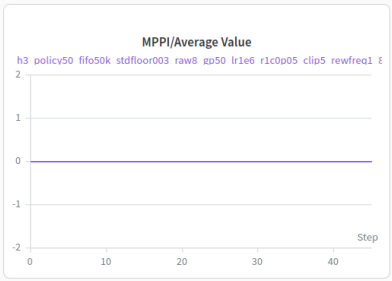

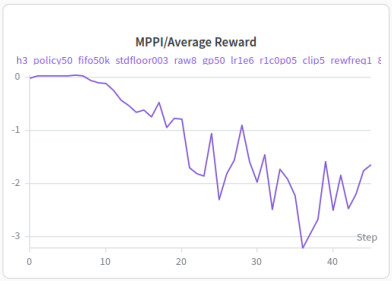

Perf 3

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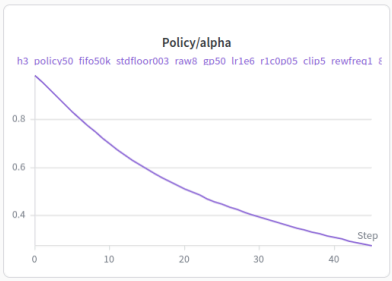
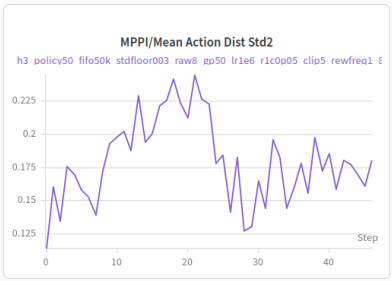
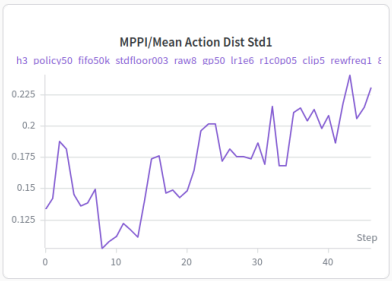
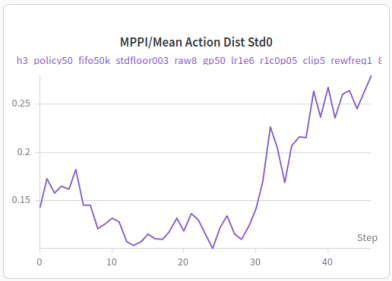
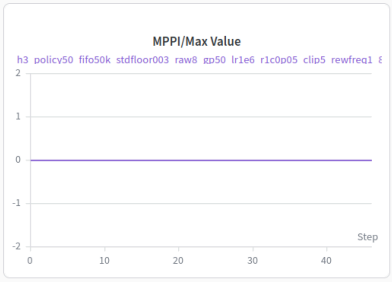


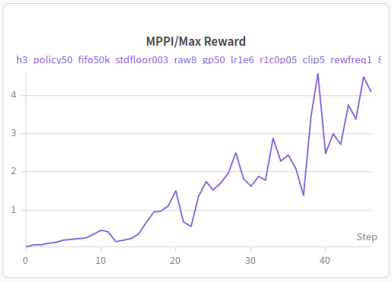
MPPI 20 13-18 of 20

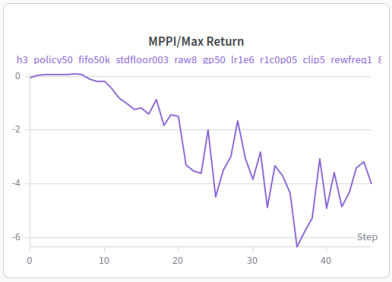












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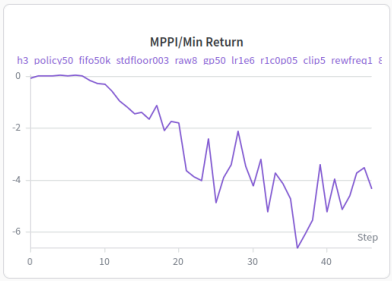

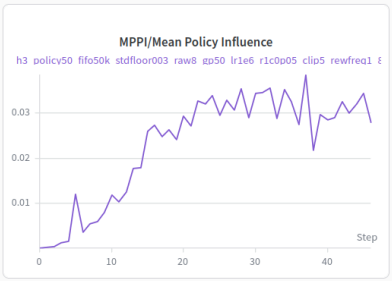

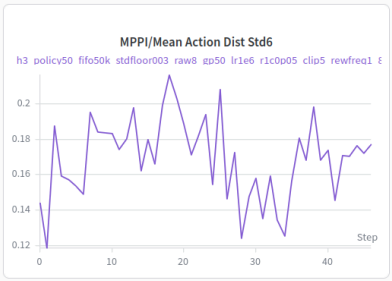

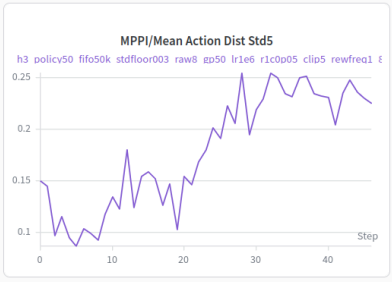

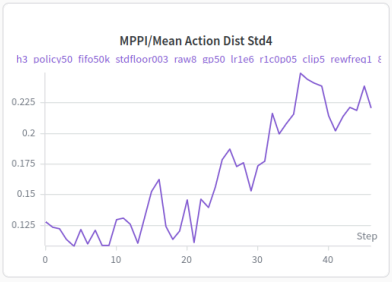

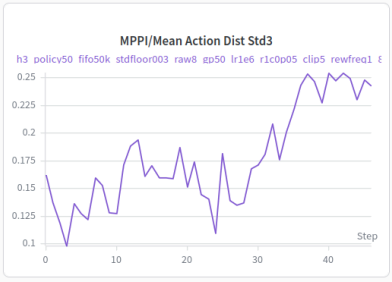

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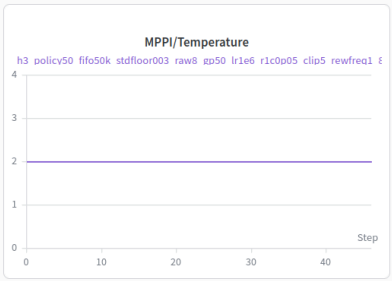
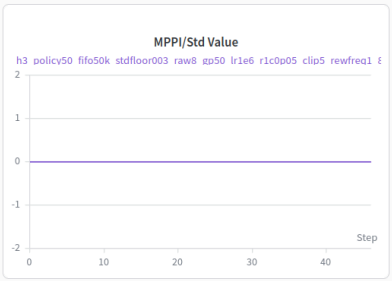
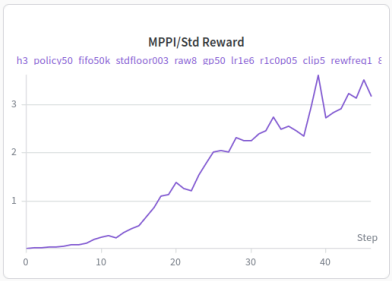
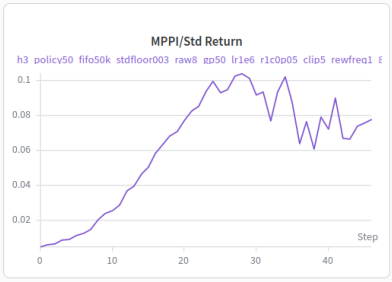
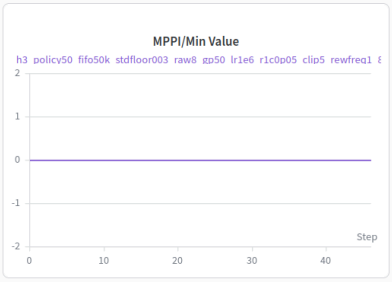
MPPI 20 1-6 of 20

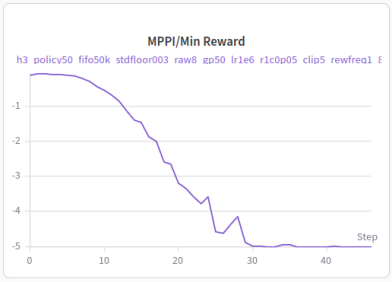












Perf 3

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Perf 3

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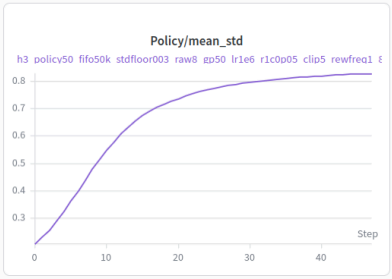
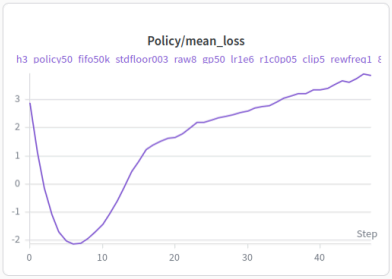
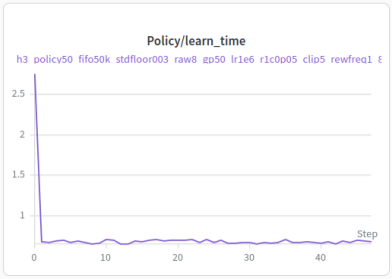
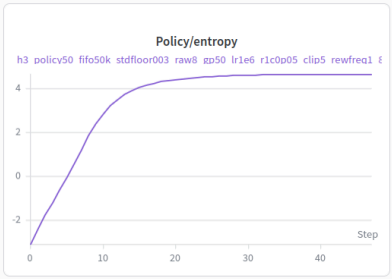
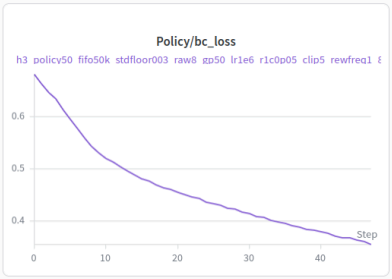
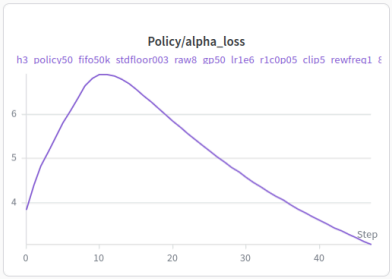






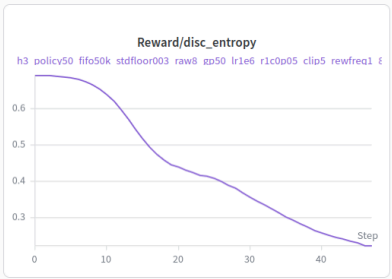


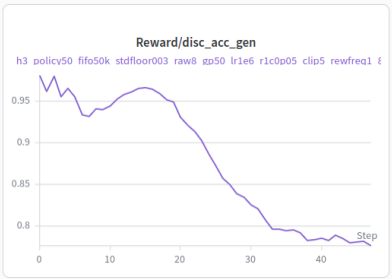


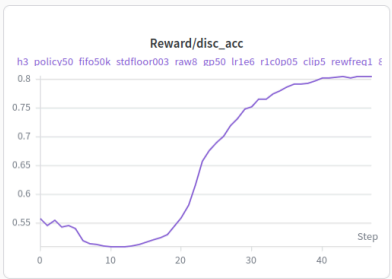


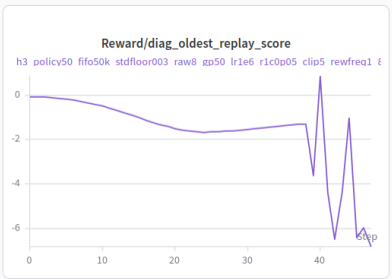
Reward 22 13-18 of 22

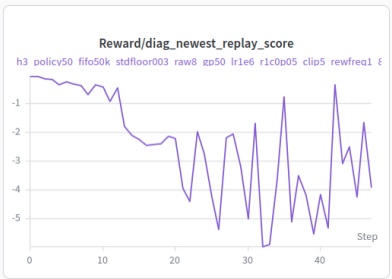
Reward 22 13-18 of 22



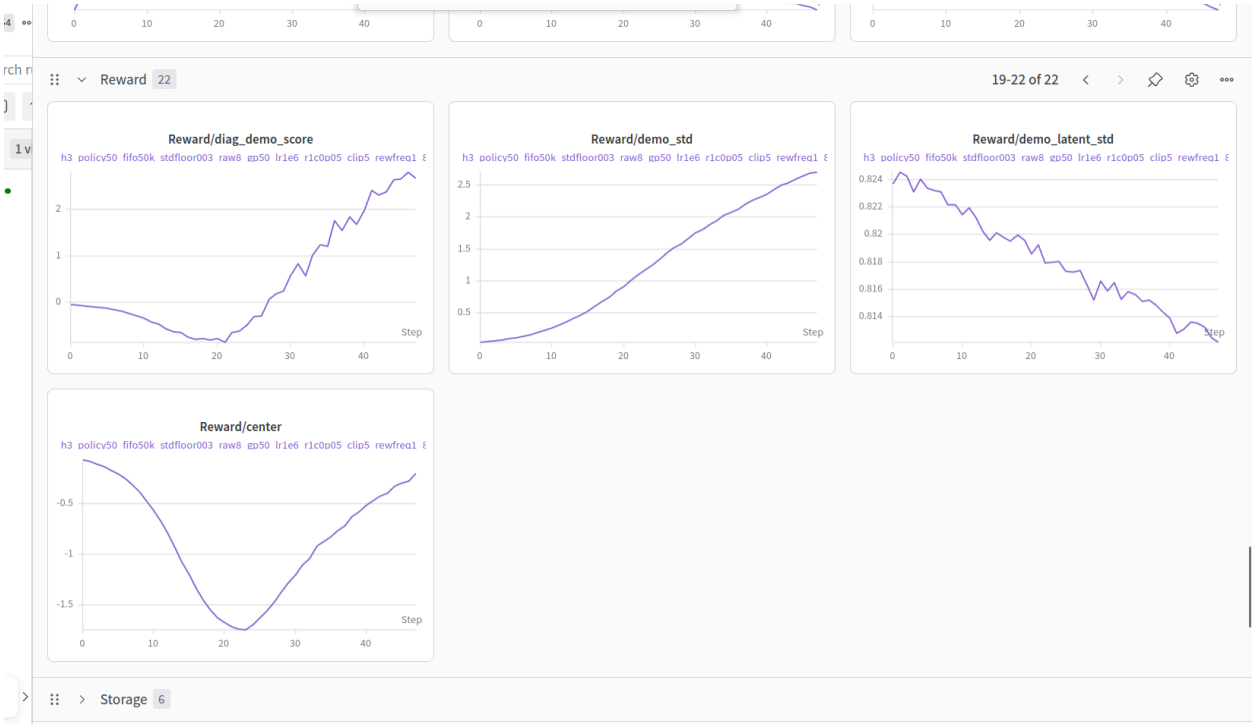


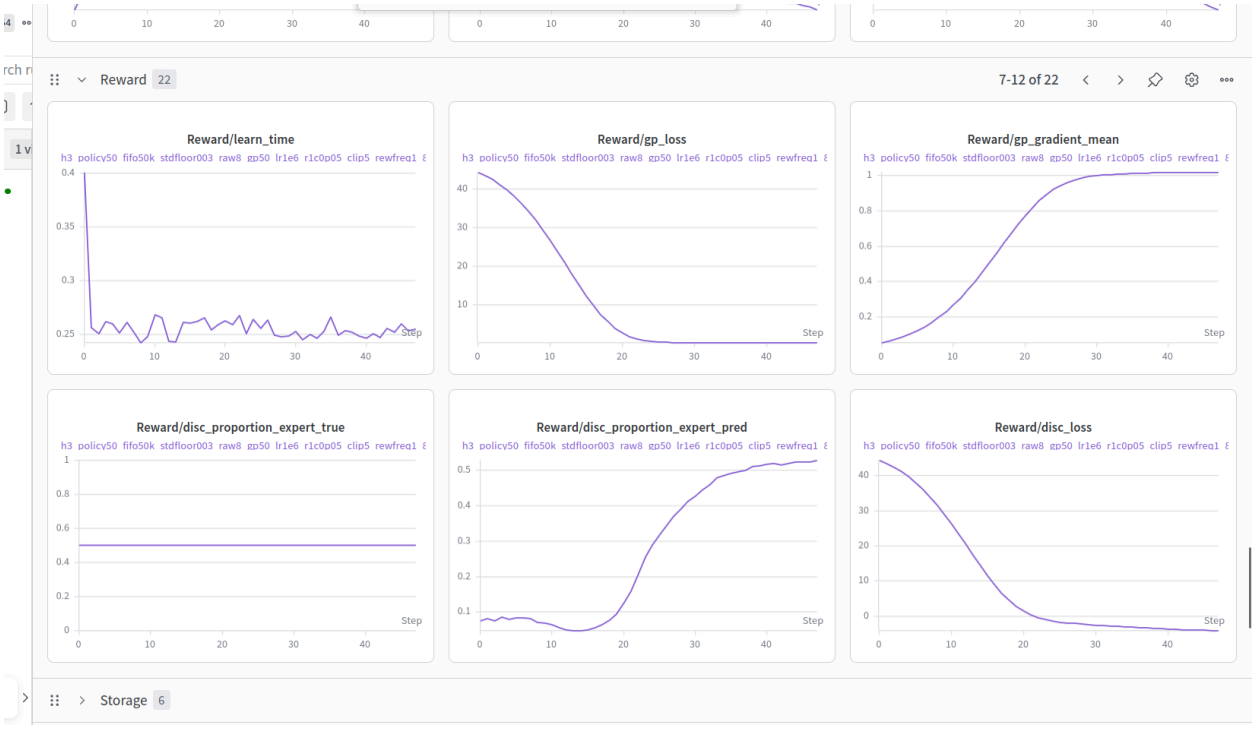


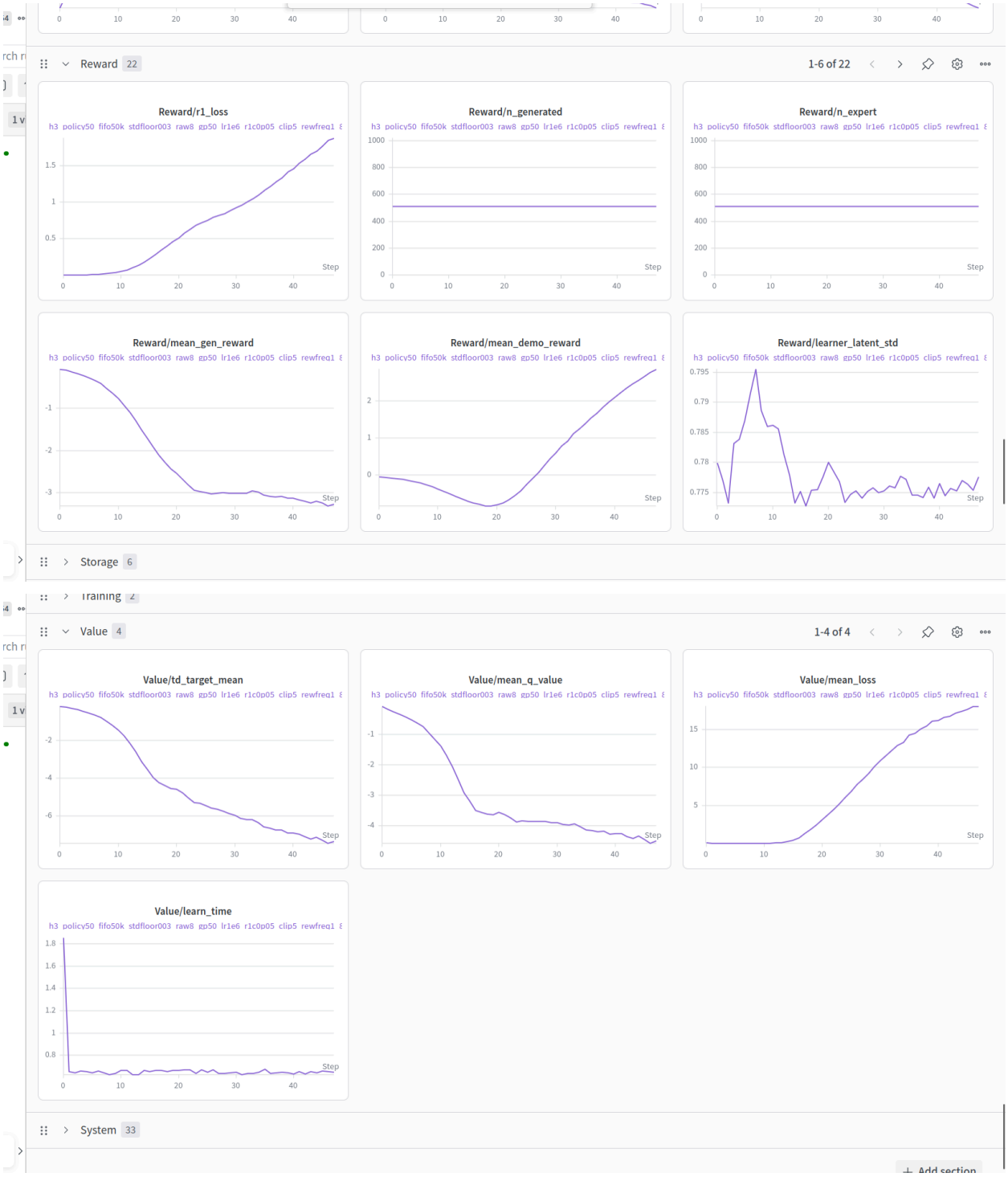




Storage 6


Reward 22
19-22 of 22
Reward/diag_demo_score
Reward/demo_std
Reward/demo_latent_std
Reward/center
Storage 6


Reward 22
7-12 of 22
Reward/learn_time
Reward/gp_loss
Reward/gp_gradient_mean
Reward/disc_proportion_expert_true
Reward/disc_proportion_expert_pred
Reward/disc_loss
Storage 6



深入的根据算法训练逻辑分析这次的现象，证明了什么，现在这个实验结果把整个训练算法的逻辑链条推到了哪一步？

Monday 11:33 AM

这次 S21 的现象非常关键，它不是“又失败了一次”这么简单，而是把问题链条又往后推了一层。

核心结论先说：S21 证明了 policy action prior 和 demo-action dynamics supervision 都能被优化，但它们没有把 MPPI 的规划目标变成可执行的专家方向。也就是说，问题已经不是 policy 不会输出类似 demo 的动作，而是 learned dynamics / latent transition / reward-planning interface 没有形成“专家动作优于零动作/随机动作”的可规划结构。

从训练逻辑逐层看：

1. Reward 学会区分 demo 和 replay 了

截图里 reward 信号非常明显：

`Reward/mean_demo_reward` 从负值一路升到正值，接近 `2.5+`。

`Reward/mean_gen_reward` 掉到约 `-3`。

`Reward/diag_demo_score` 也从负转正，后面升到 `2+`。

`disc_acc` 升到约 `0.8`，`disc_entropy` 持续下降。

这说明 reward/discriminator 不是完全坏掉的。它至少在真实 demo latent transition 和当前 replay/generated transition 之间建立了区分。

所以现在不能再说“reward 学不会 demo”。它会分。

2. 但是 reward 分得越开，MPPI/Average Reward 反而下降

MPPI 图里：

`MPPI/Average Reward` 从接近 0 降到约 `-2 ~ -3`。

`MPPI/Max Reward` 上升，说明采样里偶尔有高分 trajectory。

但 `MPPI/Average Reward`、`Min Reward`、`Return` 大面积变差。

这表示 planner 的候选轨迹分布整体没有进入 reward 认为好的区域。它只是偶尔采到高 reward，但 MPPI 加权更新没有稳定把动作推过去。

这一步非常重要：reward 已经有判别能力，但 MPPI 不能利用它形成稳定行动。

3. Value 不是这次的主因

这次依然是 `terminal_value_scale=0`，所以 MPPI 规划里 terminal value 被关掉了。

但 `Value/td_target_mean` 和 `Value/mean_q_value` 仍然一路变负，`Value/mean_loss` 上升。这说明 value 训练还是在崩，但由于 MPPI 这次不靠 terminal value，这不是导致 MPPI reward 下降的直接原因。

因此链条里 value bootstrap 已经基本排除为当前主根因：

不是 `value terminal` 把 `planner` 拉坏

value 现在更像是被坏 replay / 坏 learned transition 拖着走的结果，而不是第一推动因。

4. Policy prior 确实生效，但没有解决执行

Policy/bc_loss 持续下降，说明 action prior 真的把 policy sampler 往 expert action 拉了。离线 evaluator 也显示：

```
policy_teacher_raw_cos:
S20/S21 逐步升到约 0.56
```

也就是说，policy 不是完全乱动了。它确实更像 teacher action。

但环境成功率仍然是 0，Episode_Reward/success_reward 和 progress_context 都是 0。
这说明“动作长得像 demo”不足以产生任务进展。

所以链条进一步排除：

```
不是单纯 policy 输出不在 expert action manifold
```

5. S21 最关键：demo_action_loss 降了，但 teacher action 仍没有被 dynamics/reward/Q 明显偏好

这次新增的 Dyn/demo_action_loss 从约 1.9 降到 0.4 左右，Dyn/mean_loss 也从 3.7+ 降到 0.6+。表面看，demo-action dynamics supervision 在优化。

但离线 checkpoint evaluator 直接揭穿了关键问题：

```
model50:
teacher_dyn_demo_delta_cos = 0.1388
zero_dyn_demo_delta_cos    = 0.1364
random_dyn_demo_delta_cos  = 0.1360
policy_dyn_demo_delta_cos  = 0.1373
```

这四个几乎一样。

这句话的含义非常重：即使给 dynamics 输入专家动作，它预测出来的 latent delta 方向也没有比 zero/random/policy 更接近真实 demo next latent delta。

再看 reward/Q：

```
teacher_reward_dyn - zero_reward_dyn ≈ +0.015
teacher_q - zero_q ≈ -0.010
```

差距小到 planner 根本没法稳定利用，Q 甚至不认为 teacher 比 zero 好。

所以 S21 最硬的证明是：

```
demo-action dynamics loss 能下降，
但没有学出“专家动作 -> 专家方向”的可规划 latent 几何。
```

6. 这把算法链条推到了哪里

现在链条大概是这样：

```
demo data 正常存在
  ↓
reward 能区分 demo transition 和 replay transition
  ↓
```

```
policy prior 能让 policy action 更像 expert action
  ↓
demo-action dynamics loss 也能下降
  ↓
但是 learned dynamics 下：
teacher / policy / random / zero action 产生的 predicted next latent
在"是否接近真实 demo next latent"上几乎不可分
  ↓
reward/Q 在 planner 的 imagined transition 上无法稳定偏好 expert action
  ↓
MPPI 只能看到噪声化/扭曲的 planning score
  ↓
执行没有 progress，success=0
```

所以当前根因已经不在 reward/value/policy 的单独模块，而是在更底层的接口：

```
encoder + dynamics 的 latent transition geometry
```

尤其是这两个方向：

```
1. encoder 学出的 latent 空间不保证真实任务进展方向可由 action-conditioned dynamics 表达
2. dynamics loss 虽然数值下降，但可能主要学了状态分布/平均漂移，而不是动作条件下的真实 demo delta
```

一句话总结：S21 证明了"把专家动作塞进 policy 和 dynamics"还不够，因为 planner 看到的模型世界里，专家动作并不比零动作更像专家转移。现在要查的是 latent dynamics 的可控性和动作因果性，而不是继续调 value 或 reward 权重。

Monday 11:33 AM

ok 接下来的实验怎么做（详细的理解训练算法的代码逻辑之后再给出）？（因为我要回家了10个小时之后回来）你可以借这个机会多跑几个需要长时间的实验来做更进一步的排查

Monday 11:42 AM

Worked for 9m 50s

可以回家了。

现在 S22 已经正常跑起来，关键配置也确认生效了：`dyn_agent_coeff=0`，所以它确实是在测"只用 demo-action dynamics，不让 replay dynamics 污染"的分支。队列会自动继续跑：

```
S22 -> S23 -> S24
```

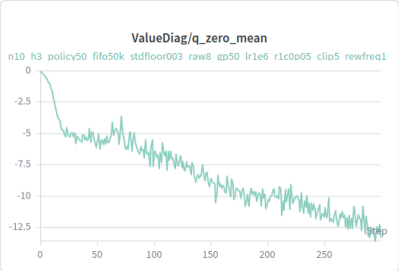
并在每个实验后自动评估 checkpoint。你 10 小时后回来，我们直接看队列日志和离线评估结果：

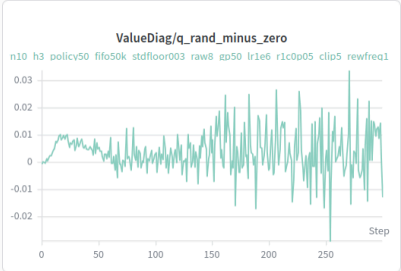
```
logs/long_dynamics_queue/s22_s24_queue_20260525_114800.log
logs/offline_checkpoint_eval_s22_model*_teacher_actions.csv
logs/offline_checkpoint_eval_s23_model*_teacher_actions.csv
logs/offline_checkpoint_eval_s24_model*_teacher_actions.csv
```

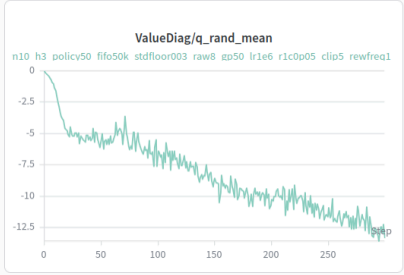
目前不用守着。

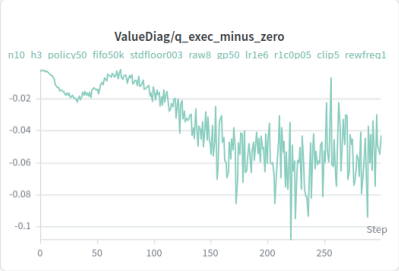
Search panels with regex

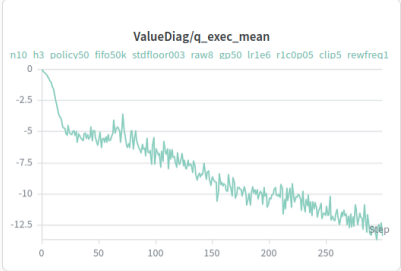
ValueDiag 7 1-6 of 7

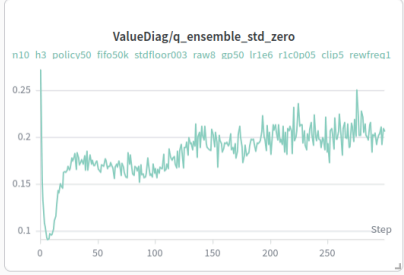












ActionDiag 2

Search panels with regex

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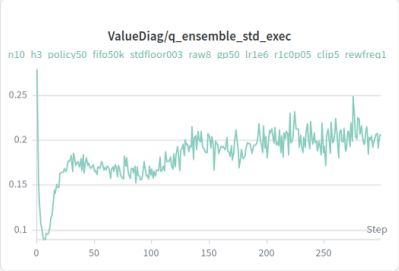
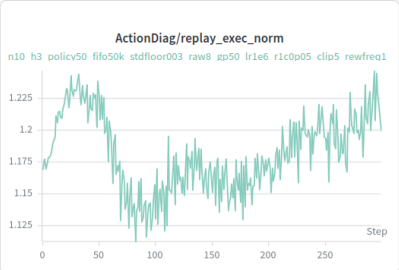
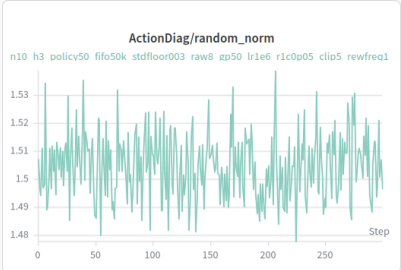


ActionDiag 2





Action 30 25-30 of 30

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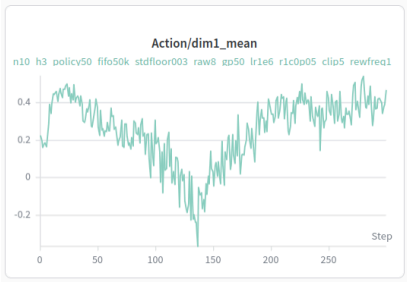

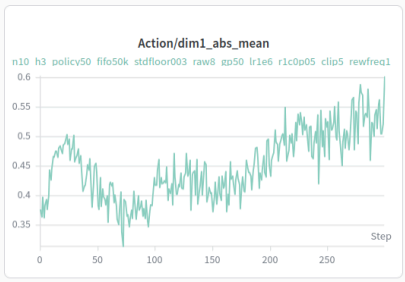

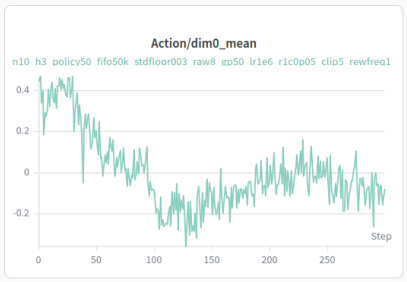

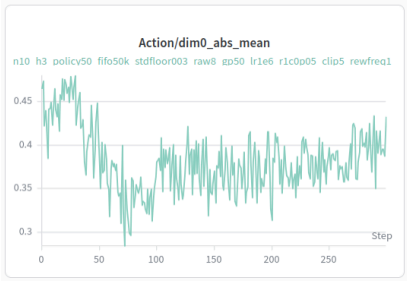

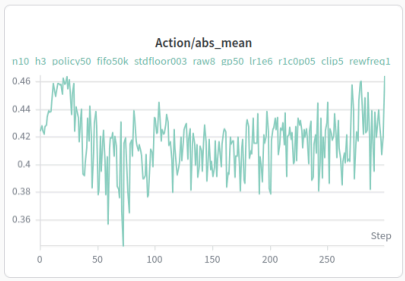

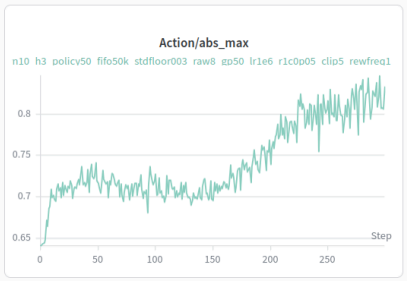

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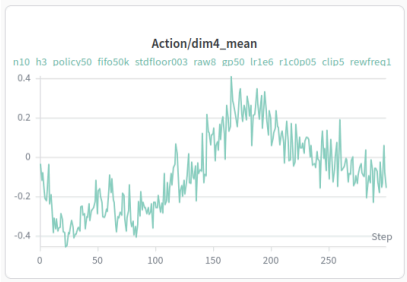

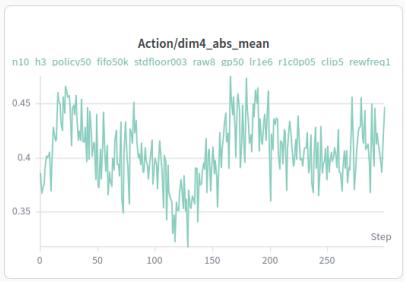

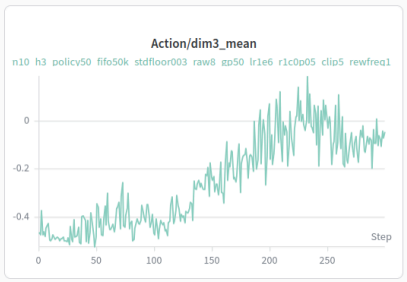

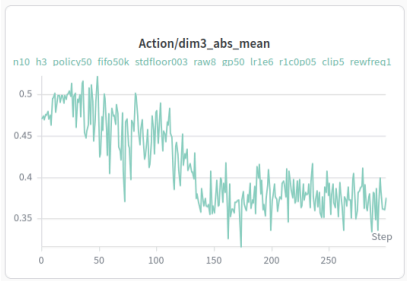

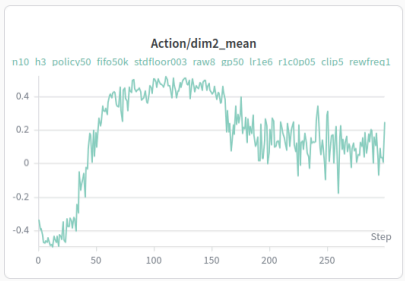

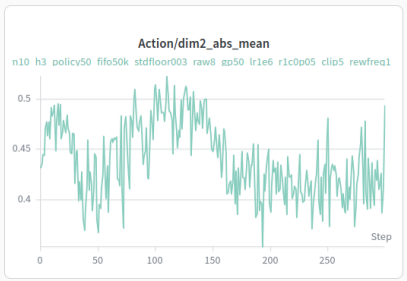

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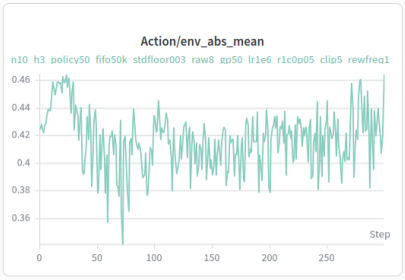

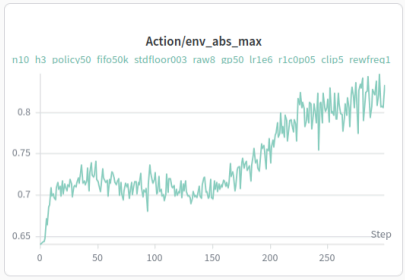

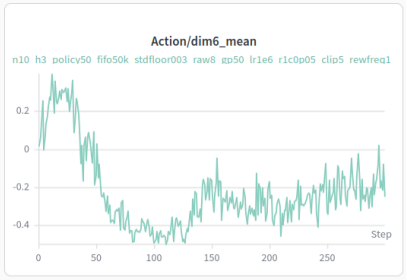

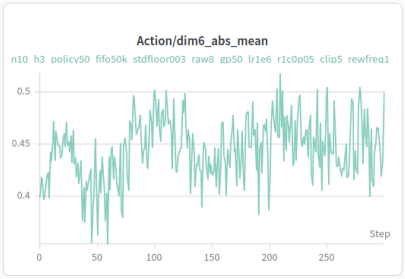

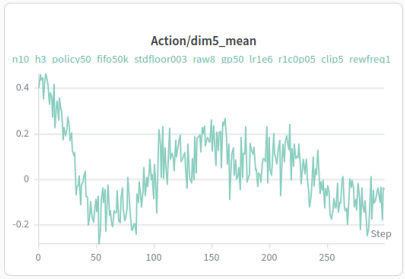

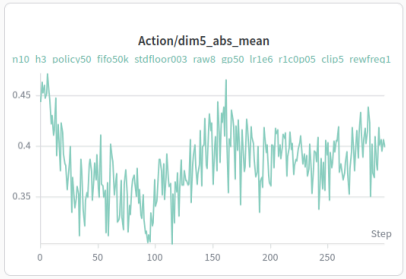

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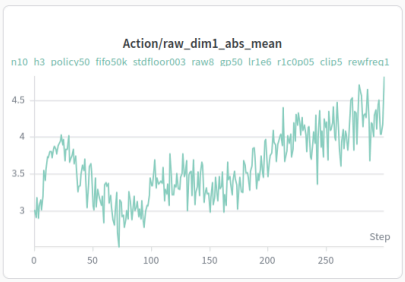

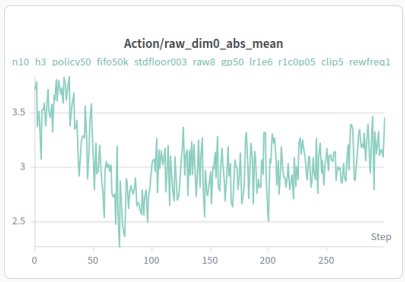

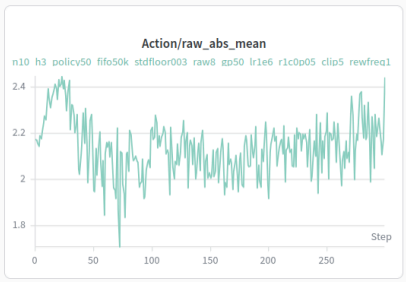

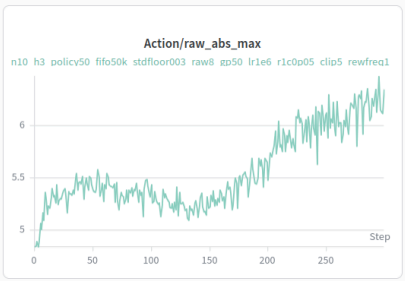

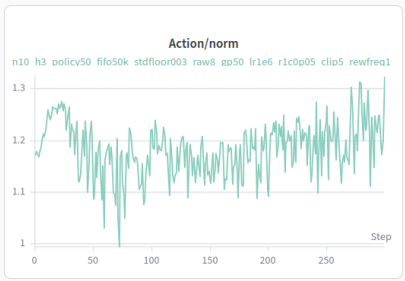

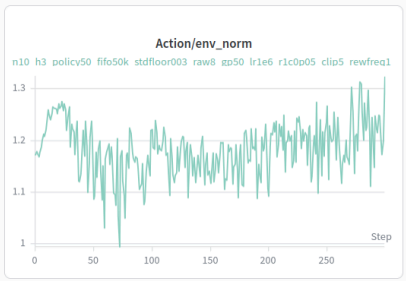

Action 30

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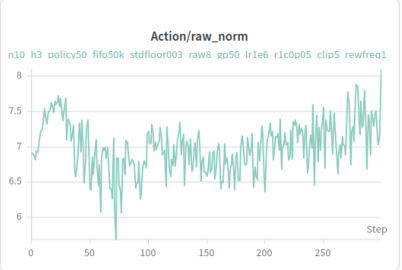

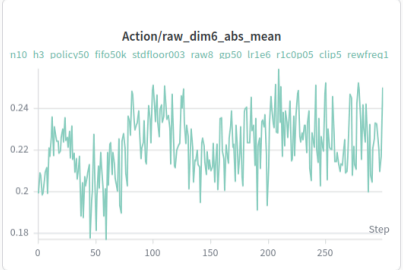

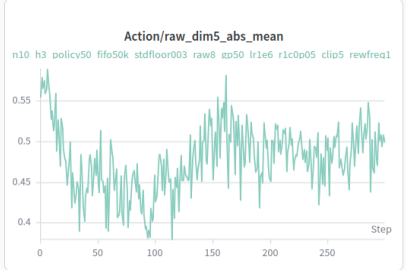

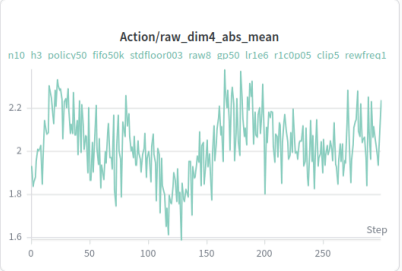

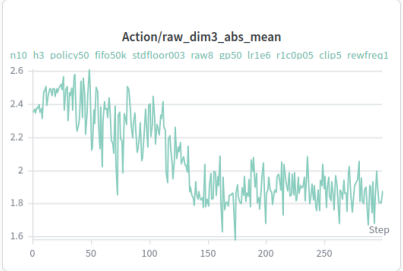

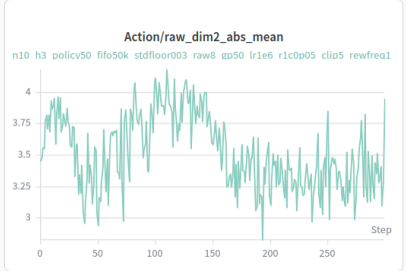

Data 2

Dyn 7

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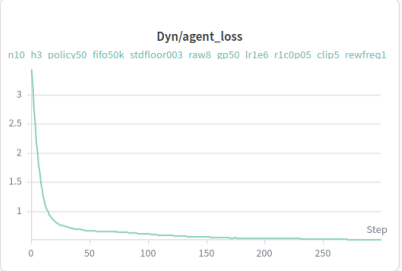
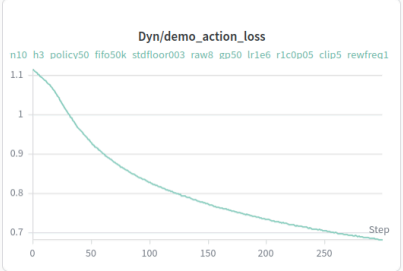
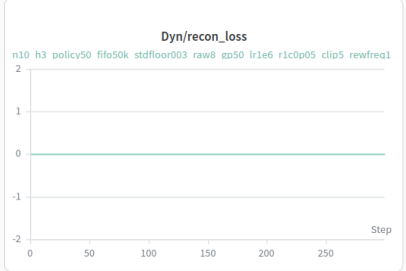
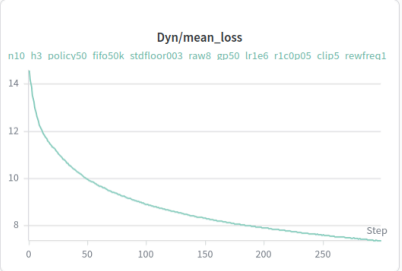
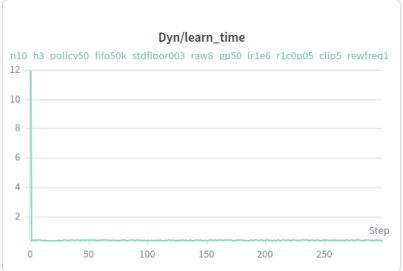
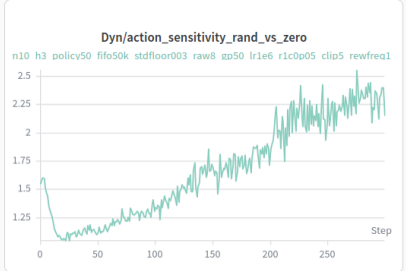








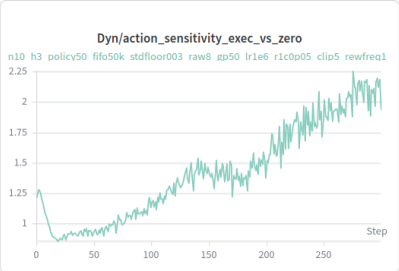




Env 11

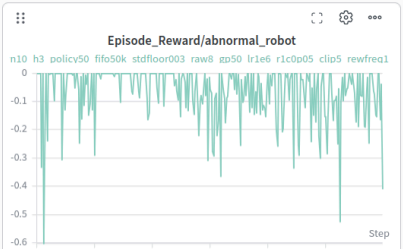
Data 2

Dyn 7 7-7 of 7

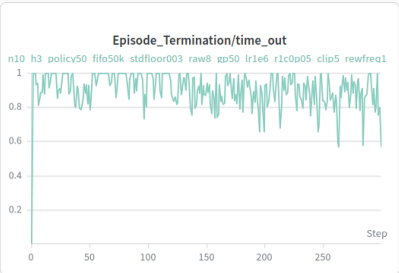


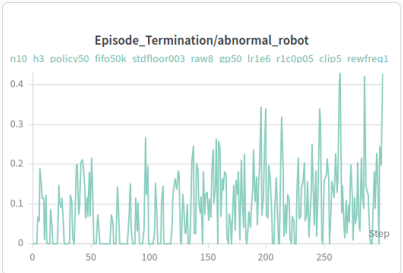
Env 11

Episode_Reward 8 7-8 of 8



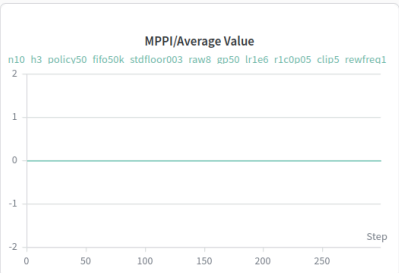
Episode_Termination 2

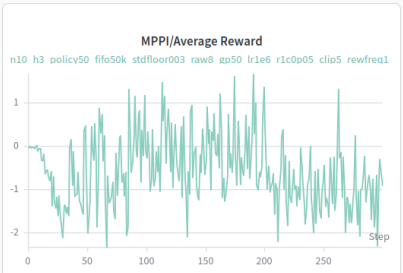




Metrics 5

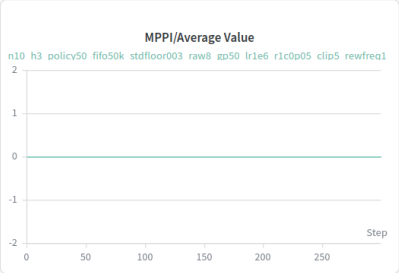
MPPI 20 19-20 of 20

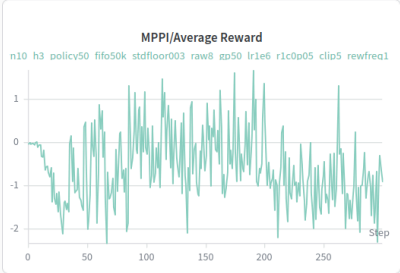




Metrics 5

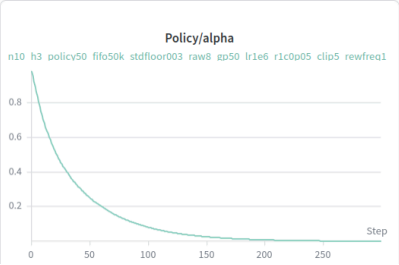
MPPI 20 19-20 of 20





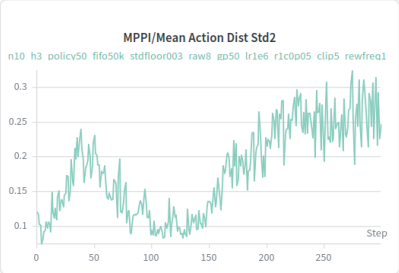
Perf 3

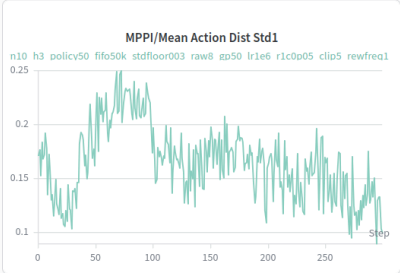
Policy 7 7-7 of 7

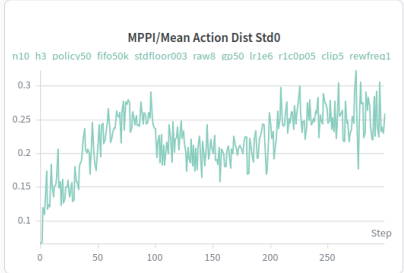


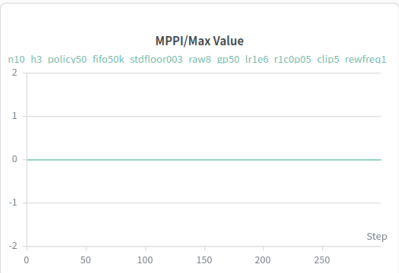
Metrics 5

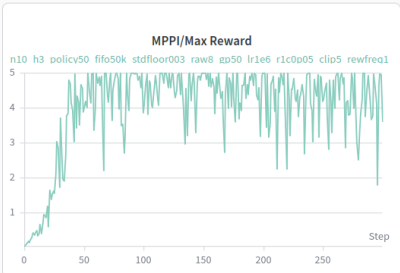
MPPI 20 13-18 of 20

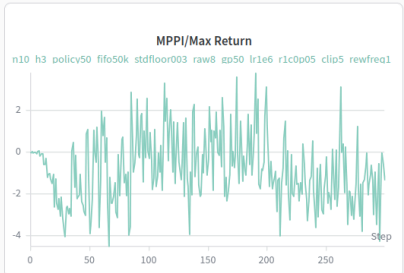








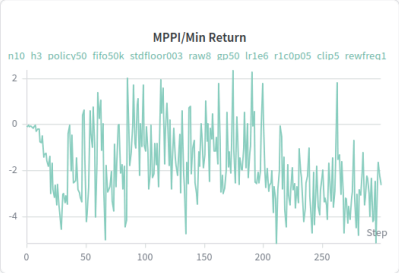


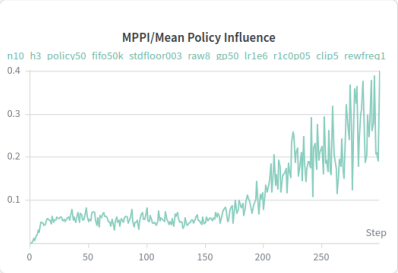


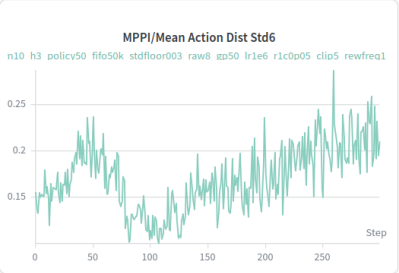
Perf 3

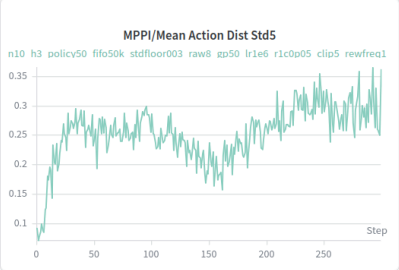
Metrics 5

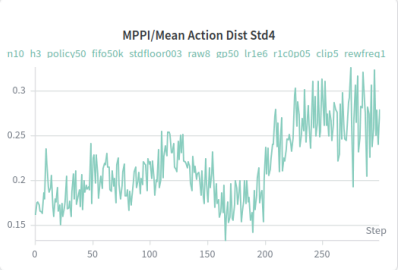
MPPI 20 7-12 of 20

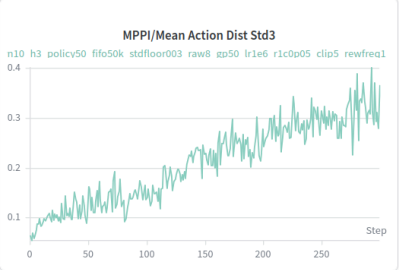








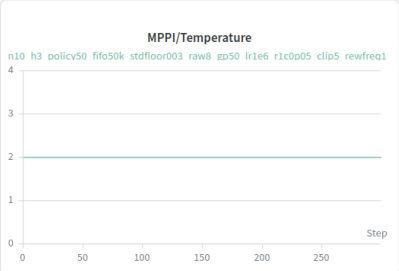


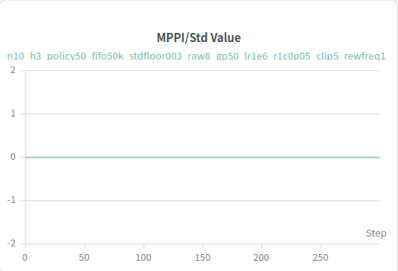


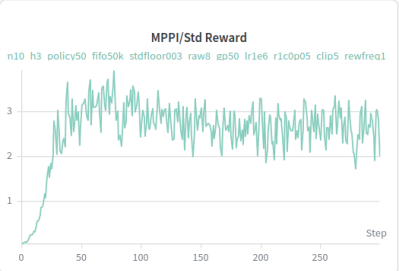
Perf 3

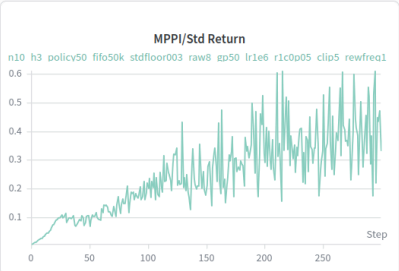
Metrics 5

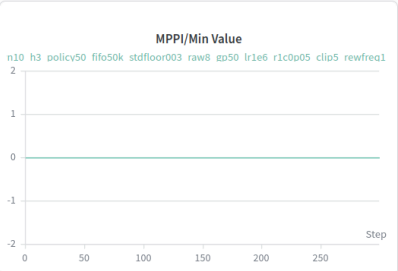
MPPI 20 1-6 of 20

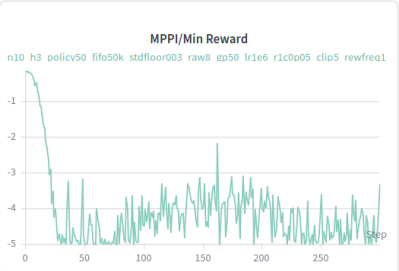




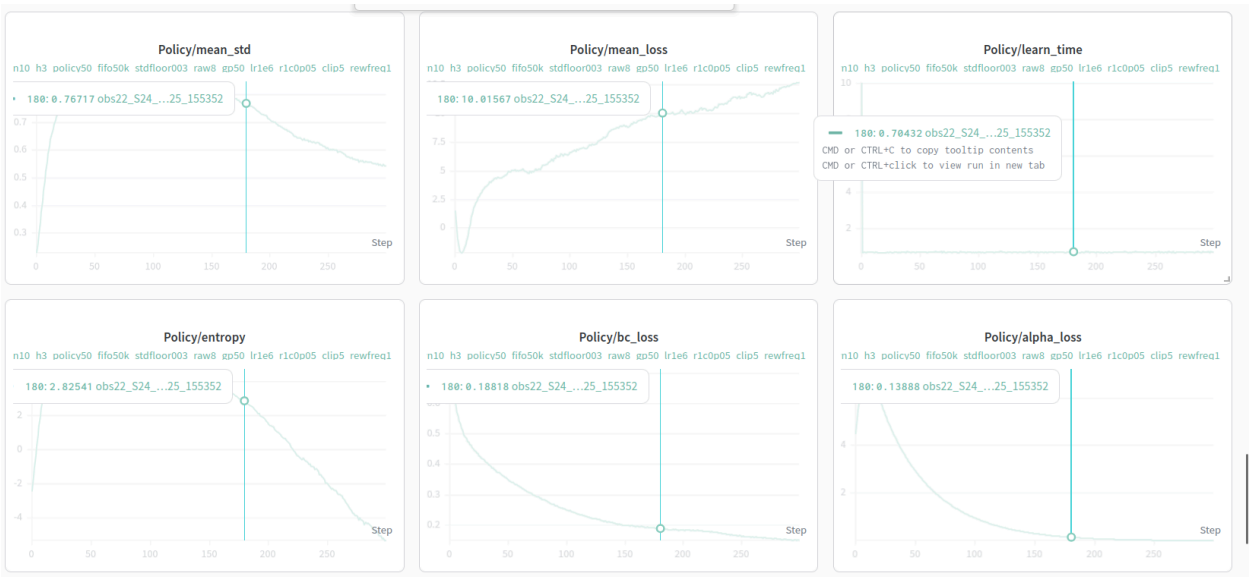




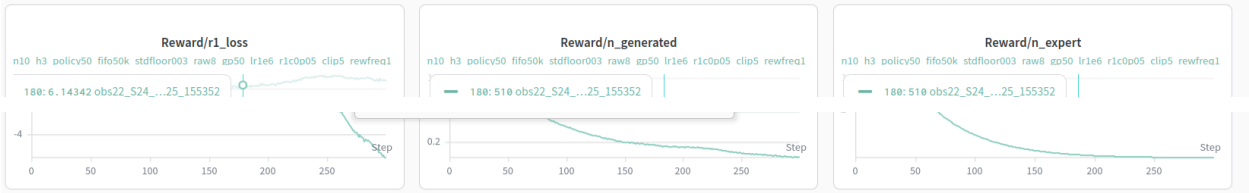




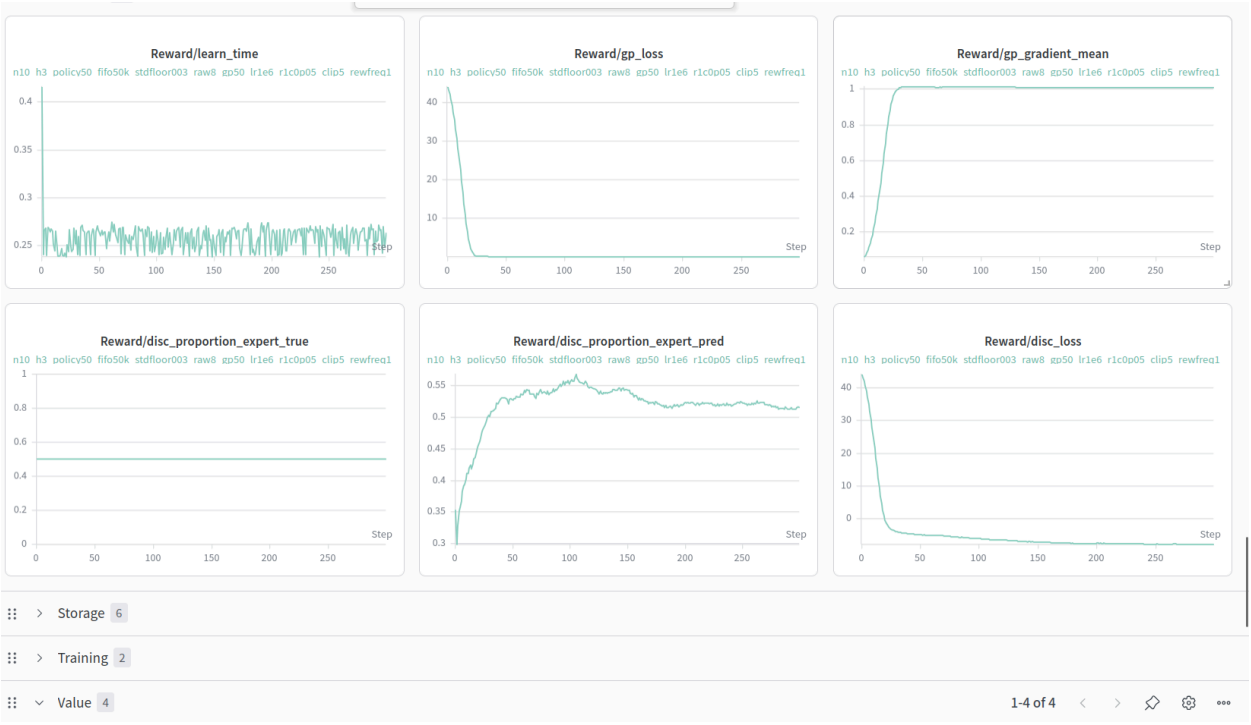
Perf 3



Reward 22 1-6 of 22



Reward 22 1-6 of 22



Storage 6

Training 2

Value 4

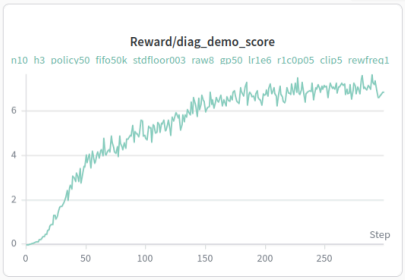
1-4 of 4

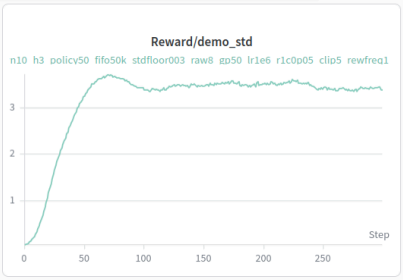
Storage 6

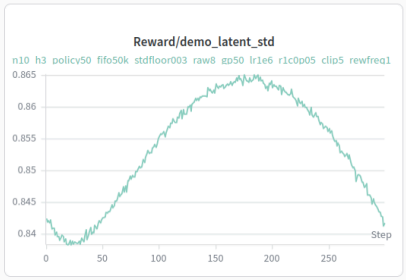
Training 2

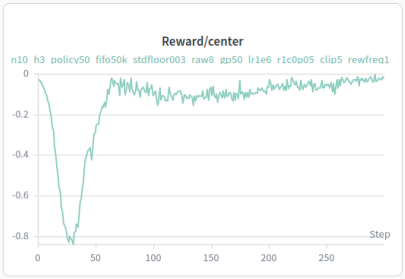
Value 4

1-4 of 4









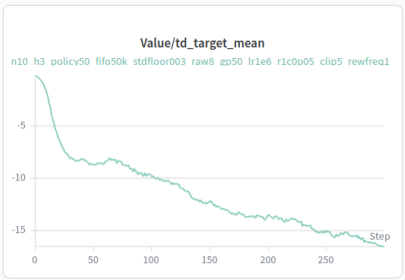
Storage 6

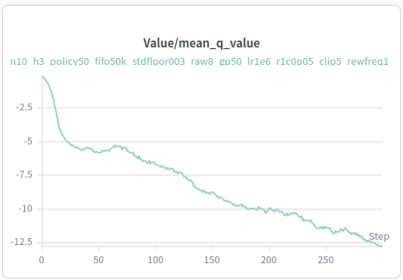
Training 2

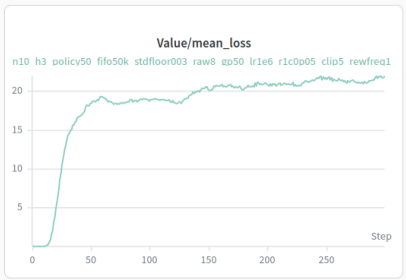
Value 4 1-4 of 4

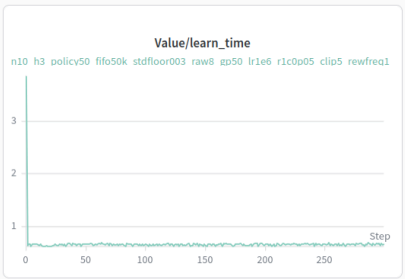
Training 2

Value 4 1-4 of 4







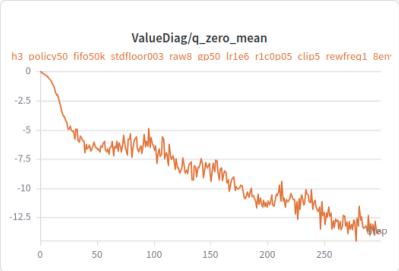


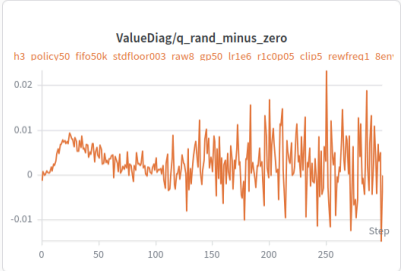
System 33

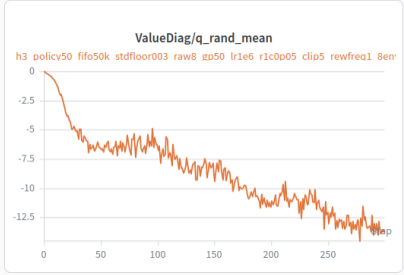
Search panels with regex

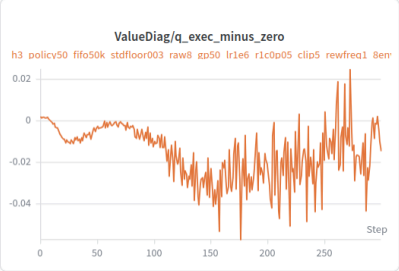
Settings | New report | Add panels

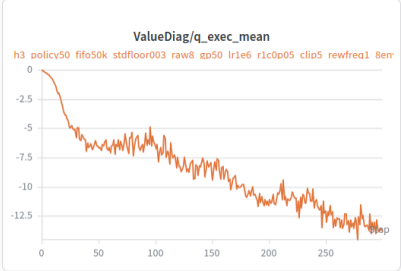
ValueDiag 7 | 1-6 of 7

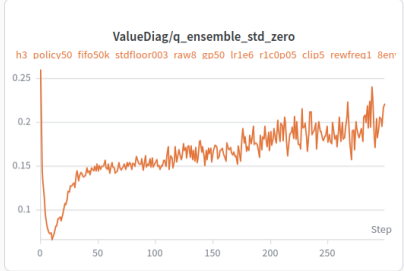










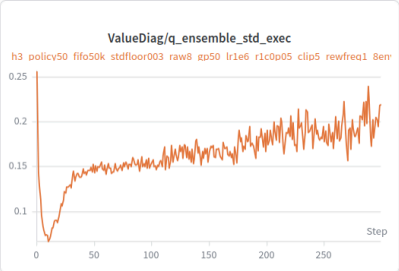


ActionDiag 2

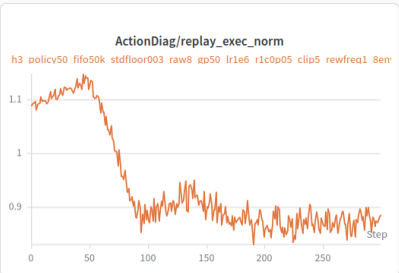
Search panels with regex

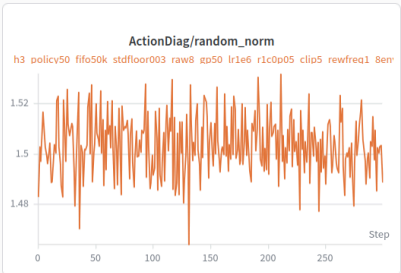
Settings | New report | Add panels

ValueDiag 7 | 7-7 of 7



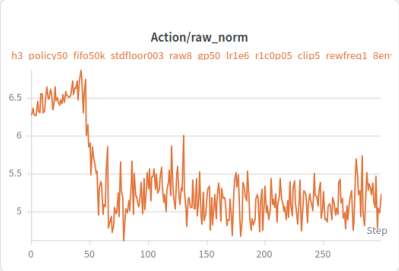
ActionDiag 2

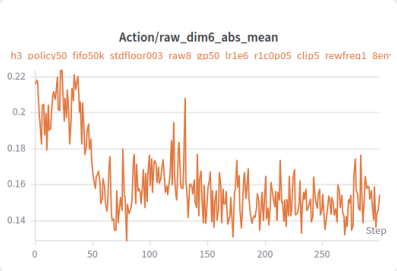


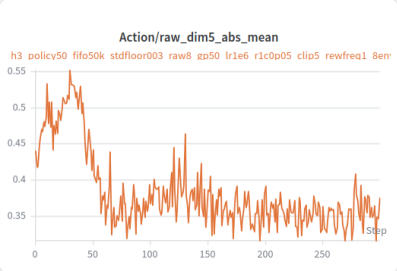


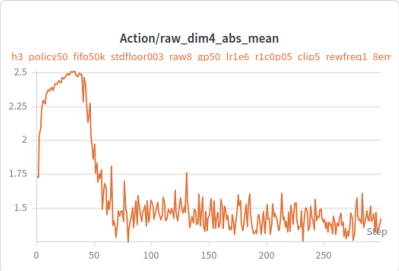
Action 30 | 1-6 of 30

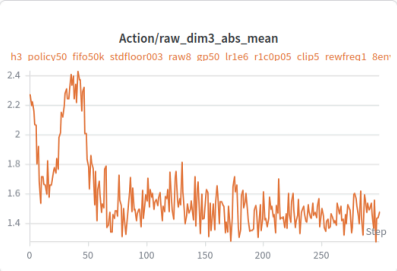
Action 30 1-6 of 30

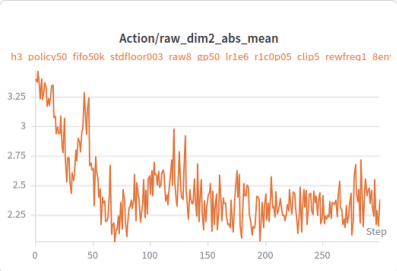




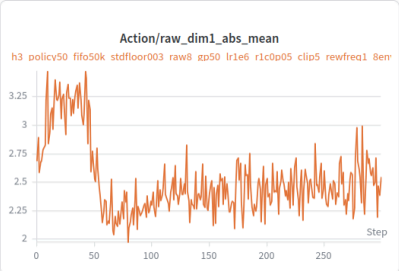


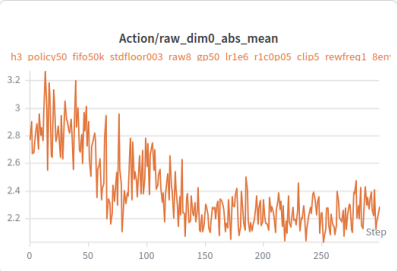


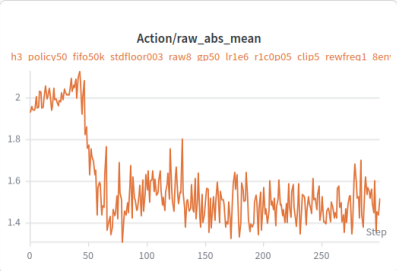


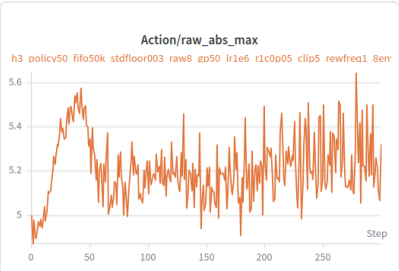


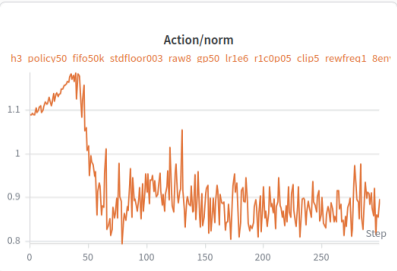
Action 30 7-12 of 30

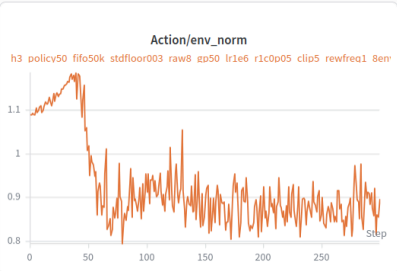












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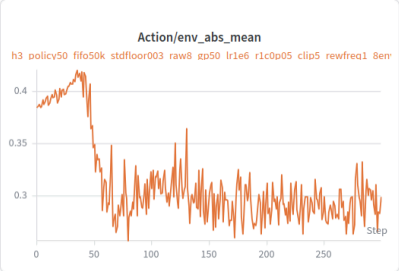

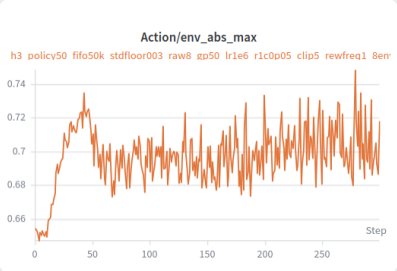

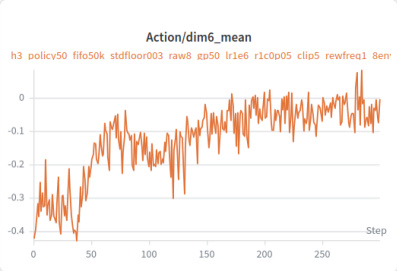

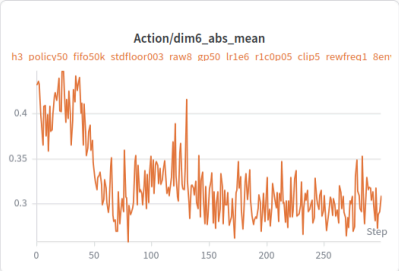

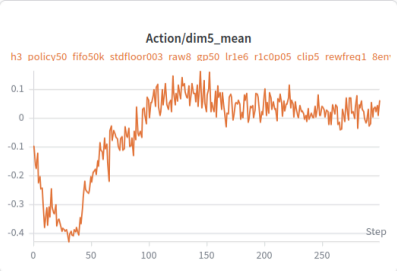

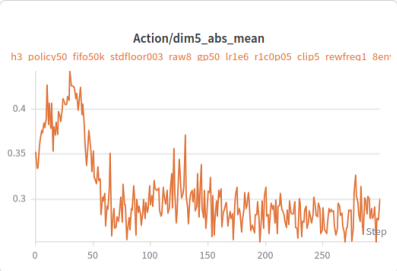

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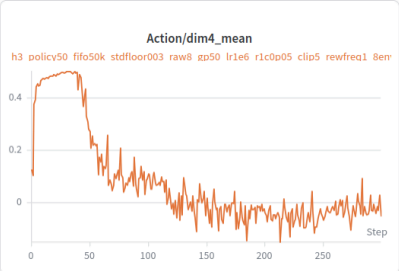

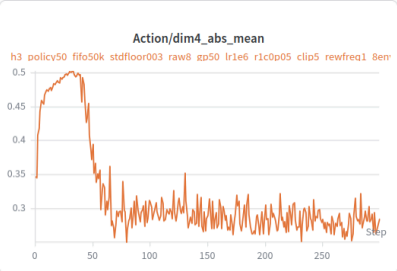

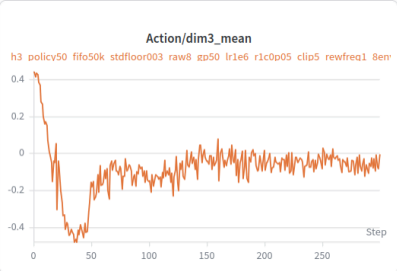

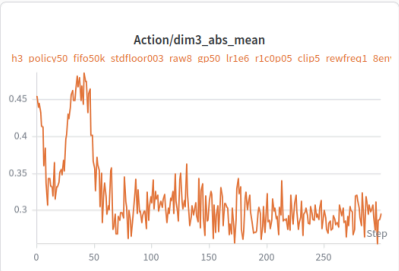

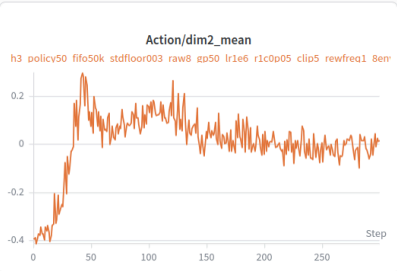

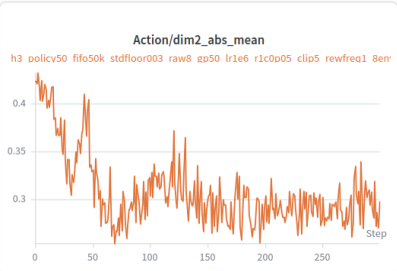

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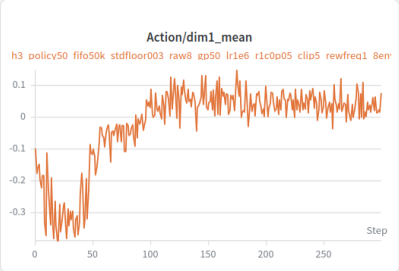

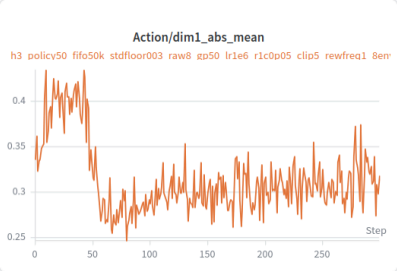

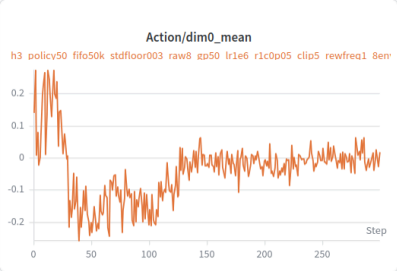

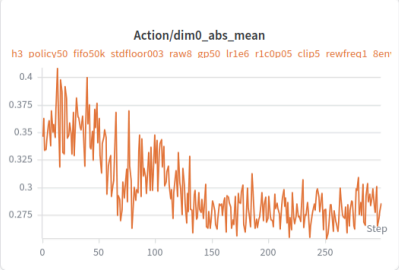

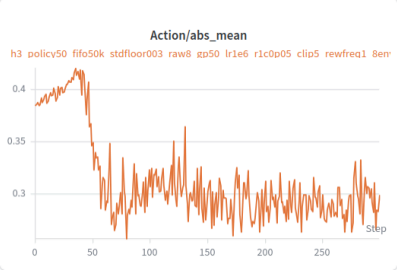

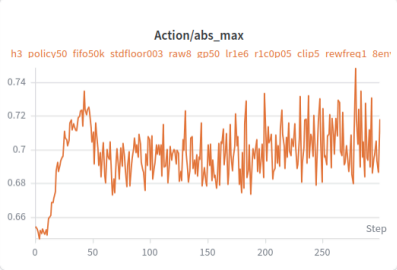

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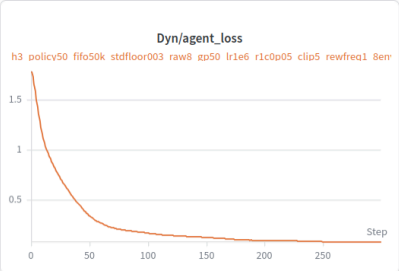
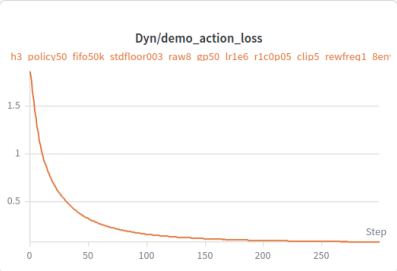
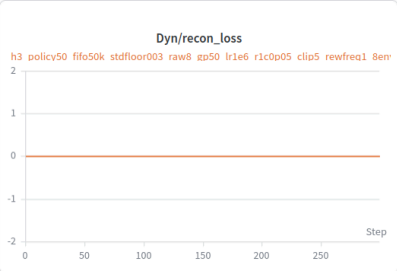
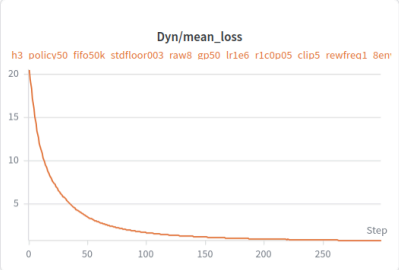
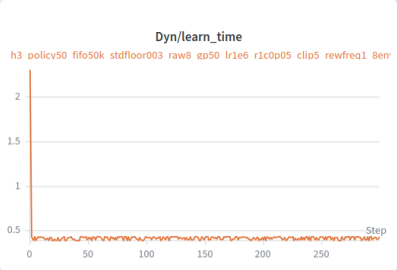
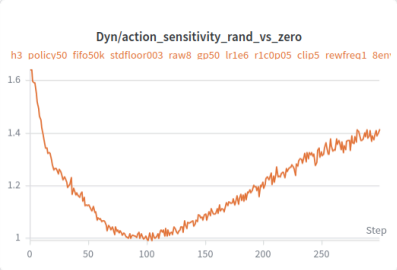




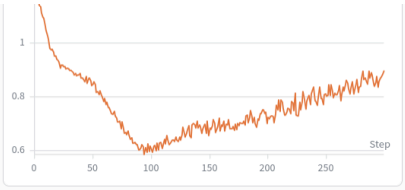








Env 11



Env 11

Episode_Reward 8

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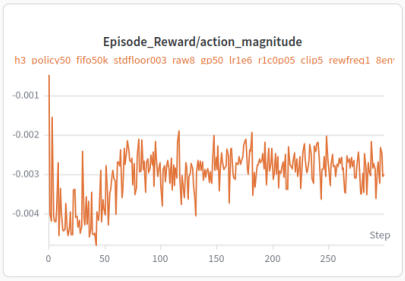
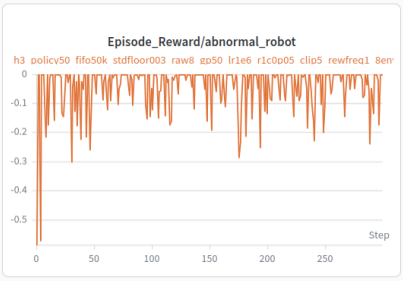
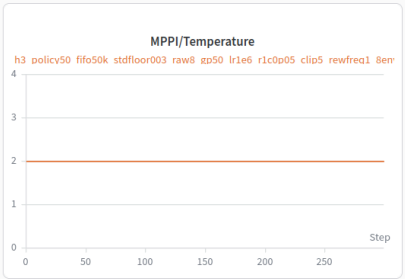
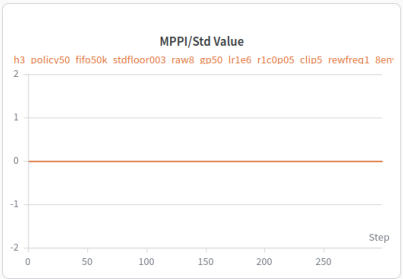
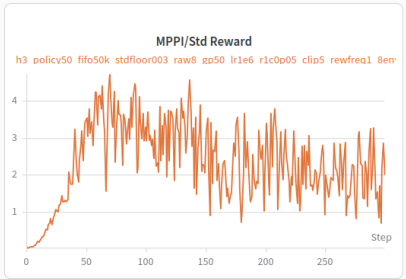


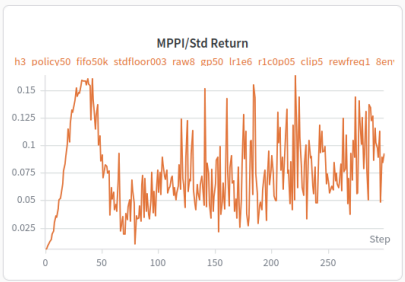


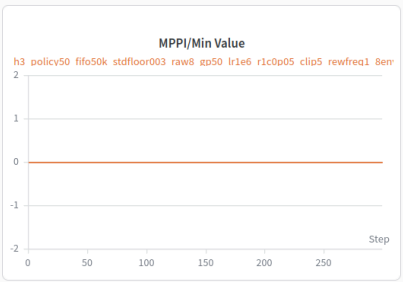
Episode_Termination 2

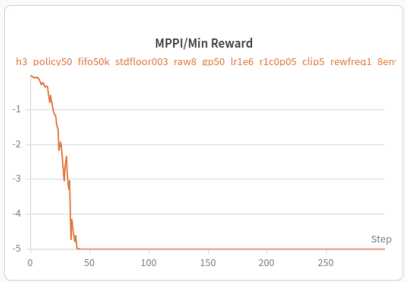




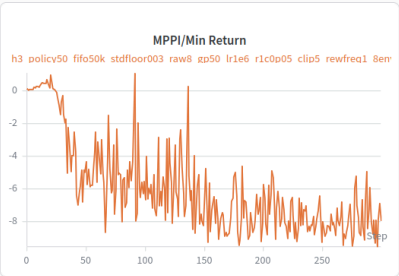


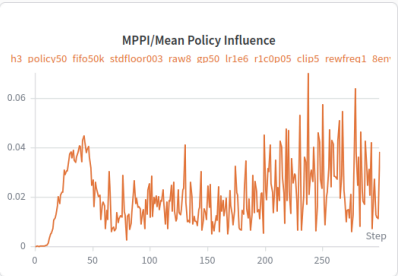


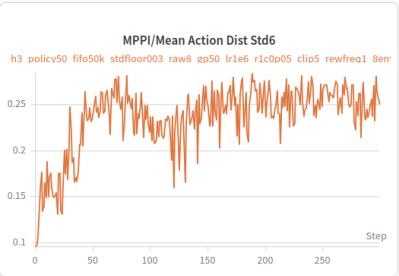


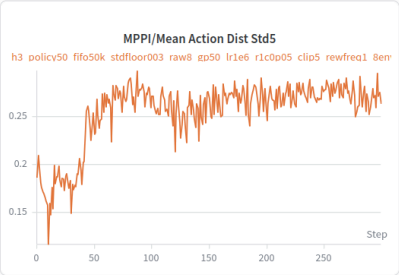


Perf 3


MPPI/Min Return
h3 policy50 fifo50k stdfloor003 raw8 gp50 lr1e6 r1c0p05 clip5 rewfreq1 8en
Step


MPPI/Mean Policy Influence
h3 policy50 fifo50k stdfloor003 raw8 gp50 lr1e6 r1c0p05 clip5 rewfreq1 8en
Step


MPPI/Mean Action Dist Std6
h3 policy50 fifo50k stdfloor003 raw8 gp50 lr1e6 r1c0p05 clip5 rewfreq1 8en
Step


MPPI/Mean Action Dist Std5
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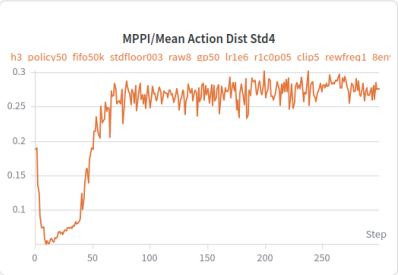
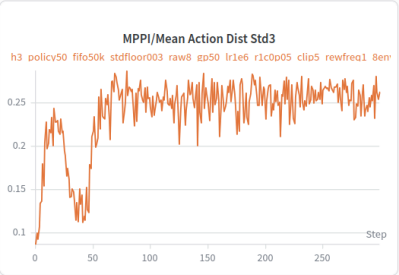
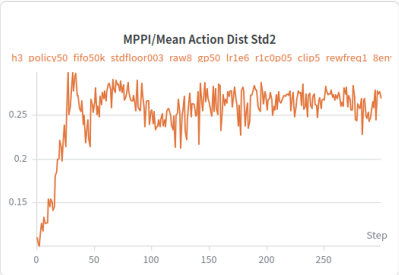
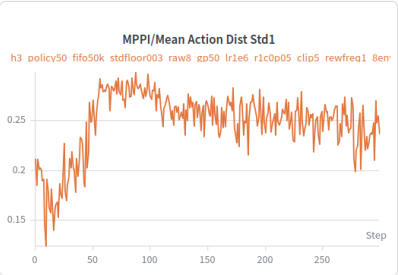
Step

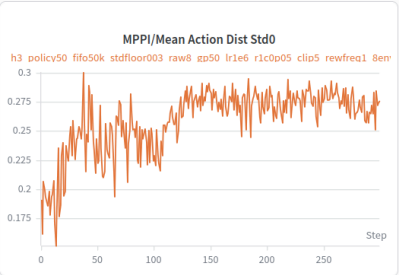

MPPI/Mean Action Dist Std4
h3 policy50 fifo50k stdfloor003 raw8 gp50 lr1e6 r1c0p05 clip5 rewfreq1 8en
Step


MPPI/Mean Action Dist Std3
h3 policy50 fifo50k stdfloor003 raw8 gp50 lr1e6 r1c0p05 clip5 rewfreq1 8en
Step

Perf 3


MPPI/Mean Action Dist Std2
h3 policy50 fifo50k stdfloor003 raw8 gp50 lr1e6 r1c0p05 clip5 rewfreq1 8en
Step


MPPI/Mean Action Dist Std1
h3 policy50 fifo50k stdfloor003 raw8 gp50 lr1e6 r1c0p05 clip5 rewfreq1 8en
Step


MPPI/Mean Action Dist Std0
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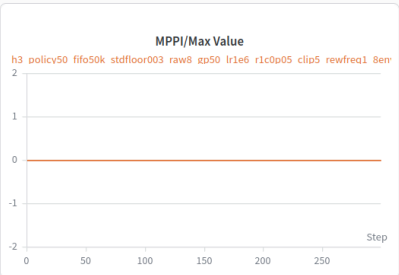
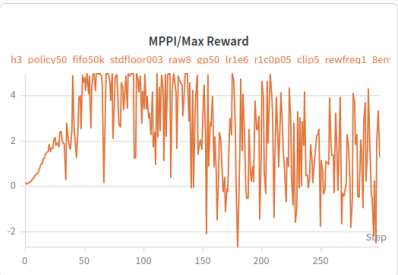
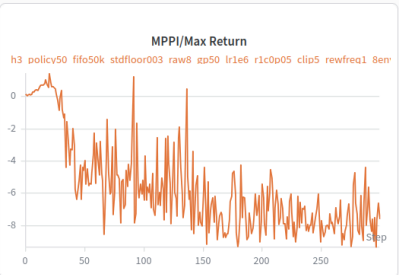
Step

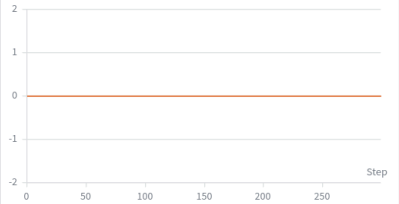

MPPI/Max Value
h3 policy50 fifo50k stdfloor003 raw8 gp50 lr1e6 r1c0p05 clip5 rewfreq1 8en
Step


MPPI/Max Reward
h3 policy50 fifo50k stdfloor003 raw8 gp50 lr1e6 r1c0p05 clip5 rewfreq1 8en
Step

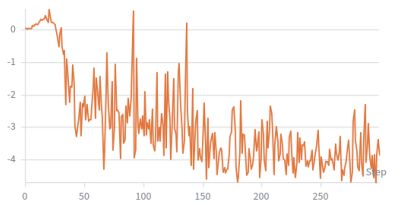

MPPI/Max Return
h3 policy50 fifo50k stdfloor003 raw8 gp50 lr1e6 r1c0p05 clip5 rewfreq1 8en
Step

Perf 3

MPPI/Average Value



MPPI/Average Reward



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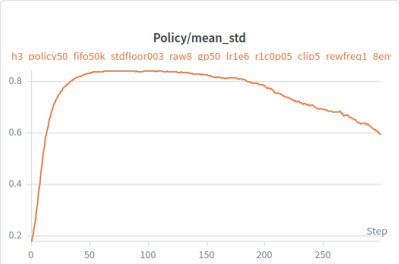
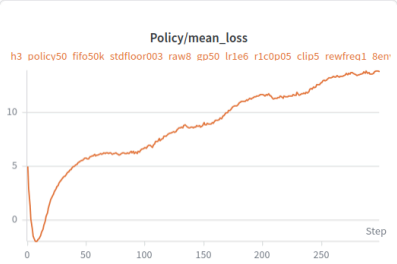
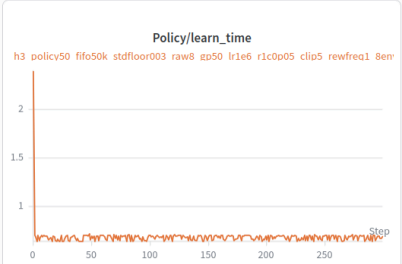
Policy/mean_std



Policy/mean_loss

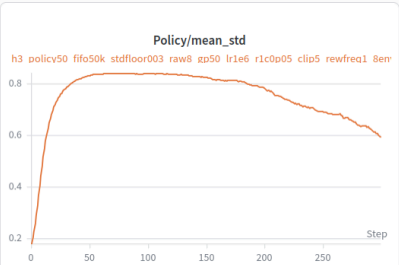


Policy/learn_time

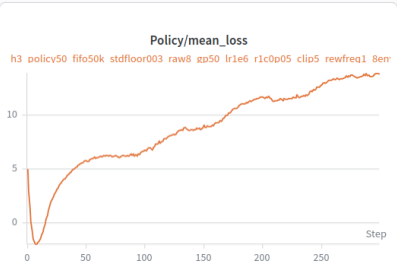


Pin section

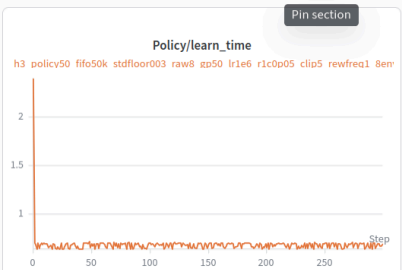
Policy/mean_std



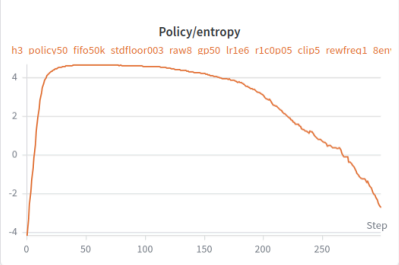
Policy/mean_loss



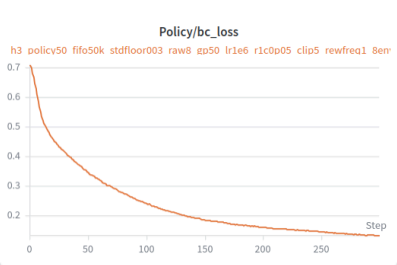
Policy/learn_time



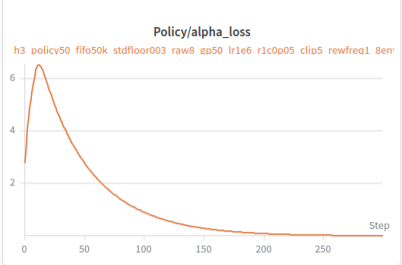
Policy/entropy



Policy/bc_loss



Policy/alpha_loss



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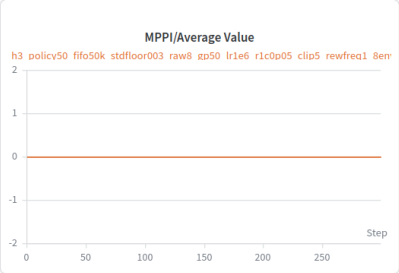

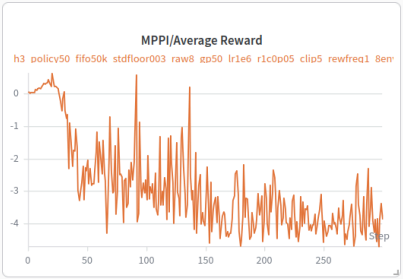

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Policy 7

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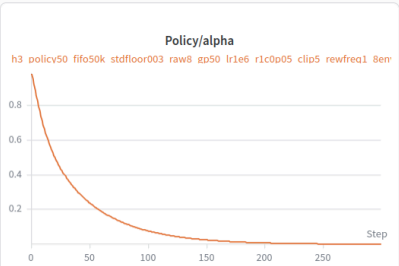
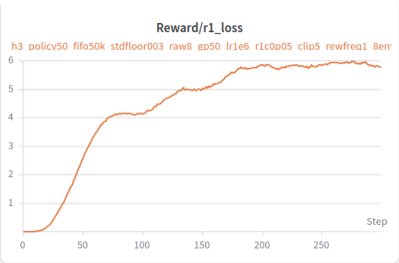
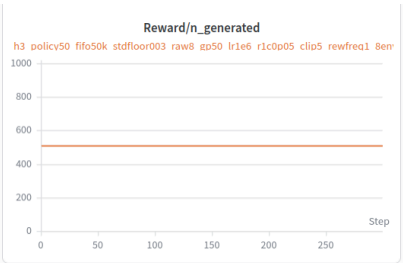
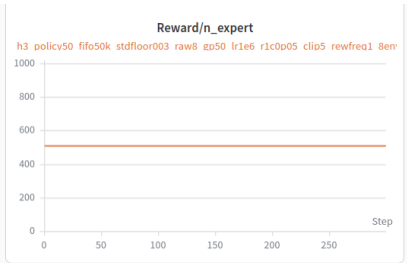
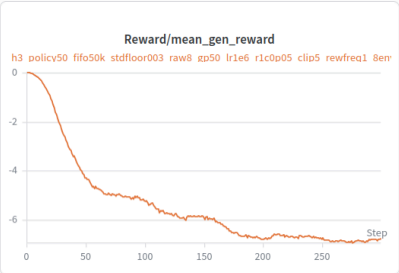
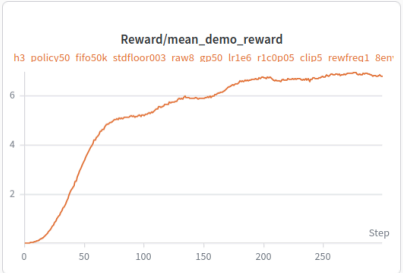


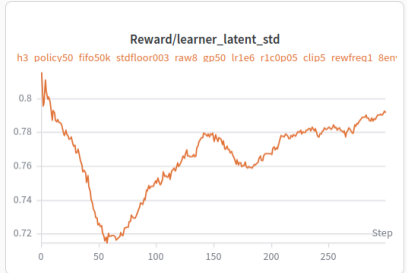






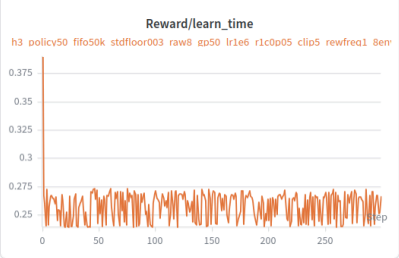


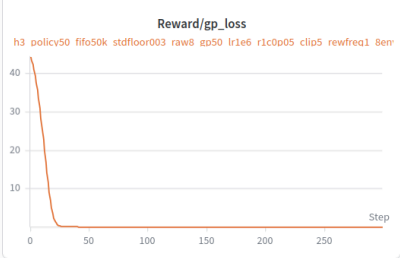


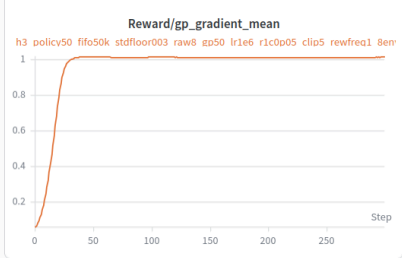


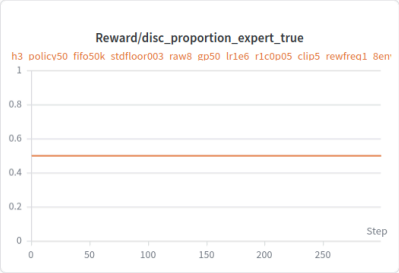
Storage 6

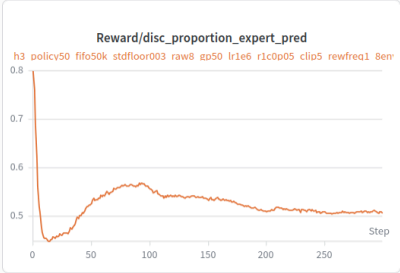
Training 2

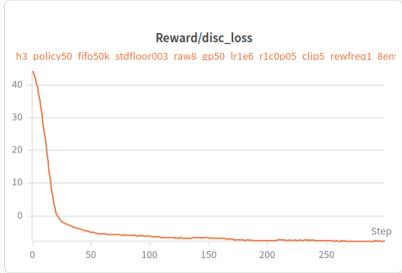






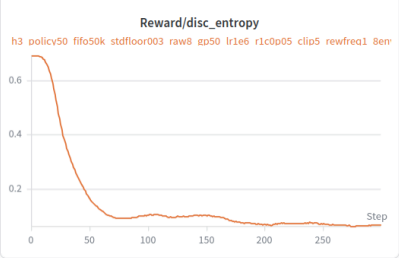


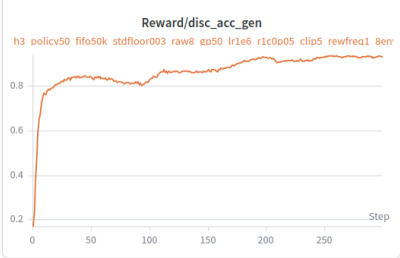


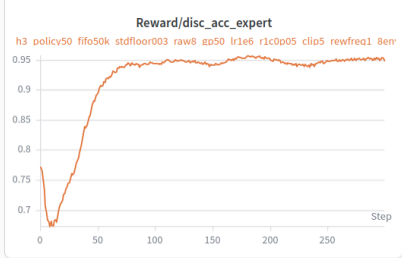


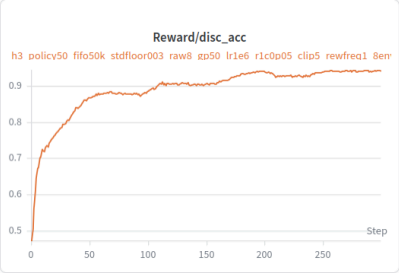
Storage 6

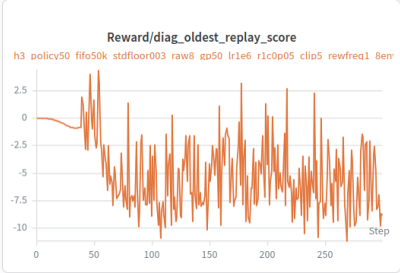
Training 2

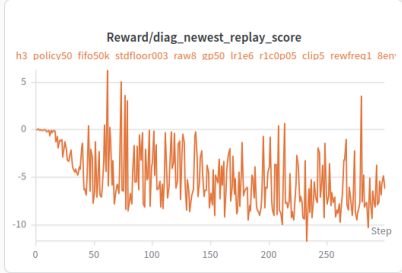






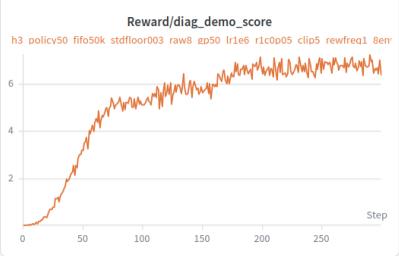


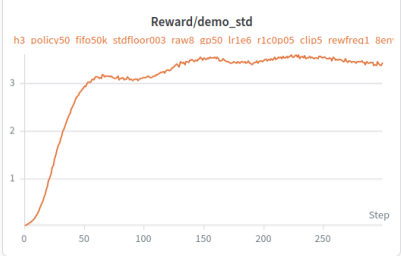


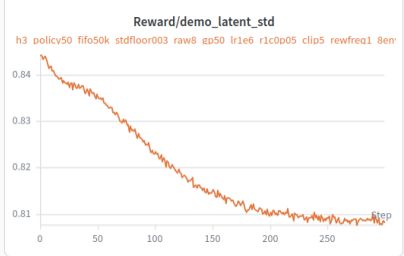


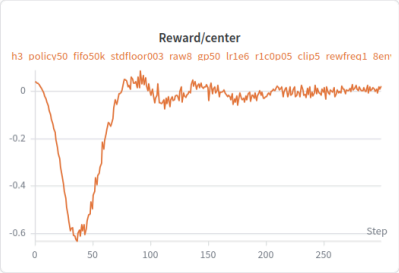
Storage 6

Training 2







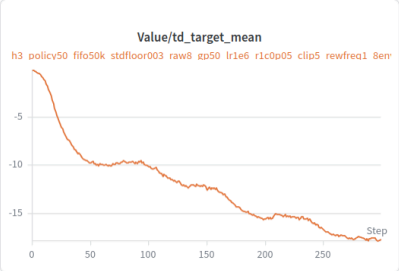


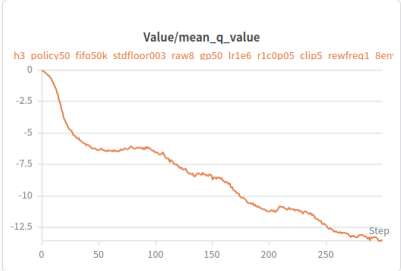
Storage 6

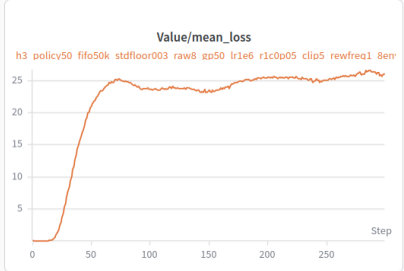
Training 2

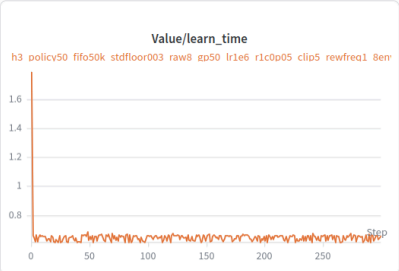
Training 2

Value 4 1-4 of 4

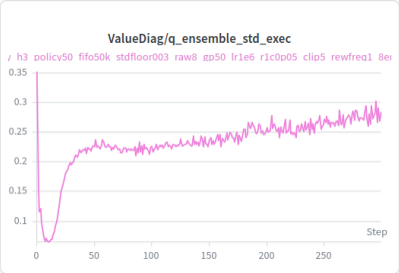




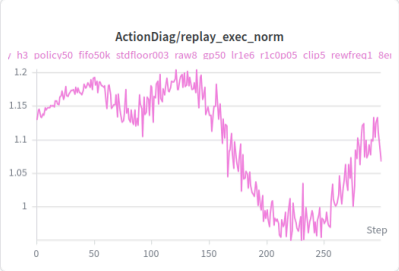


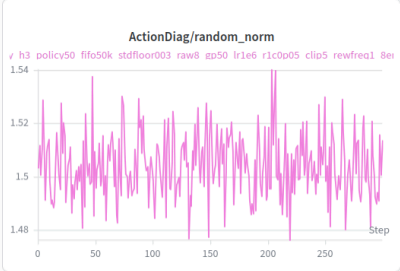


ValueDiag 7 7-7 of 7

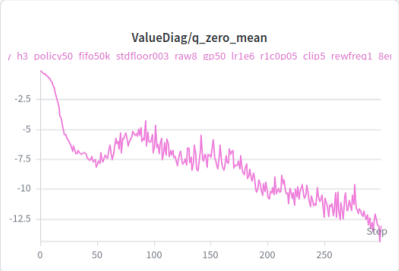


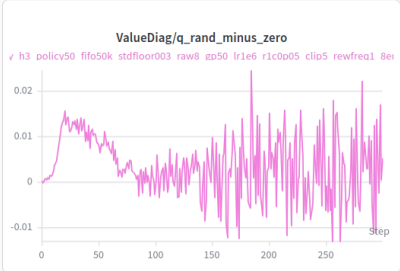
ActionDiag 2

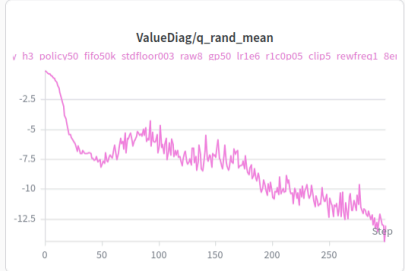


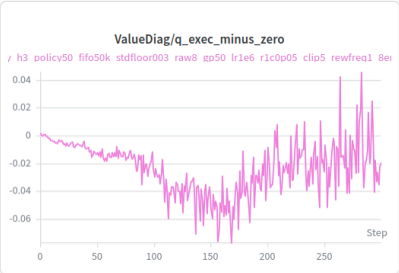


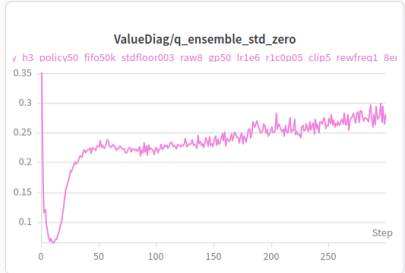
ValueDiag 7 1-6 of 7











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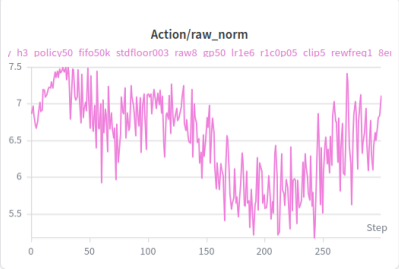
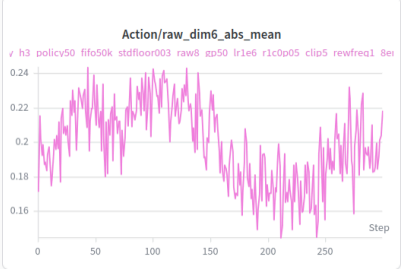
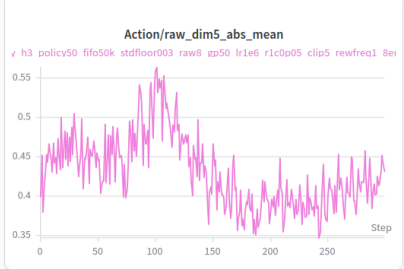
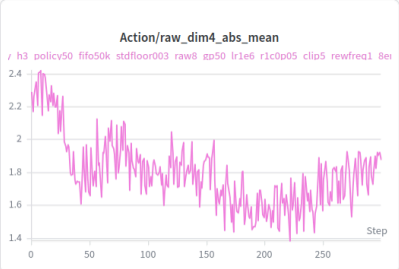
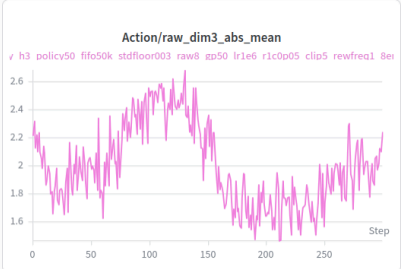
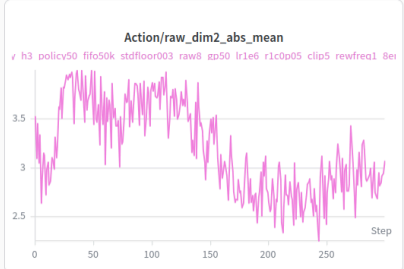






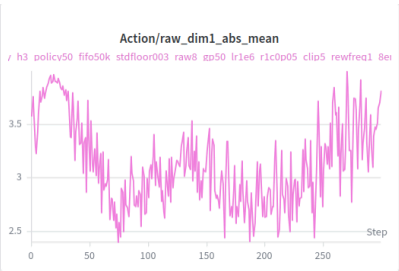


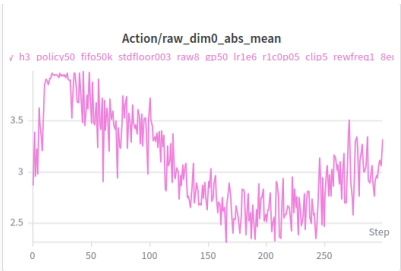


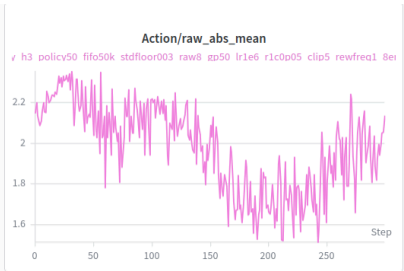


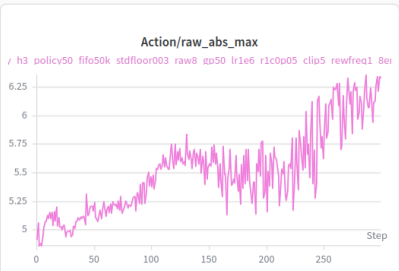
Charts 2

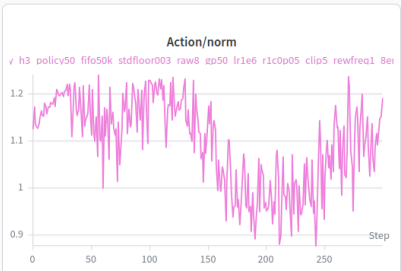
Data 2

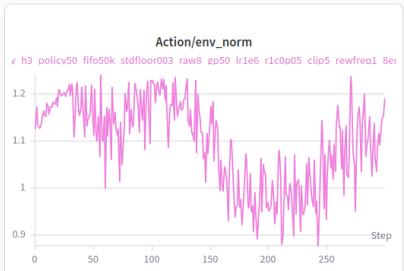






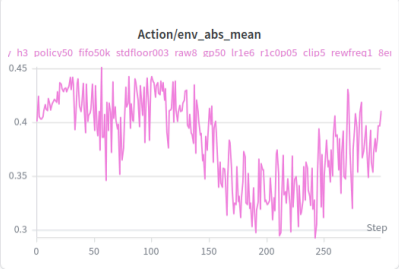


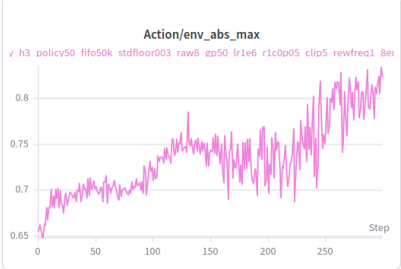


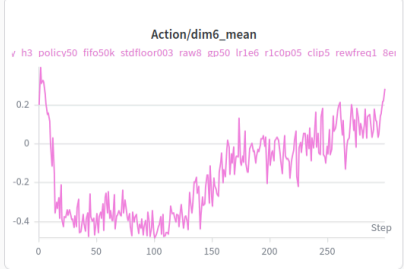


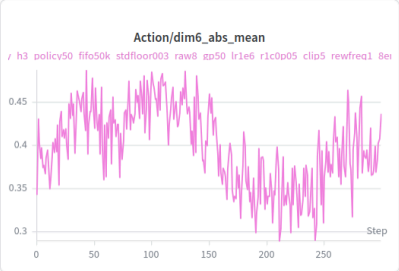
Charts 2

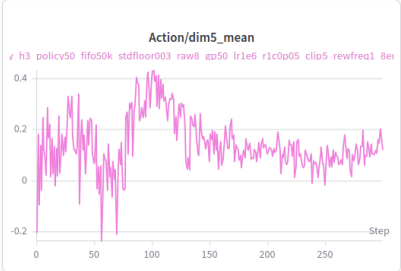
Data 2

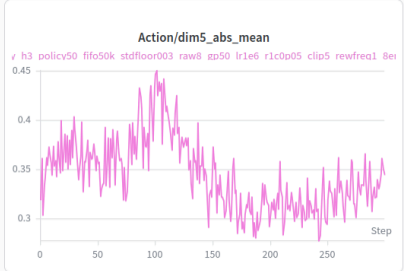






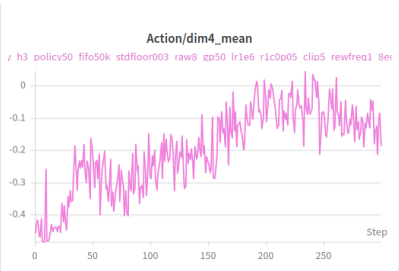


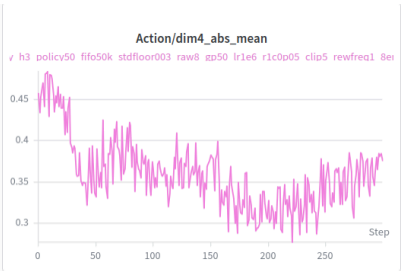


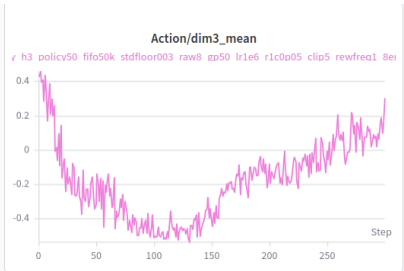


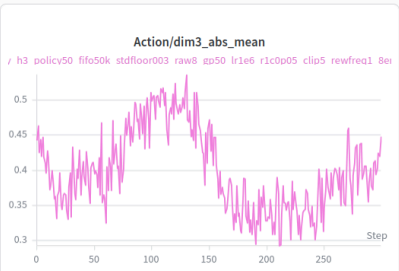
Charts 2

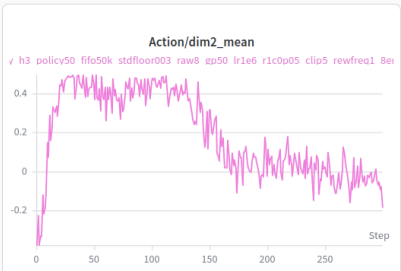
Data 2

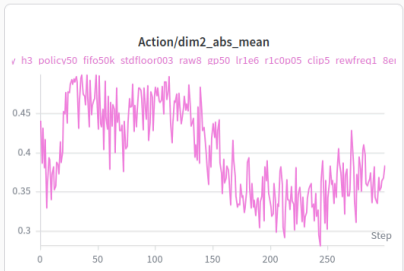






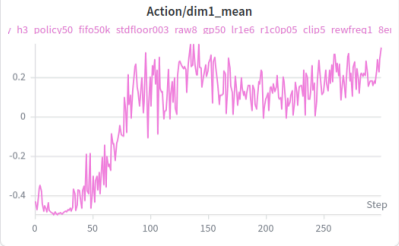


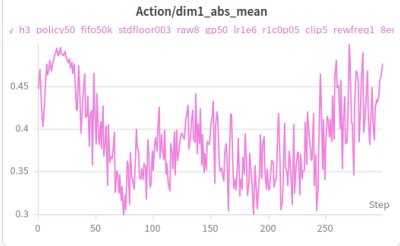


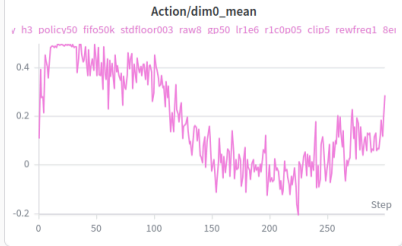


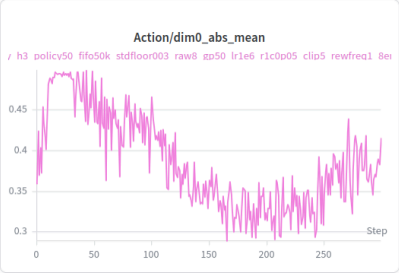
Charts 2

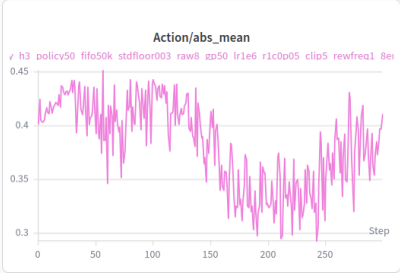
Data 2

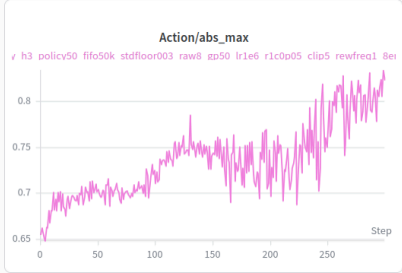






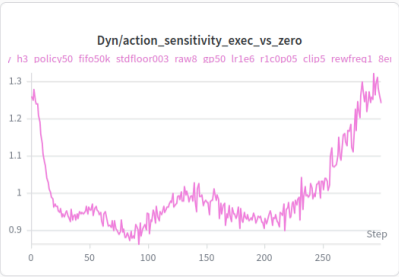






Charts 2

Data 2



Env 11

Episode_Reward 8

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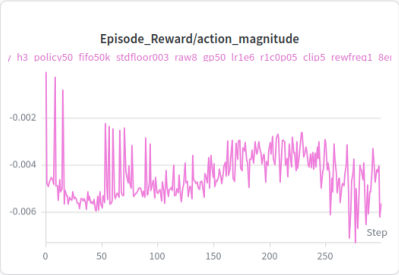

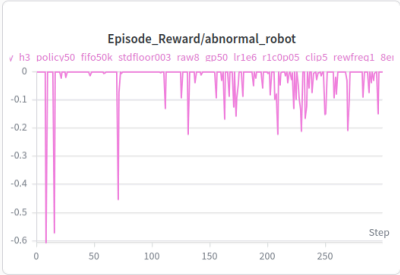

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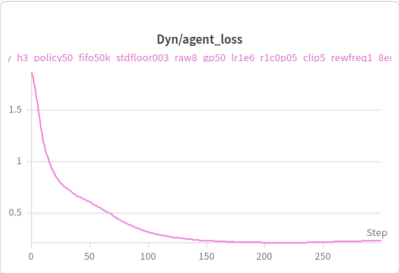

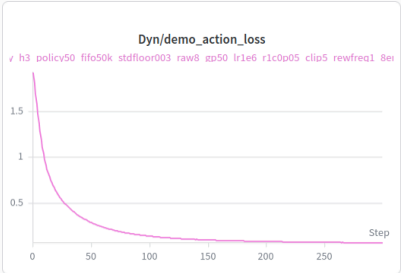

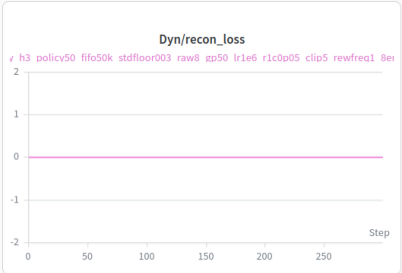

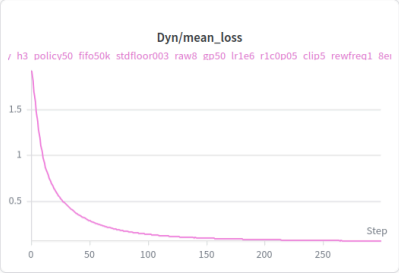

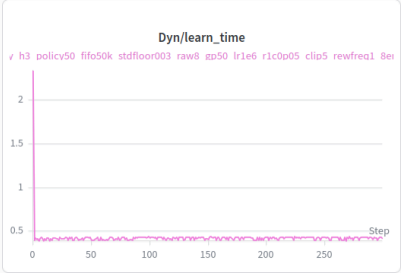

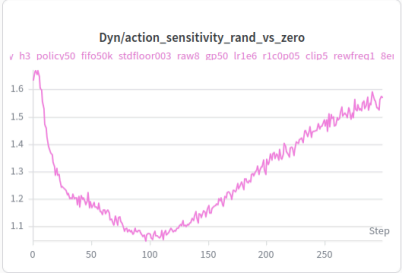

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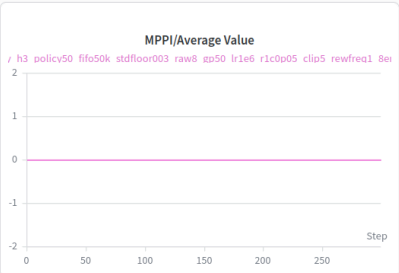
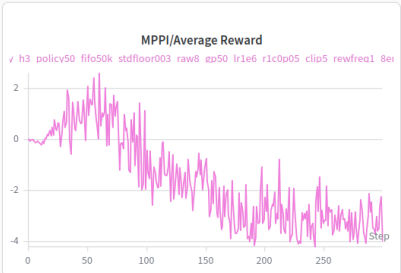
MPPI 20 19-20 of 20





Perf 3

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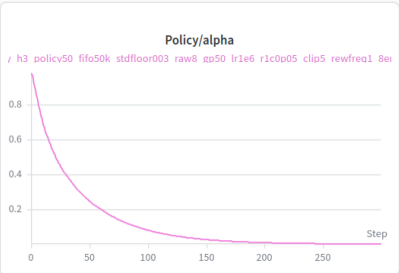
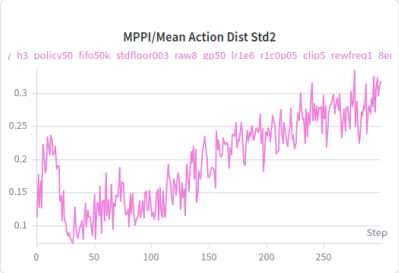
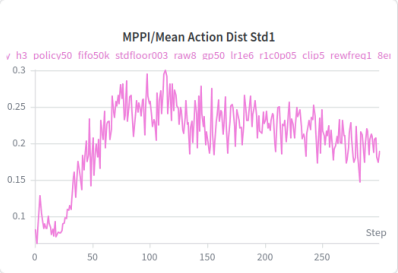
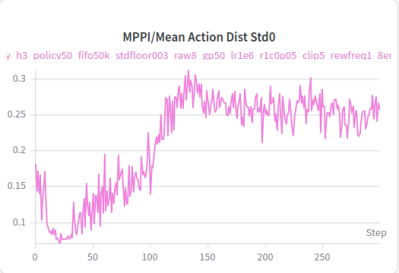
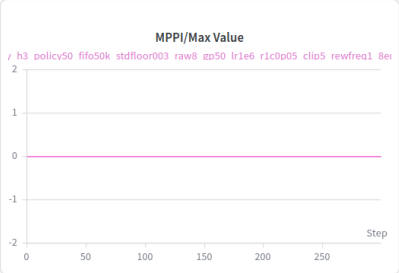


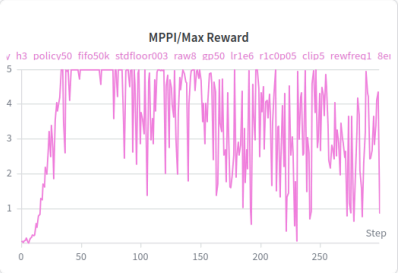
MPPI 20 13-18 of 20

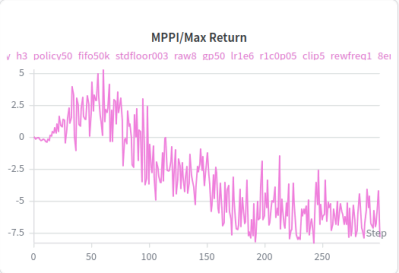












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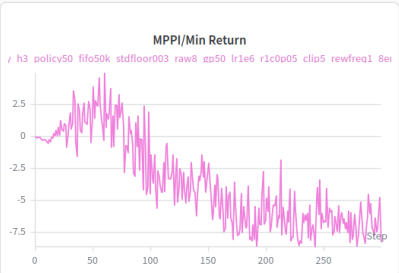

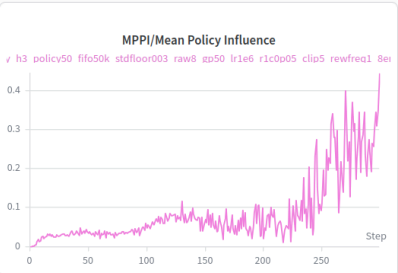

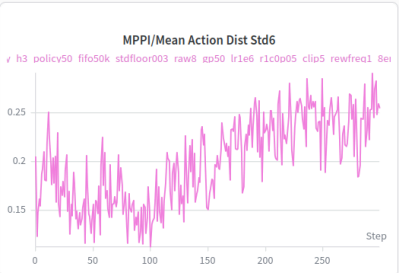

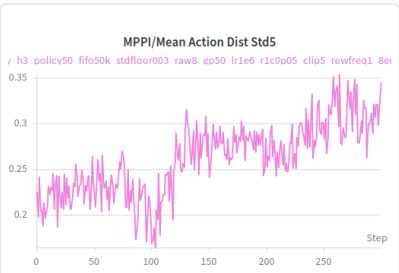

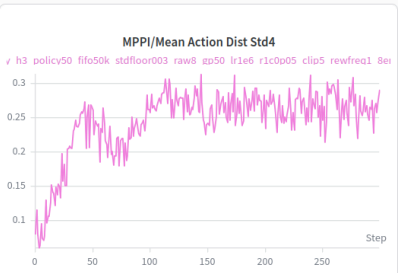

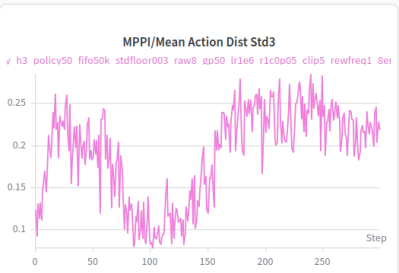

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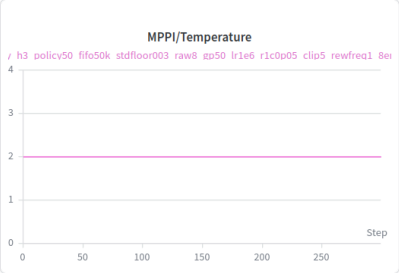
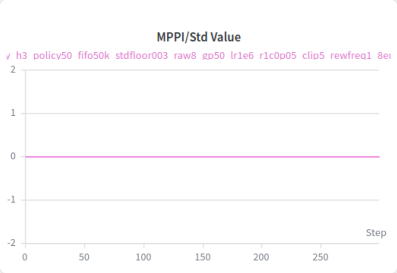
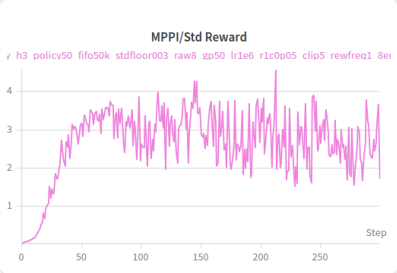
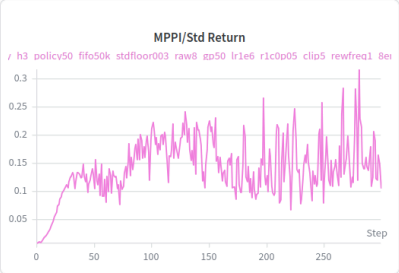
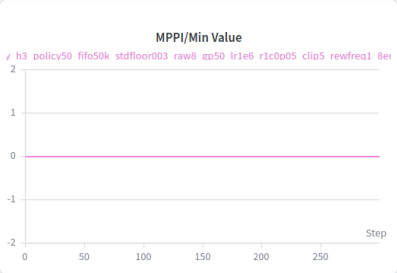
MPPI 20 1-6 of 20

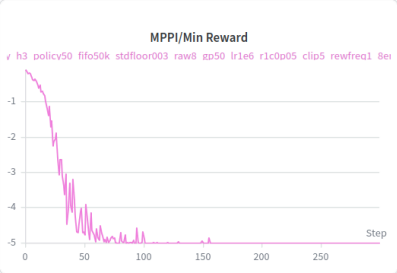












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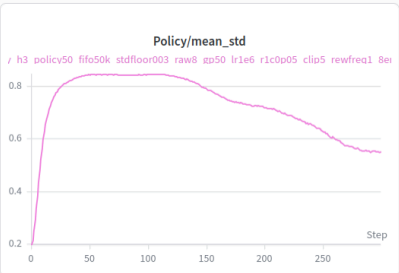

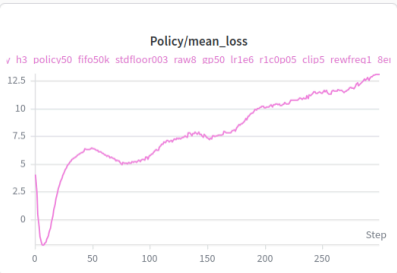

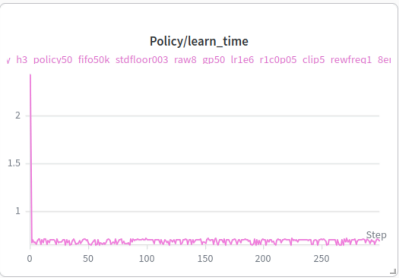

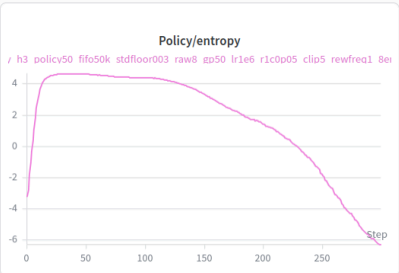

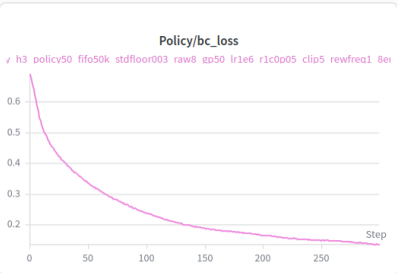

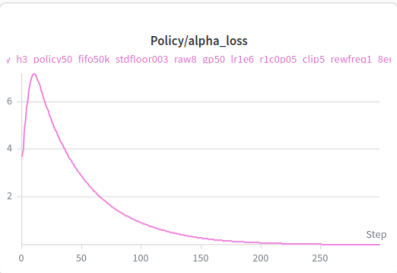

Reward 22 19-22 of 22

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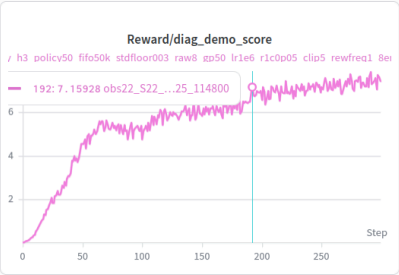

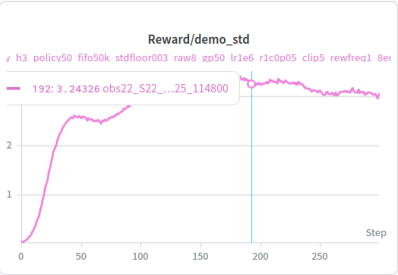

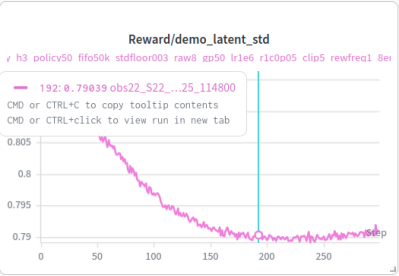

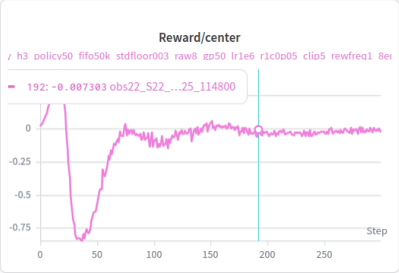

Storage 6

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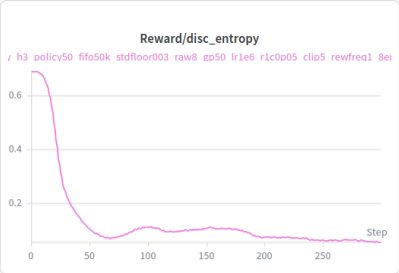

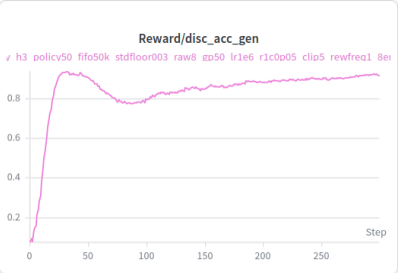

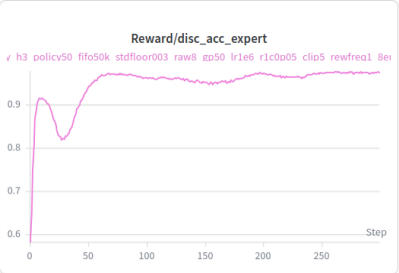

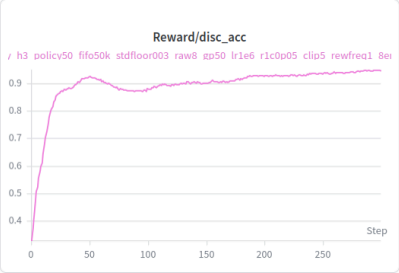

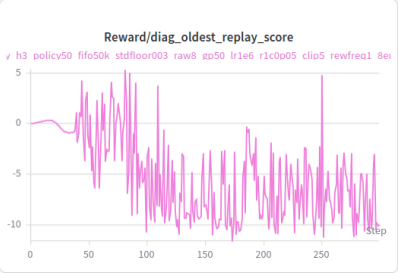

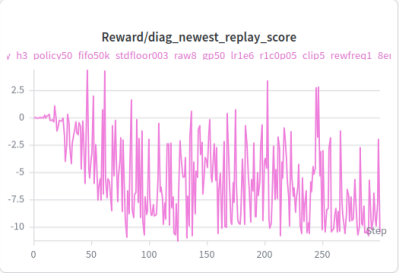

Storage 6

深入的根据算法训练逻辑分析这次的现象，证明了什么，现在这个实验结果把整个训练算法的逻辑链条推到了哪一步？

3:14 PM

Worked for 8m 8s

这批 S22/S23/S24 很关键。它基本把链条推过了“value 崩”“replay dynamics 污染”“demo action loss 太弱”“absolute/residual dynamics 形式”这几层，指向更核心的一层：

当前失败主因不是 value，也不是单纯 policy 不会模仿，而是 reward-dynamics-planning 形成的 latent transition score 不是可执行的、action-causal 的任务方向。

我按训练逻辑拆开说。

这次实验做了什么
S22/S23/S24 都关闭了 MPPI terminal value 影响，所以截图里 `MPPI/Average Value = 0`、`Std Value = 0`。也就是说这次 MPPI 只看 imagined reward，不再被 value bootstrap 拖坏。

三个实验分别排查：

- S22：`dyn_agent_coeff=0`，只用 demo action dynamics，排除 replay/agent dynamics 污染。
- S23：demo action dynamics 权重加到 10，排除“demo-action loss 太弱”。
- S24：再把 dynamics 改成 residual delta 形式，排除“absolute next latent 难学”。

代码逻辑对应位置是：dynamics loss 在 learner.py (line 518)，BC/action prior 在 learner.py (line 812)，residual dynamics 在 dynamics.py (line 37)。

最重要的现象
所有实验最终 `succ=0.000`。但是 reward 和 dynamics 看起来都“学会了”：

- discriminator 极强：demo reward 到 6-7，gen reward 到 -6/-7，`disc_acc` 接近 0.9+，`disc_entropy` 很低。
- BC 有效：offline evaluator 里 `policy_teacher_raw_cos` 到了约 `0.73`，说明 policy 确实被拉向 teacher action。
- demo-action dynamics 也有效：S22 的 `dyn_loss` 最后约 `0.0676`，S23 约 `0.814`，S24 residual 因尺度不同约 `7.35`，但趋势都按设定优化了。
- 但是 MPPI 仍然崩：`MPPI/Average Reward` 从短暂上升后掉到负区间，`Min Reward` 经常贴近 -5，`Max Reward` 却很高、很抖。

这个组合很有诊断意义：
reward 能把 demo/replay 分开，但 MPPI 找不到能在真实环境里通向 demo 的动作序列。它找到的是 learned latent model 里高分/低分混杂的幻觉路径。

offline evaluator 把问题钉死了
如果 dynamics/reward 真有可执行方向，那么同一个 demo state 上：

`teacher action` 应该明显优于 `zero/random/policy action`。

但实际不是。最终 checkpoint：

- S22 model300：
  - teacher delta cosine `0.462`
  - zero delta cosine `0.402`
  - policy delta cosine `0.442`
  - teacher reward `4.617` vs zero reward `4.590`
  - 差距只有 `+0.027`
- S23 model300：
  - teacher delta cosine `0.463`
  - zero `0.398`
  - teacher reward `4.235` vs zero `4.189`
  - 差距只有 `+0.046`
- S24 model300：
  - teacher delta cosine `0.434`
  - zero `0.373`
  - teacher reward `3.634` vs zero `3.441`
  - 差距稍大，但仍远不足以稳定规划；Q 上 policy-zero 也只有约 `+0.07`

这说明：即使 teacher action 在动作空间被喂进去了，reward/dynamics 仍没有学出强烈的“teacher action 是正确动作”的排序。它只是学会了“demo latent transition 像 demo，replay latent transition 不像 demo”，但没有学会“什么 action 造成 demo-like transition”。

这证明了什么

这批实验强排除了三个旧假设：

1. 不是 value terminal bootstrap 主因
   这次 MPPI value 是 0，但 reward-only planning 仍失败。
2. 不是 replay dynamics 污染主因
   S22 只用 demo action dynamics，仍然没有成功。
3. 不是 demo-action dynamics 权重不够主因
   S23 权重 10，仍然失败。
4. 不是 absolute vs residual dynamics 这个单点主因
   S24 residual 后 reward 曲线稍有不同，但没有改变失败结构。

现在链条被推到：

```
env/action wrapper 可能有问题
    ↓ 已部分缓解：BC 能把 policy 拉向 teacher
value bootstrap 可能有问题
    ↓ 已排除：MPPI value=0 仍失败
```

replay dynamics 污染可能有问题
↓ 已排除：demo-only dynamics 仍失败
demo action supervision 不够
↓ 已排除：dyn coeff=10 仍失败
dynamics absolute/residual 形式不对
↓ 基本排除：residual 仍失败
核心剩余：
latent representation + reward/dynamics/planning score 不具备 action-causal 可执行性

现在最合理的根因

reward 是对 (z, z_next) 的判别器，dynamics 是从 (z, action) 预测 z_next。但当前 latent 表征允许一种坏解：

- reward 学会 demo/replay 的分布差异；
- dynamics 学会某种平均/捷径 latent transition；
- policy 被 BC 拉近 teacher，但 MPPI 的 imagined rollout 仍然不等价于真实环境 rollout；
- planner 追的是 discriminator artifact，而不是真实任务 progress。

所以这不是“reward 没学到 demo 高分”这么简单。恰恰相反：reward 学得太像分类器了，但分类结果没有变成可操作的控制梯度。

下一步实验应该直接打这一层

我建议下一轮不要再调 gp/r1/value/horizon 这些外围参数了。要做两个更外科式实验：

1. S25：identity/raw-compact dynamics 实验
   不用 learned latent，直接在 compact obs 空间上做 dynamics/reward/planning。
   如果 raw compact 上 teacher 明显优于 zero/random，并且 MPPI 变好，根因就是 encoder latent 表征。
   如果 raw compact 仍失败，根因更靠近 action wrapper / demo-env mismatch / planner action execution。
2. S26：action-ranking dynamics/reward 约束
   对同一个 demo state 强制：
   `score(z, teacher_action) > score(z, zero/random_action) + margin`。
   不是只让 dynamics 拟合 demo next，而是直接让 planning score 学会“teacher action 比 zero/random 好”。

这次结果已经足够明确：可以不用继续在 S22/S23/S24 这条线上等了。下一步要检查并修正的是“高 reward imagined transition 是否真由正确 action 导致”。

3:22 PM
好，同时跑25 26 50tier
3:30 PM
Worked for 22m 28s

已安排好，S25/S26 会顺序跑，每个 50 iter，并在各自 `model_50.pt` 出来后自动跑 offline evaluator。

当前真正活跃的是这组：

- 队列 PID：`82120`
- 当前训练 PID：`82188`
- 当前正在跑：`S25 identityCompact`
- 主队列日志：`logs/long_dynamics_queue/s25_s26_50_queue_20260526_155134.log`
- 启动日志：`logs/long_dynamics_queue/s25_s26_launcher_20260526_155134.log`
- 当前 S25 run dir：
  `logs/obs22_S25_identityCompact_h3_policy50_fifo50k_stdfloor003_raw8_gp50_lr1e6_r1c0p05_clip5_rewfreq1_8env_50iter_20260526_155134`

我确认过：S25 已经进训练，GPU 正在跑，identity encoder 生效了，日志里显示 `Encoder： 0`、`Identity()`、latent dim 22。

这次我也补了两处诊断代码：

- S25：支持 `identity_encoder=True`，直接用 compact obs 当 latent。
- S26：新增 `demo_action_rank_coeff/rank_margin`，让 expert action 的 dynamics next-state prediction 必须优于 zero/random action。

你可以回家了。现在机器上会先跑 S25，结束后自动跑 S26，再自动生成：

- `logs/offline_checkpoint_eval_s25_model50_teacher_actions.csv`
- `logs/offline_checkpoint_eval_s26_model50_teacher_actions.csv`

ValueDiag 7

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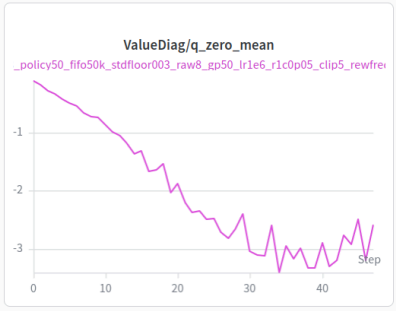
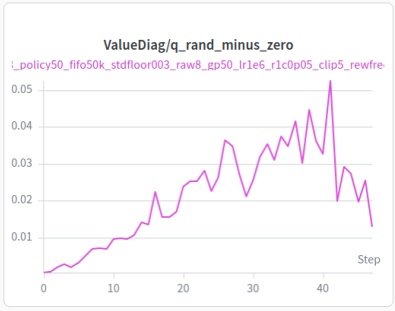
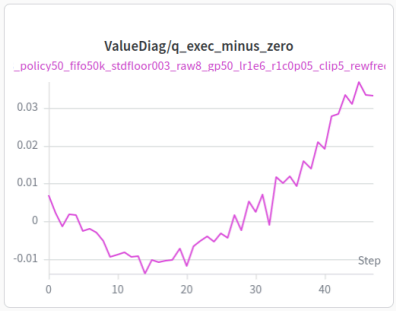
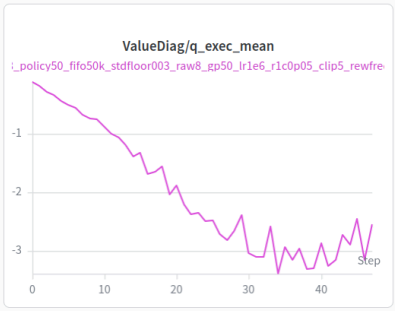
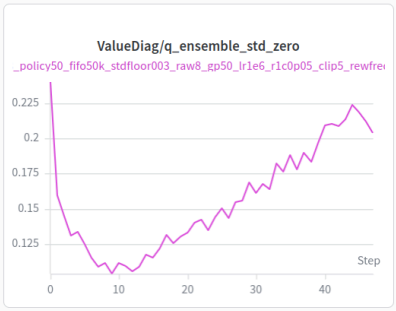
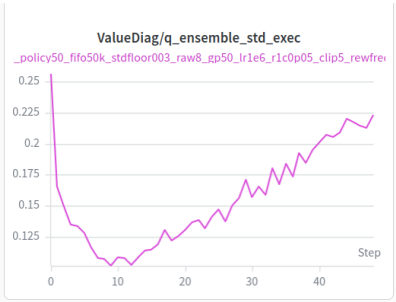




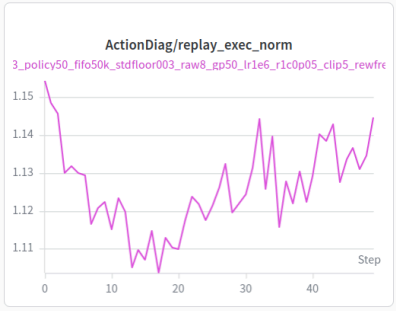


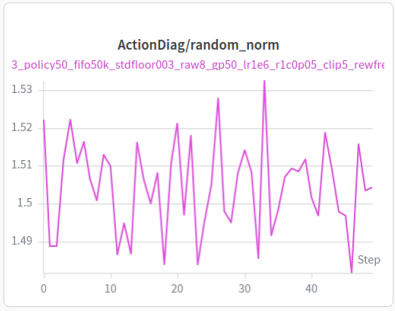






ActionDiag 2





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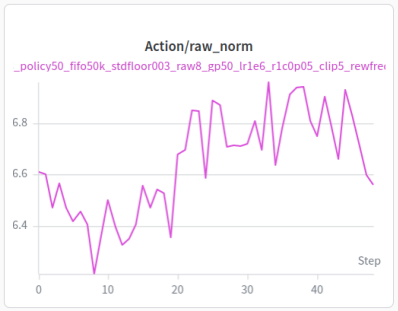

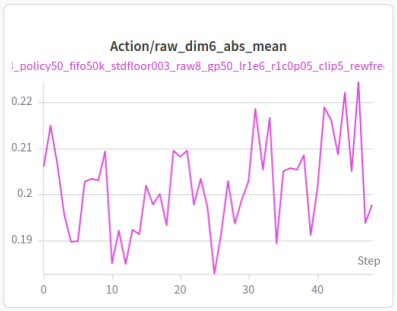

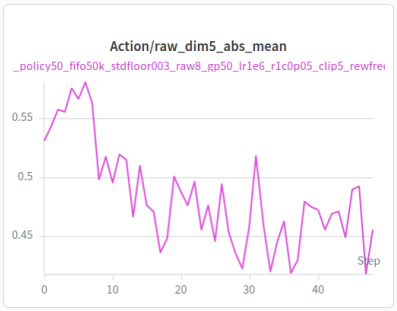

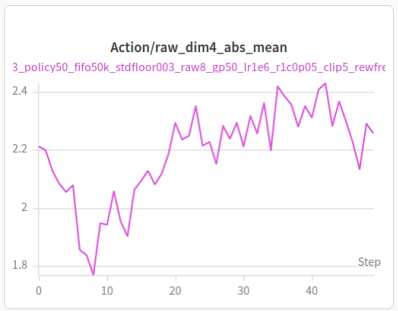

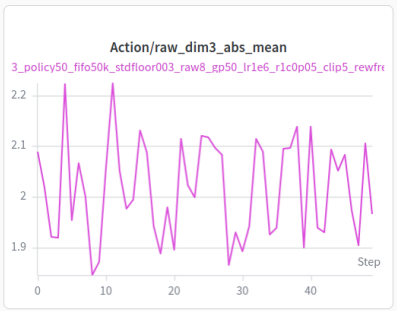

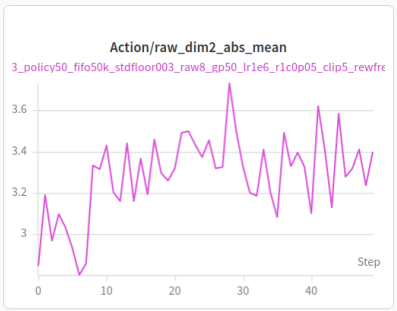

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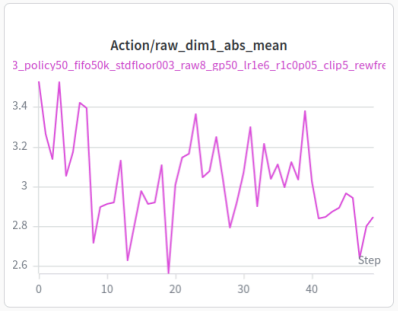

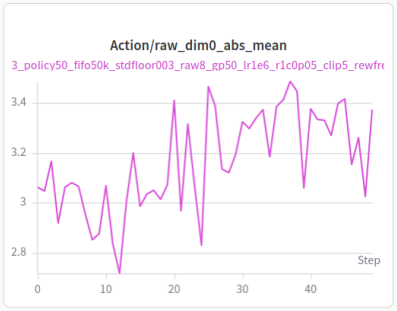

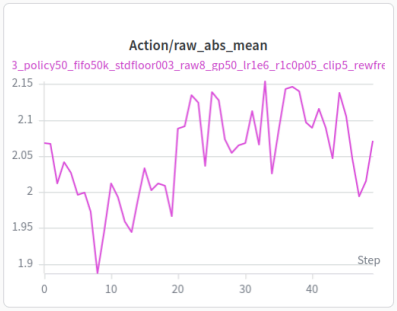

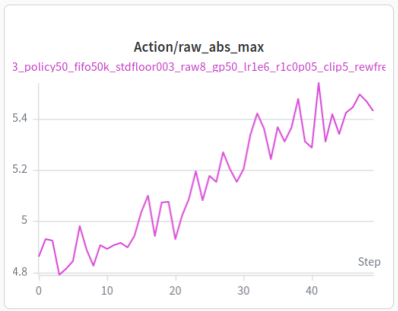

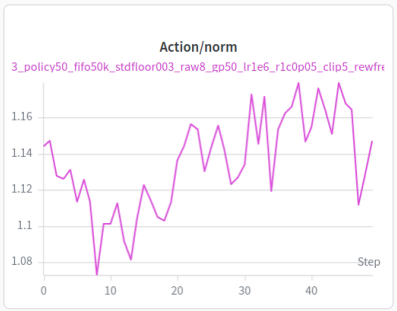

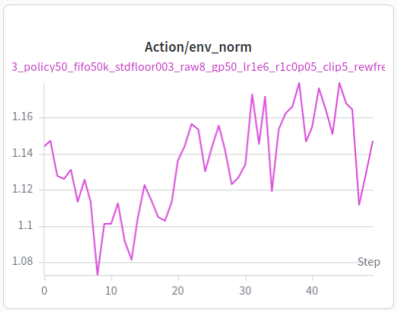

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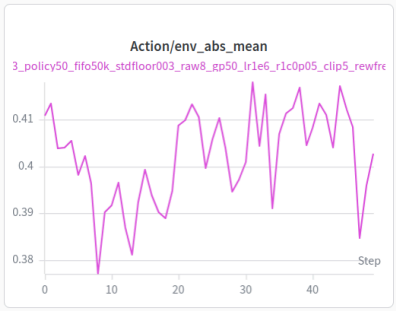

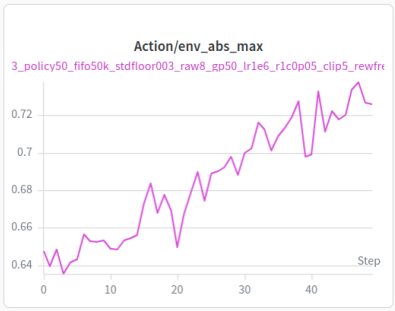

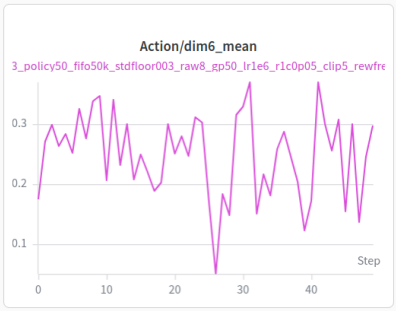

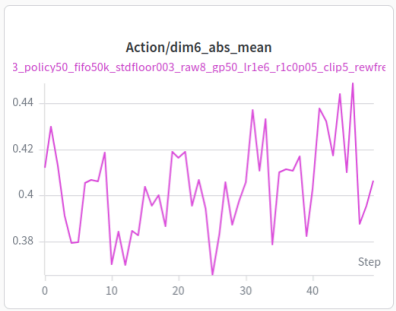

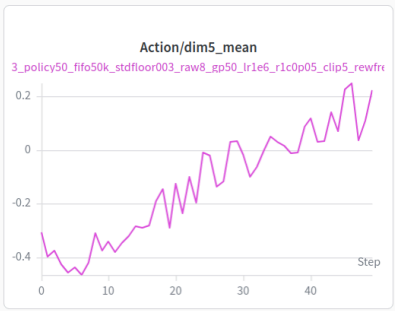

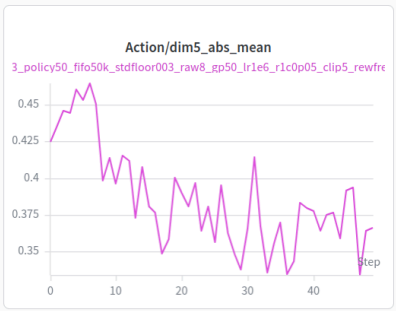

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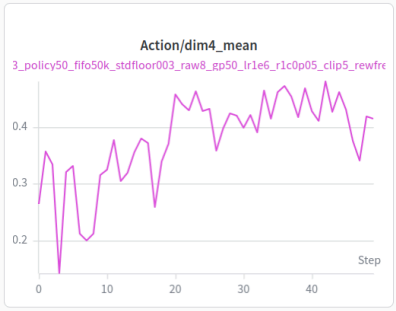

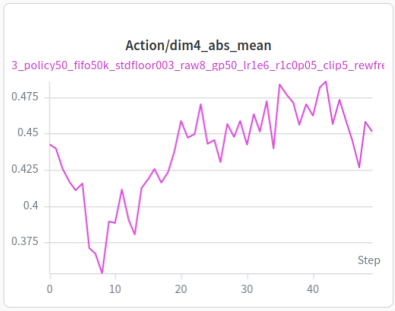

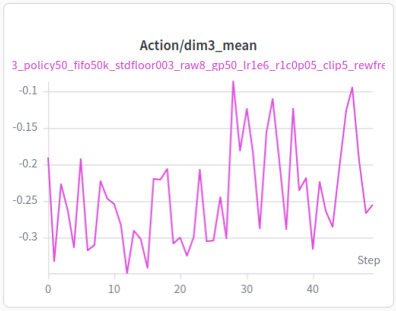

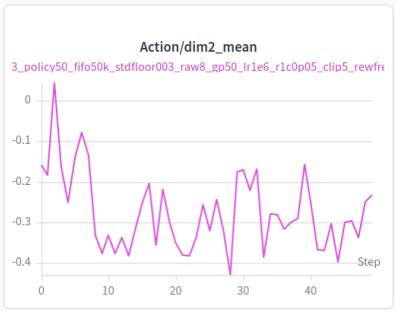

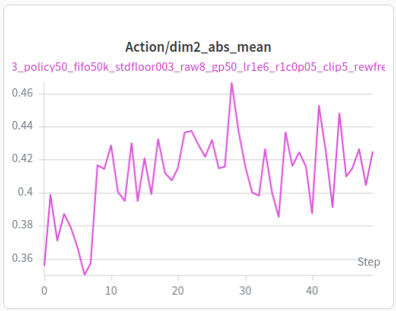

Action 30

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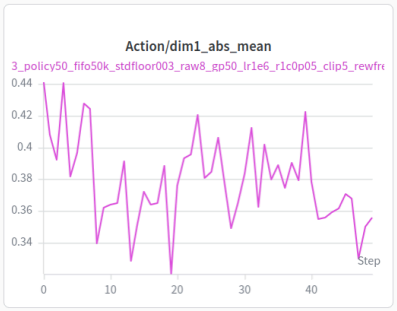
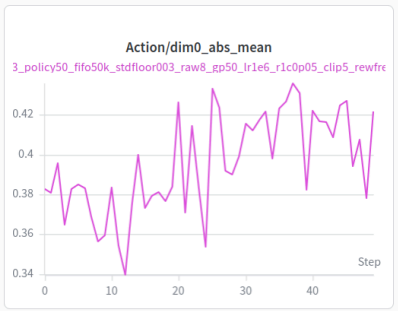
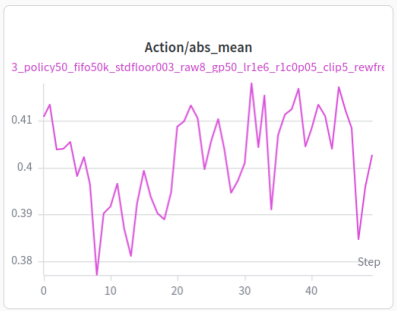
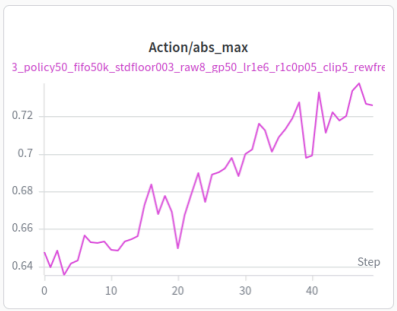
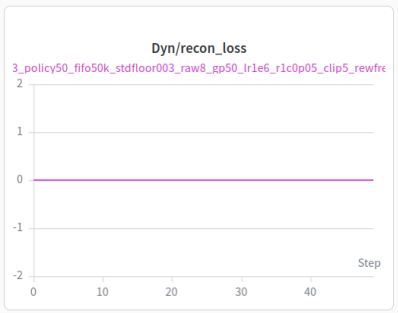
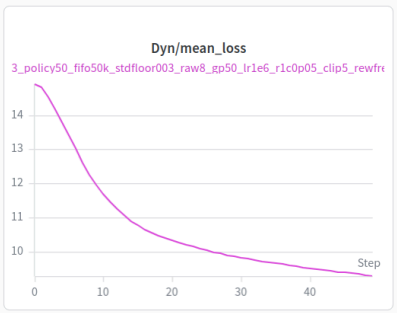


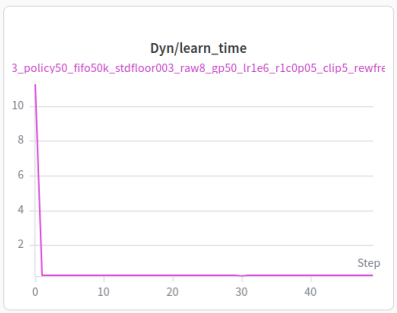


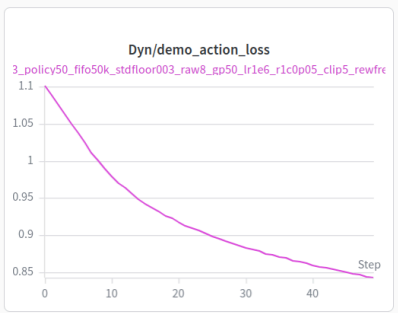


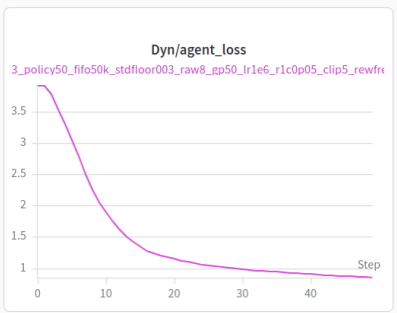


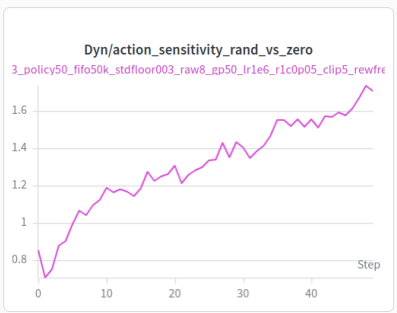




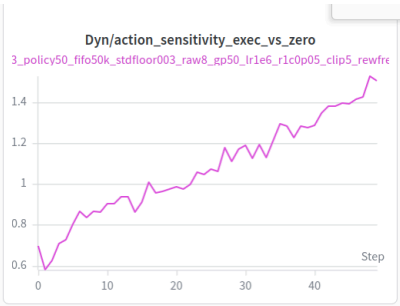








Env 11



Env 11

Episode_Reward 8

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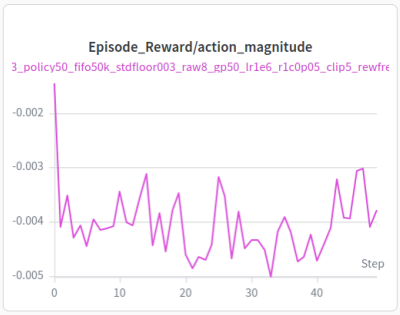

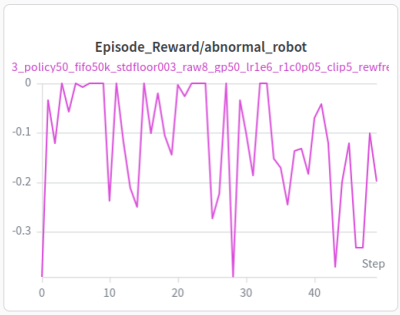

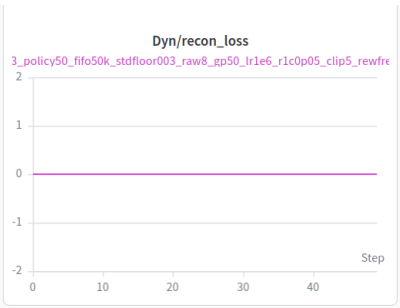

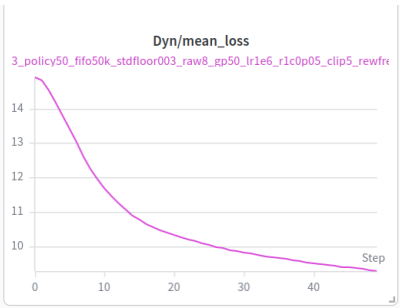

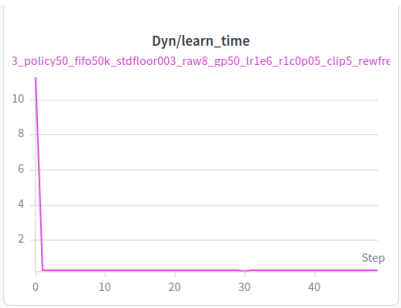

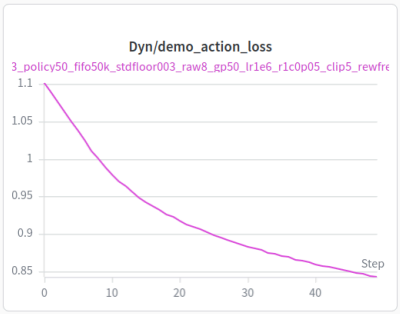

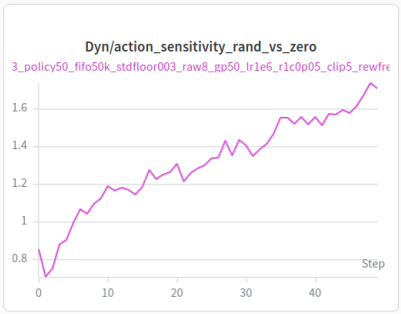

Env 11

Episode_Reward 8

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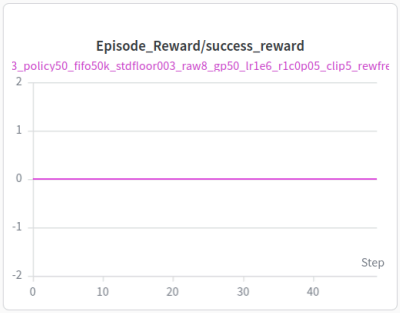
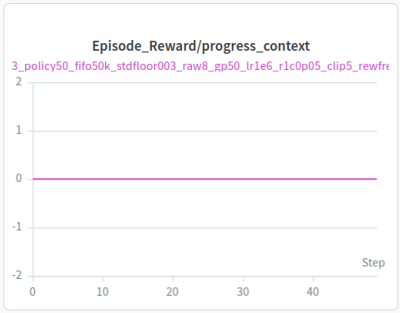
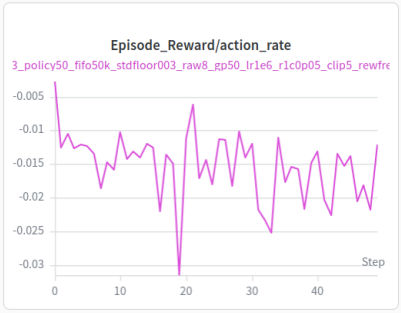
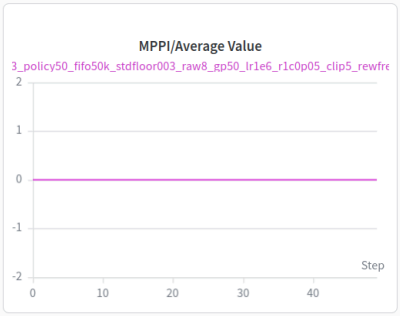
Episode_Reward 8 1-6 of 8

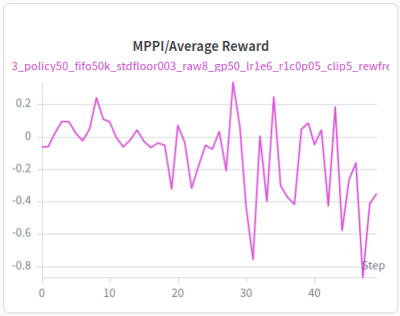






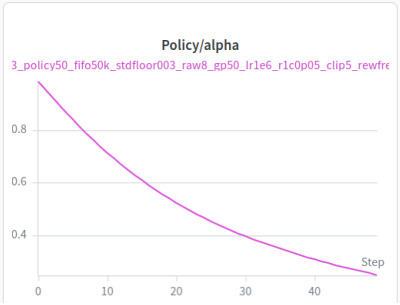
Episode_Termination 2

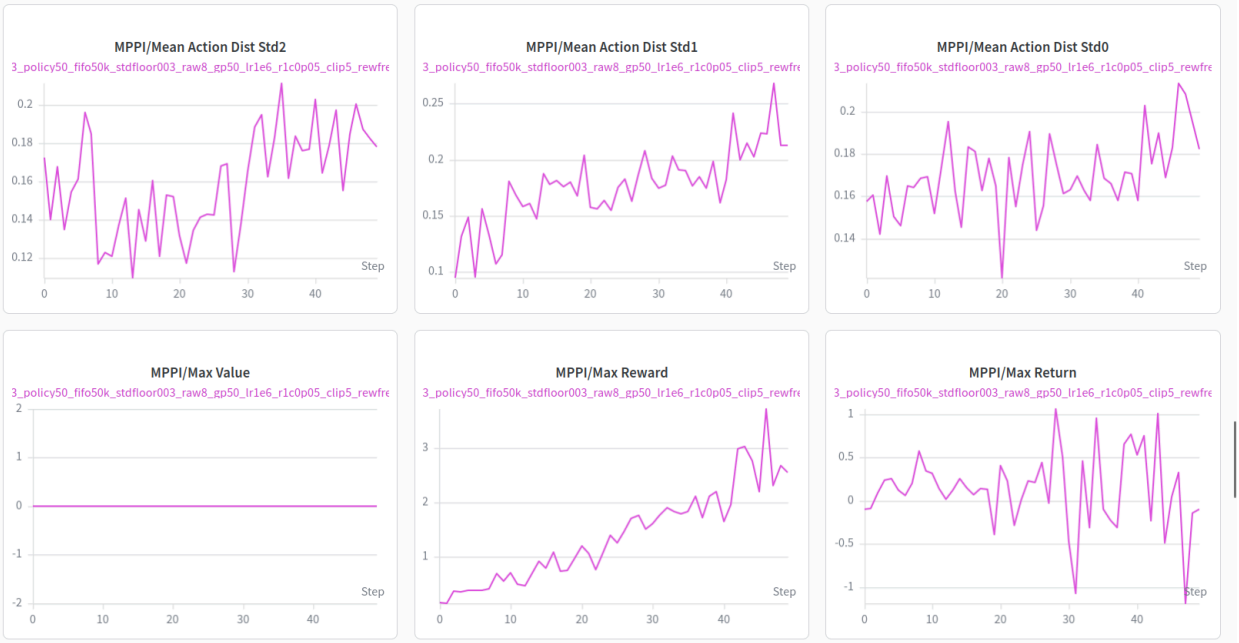




Perf 3

Policy 7 7-7 of 7

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MPPI/Mean Action Dist Std1
3_policy50_fifo50k_stdfloor003_raw8_gp50_lr1e6_r1c0p05_clip5_rewfre
MPPI/Mean Action Dist Std0
3_policy50_fifo50k_stdfloor003_raw8_gp50_lr1e6_r1c0p05_clip5_rewfre
MPPI/Max Value
3_policy50_fifo50k_stdfloor003_raw8_gp50_lr1e6_r1c0p05_clip5_rewfre
MPPI/Max Reward
3_policy50_fifo50k_stdfloor003_raw8_gp50_lr1e6_r1c0p05_clip5_rewfre
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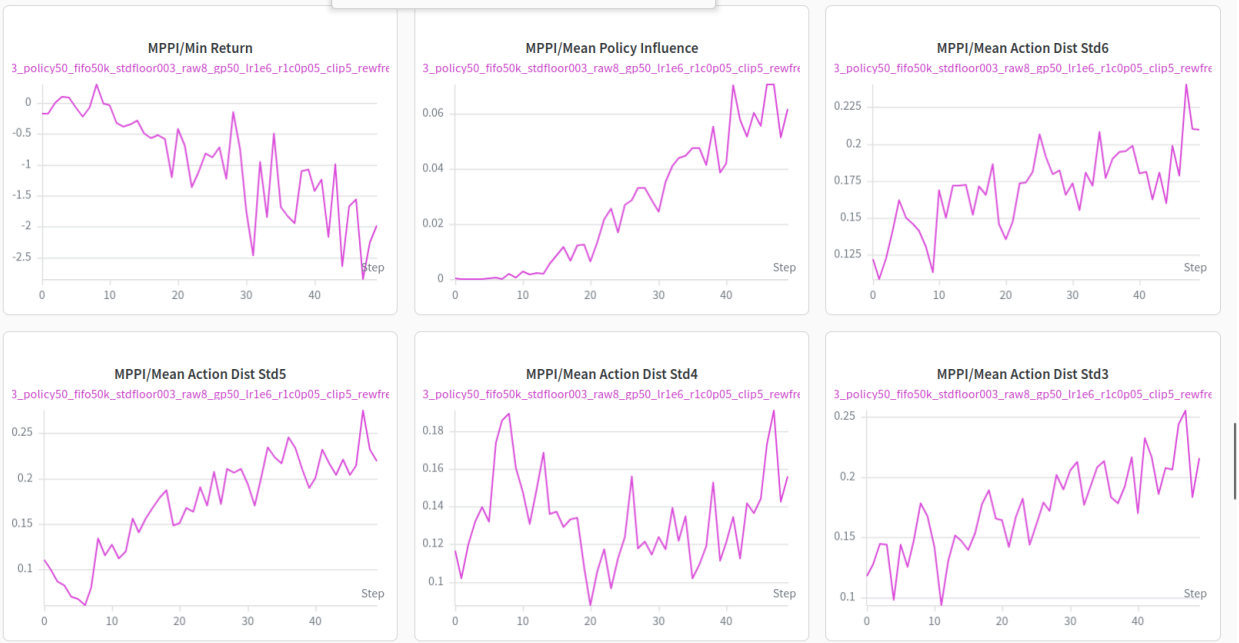
Step

Perf 3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MPPI/Mean Action Dist Std6
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MPPI/Mean Action Dist Std5
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MPPI/Mean Action Dist Std4
3_policy50_fifo50k_stdfloor003_raw8_gp50_lr1e6_r1c0p05_clip5_rewfre
MPPI/Mean Action Dist Std3
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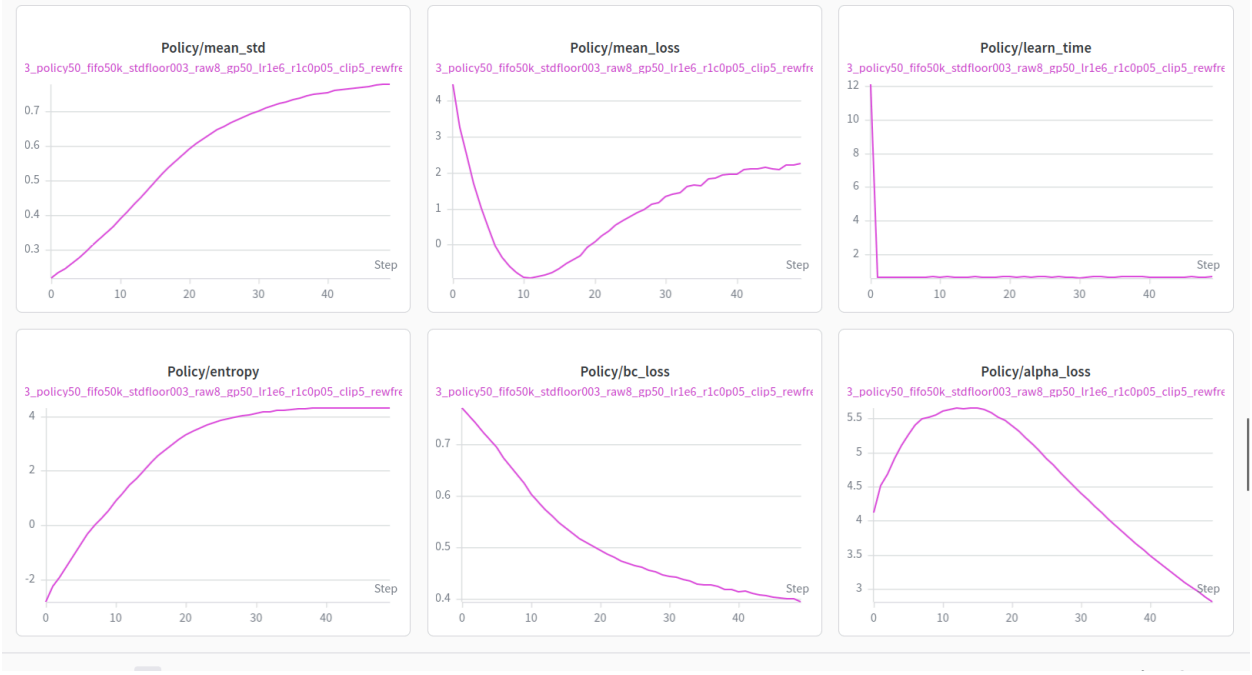
Step

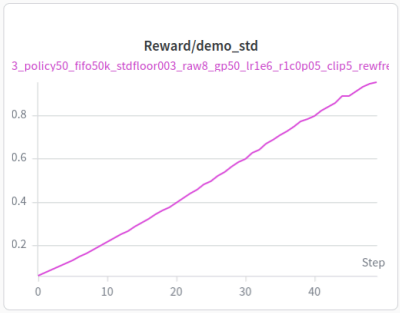
Perf 3

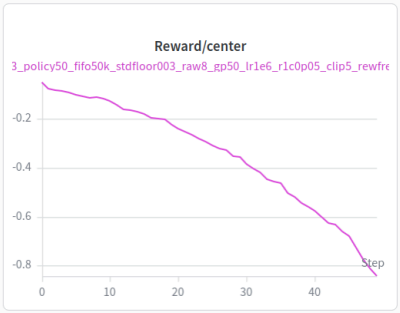
Perf 3

Policy 7 1-6 of 7

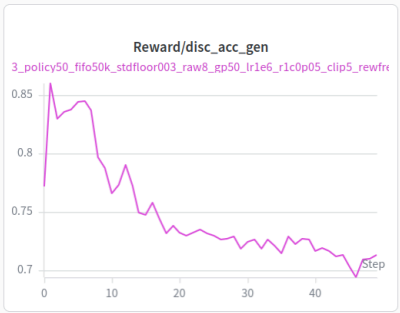


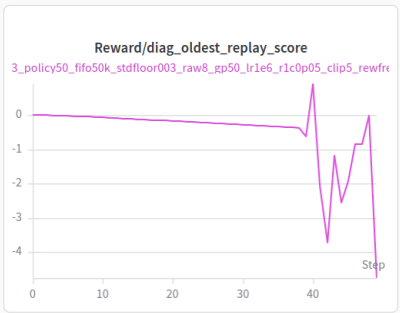
Reward 22 19-22 of 22

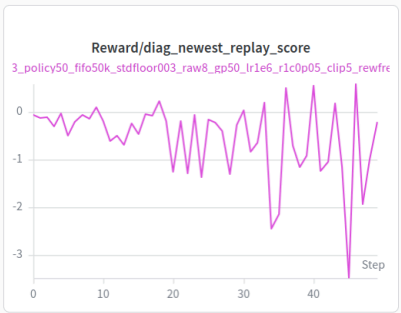




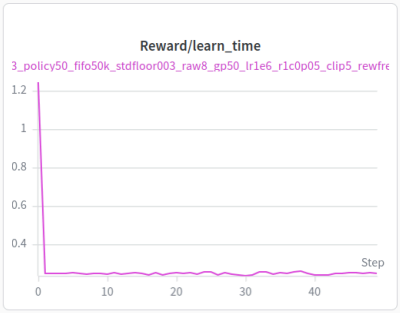
Reward 22 13-18 of 22

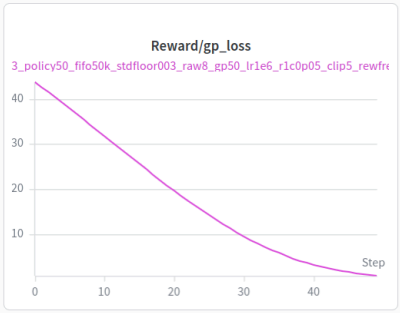


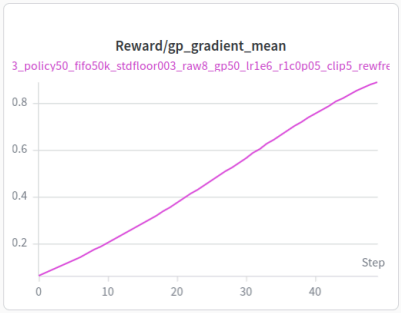


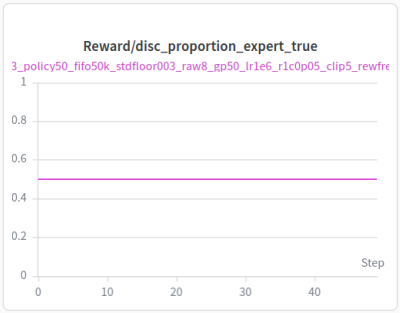


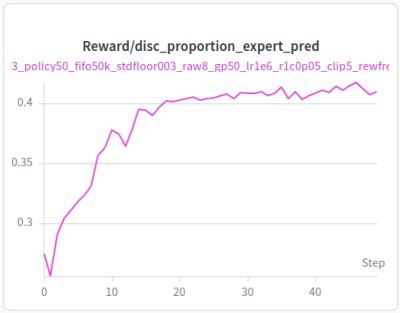
Reward 22 7-12 of 22



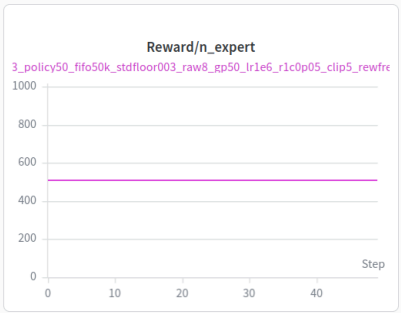


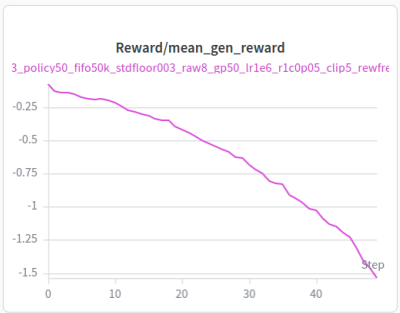


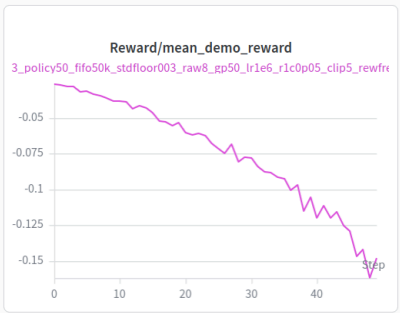


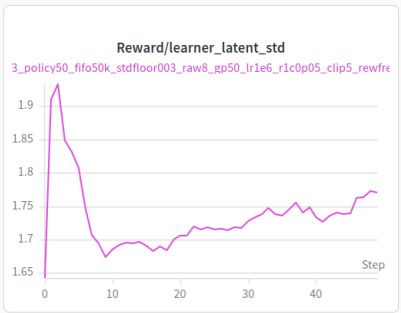


Reward 22 1-6 of 22









Value 4

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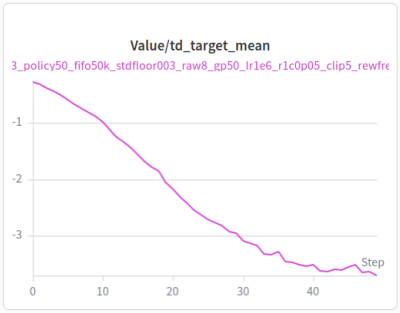
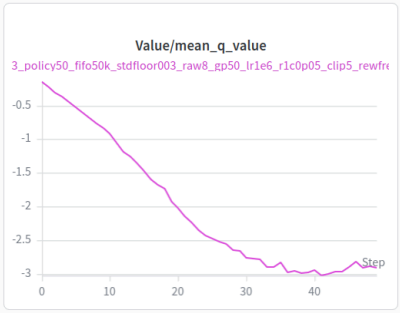
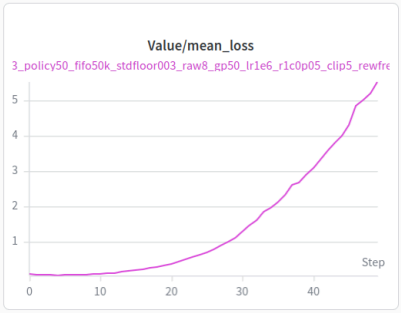
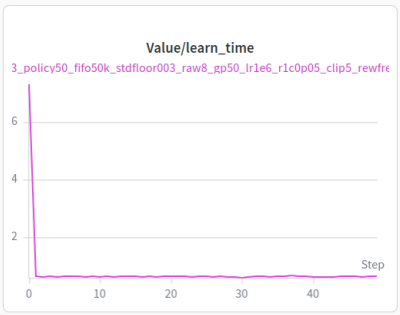
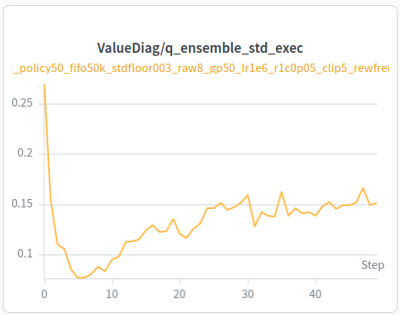




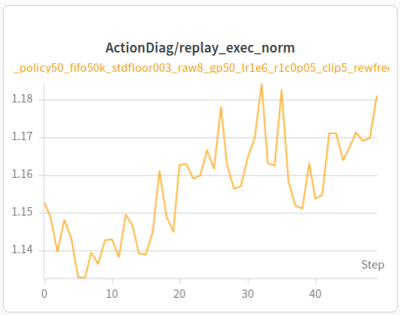


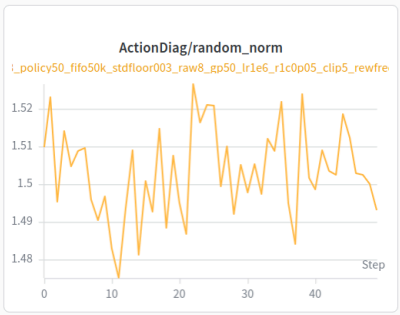


System 33



ActionDiag 2





Action 30

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Search panels with regex

Settings | New report | Add panels

ValueDiag 7

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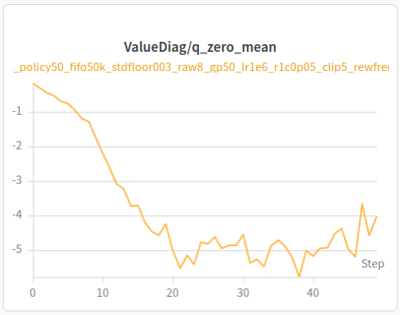

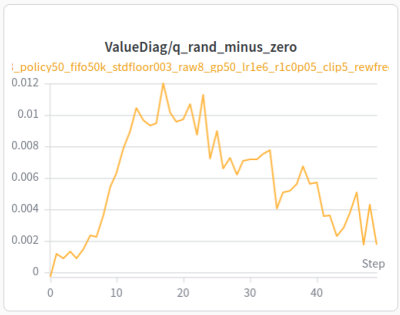

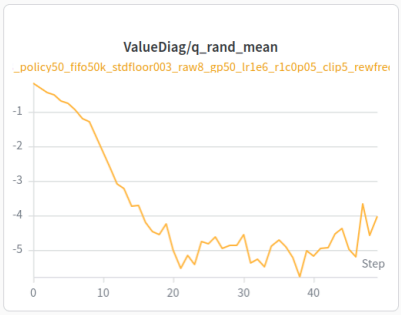

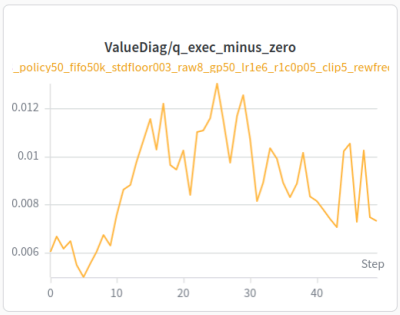

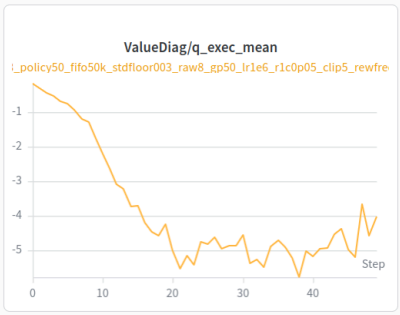

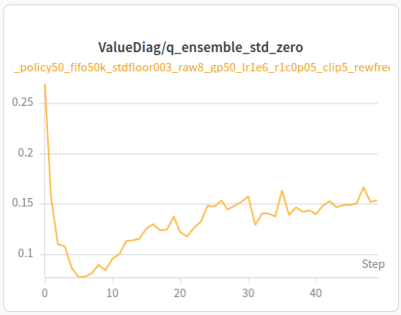

Action 30

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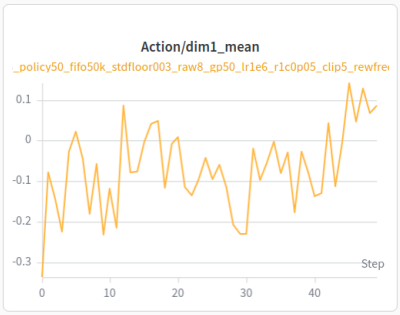
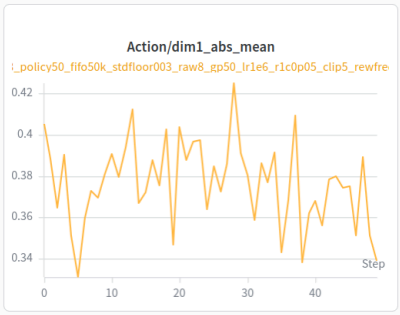
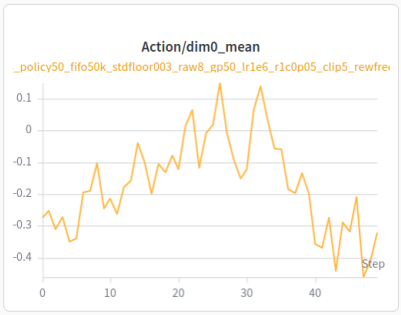
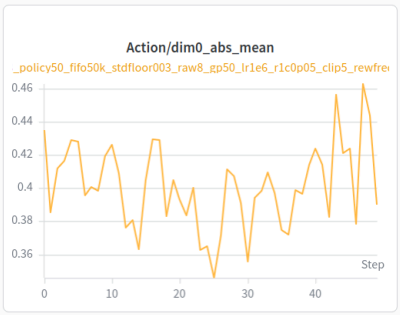
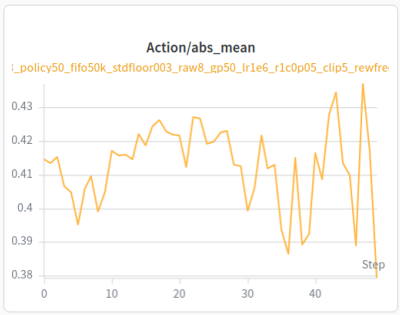
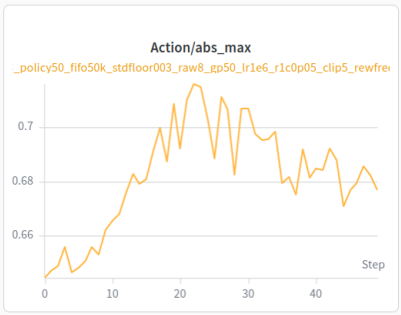






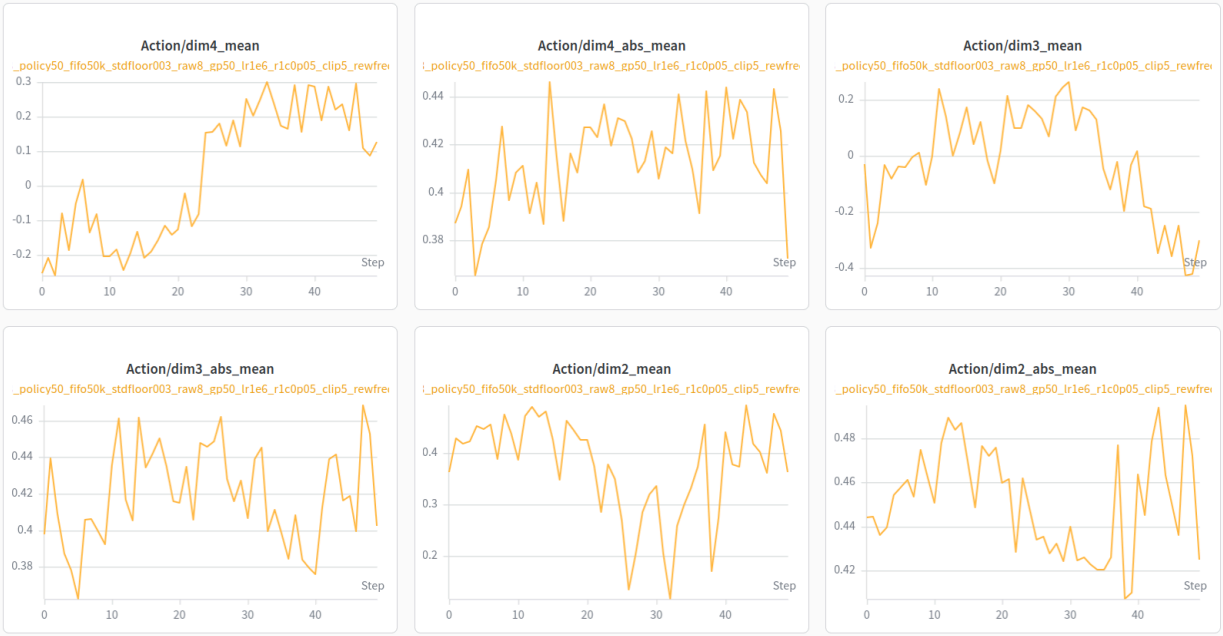






Charts 2

Action 30 19-24 of 30



Charts 2

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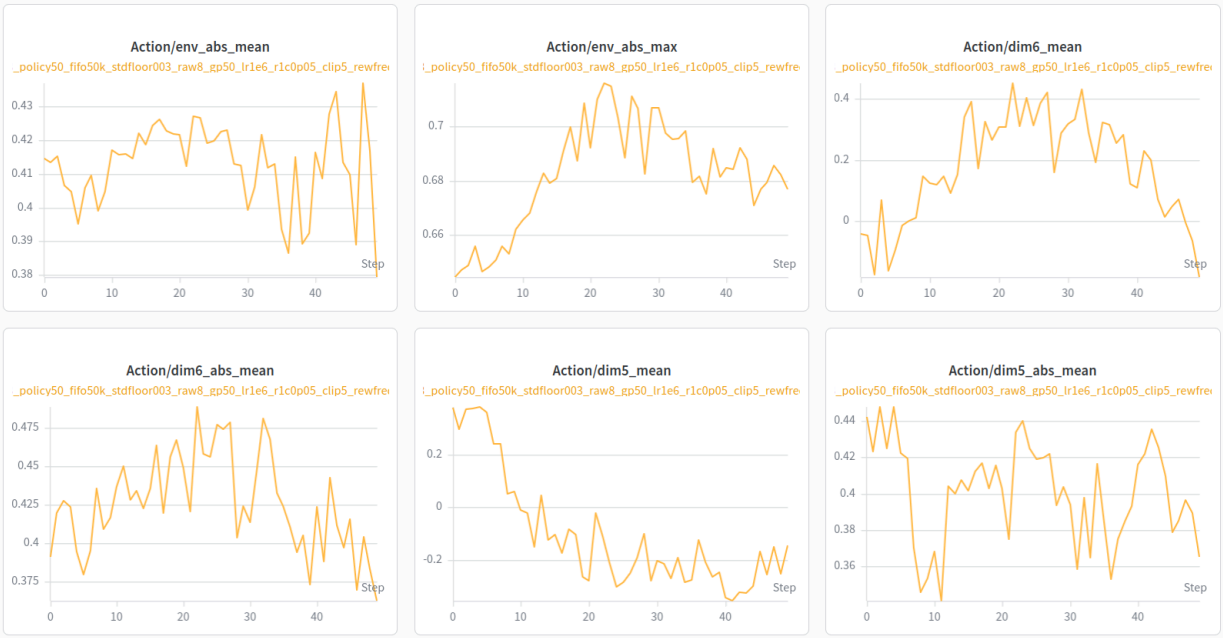
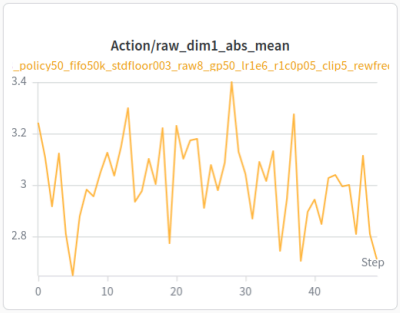
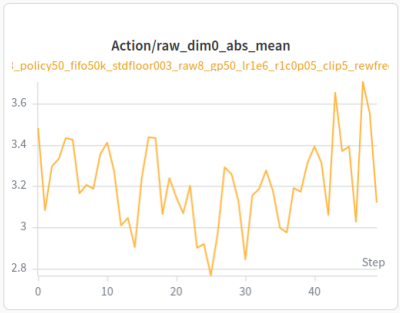
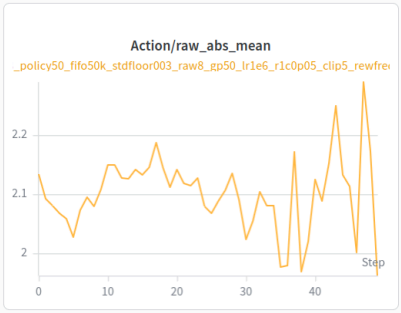


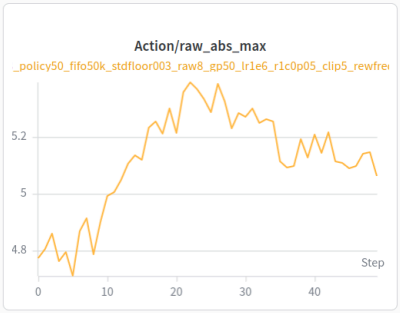
Charts 2

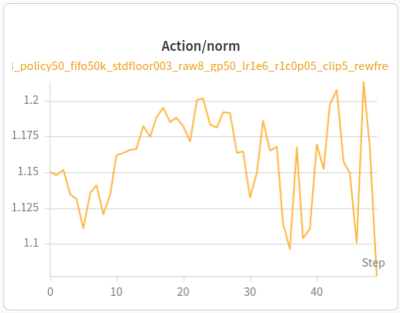
Action 30 7-12 of 30

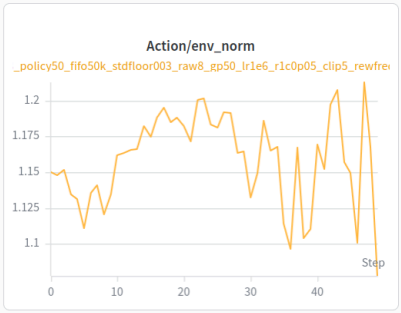












Charts 2

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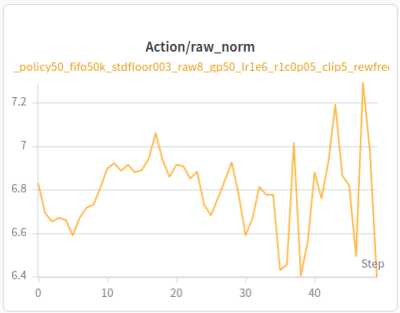

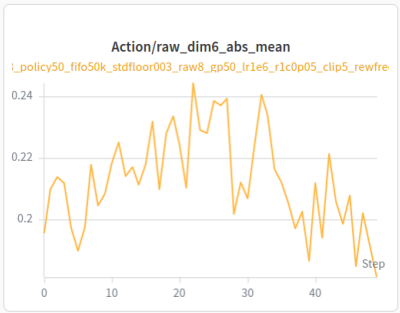

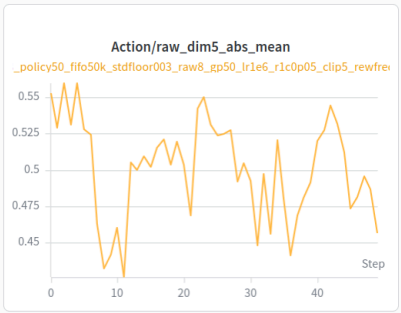

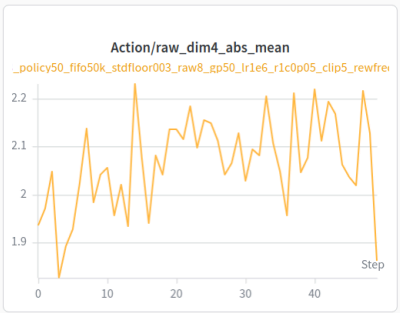

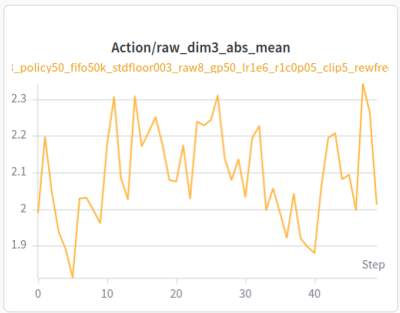

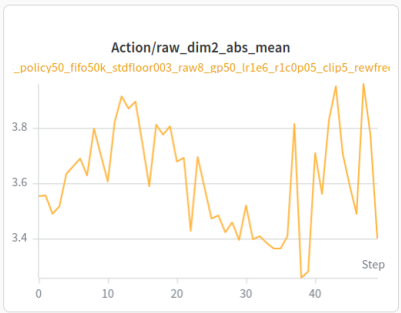

Charts 2

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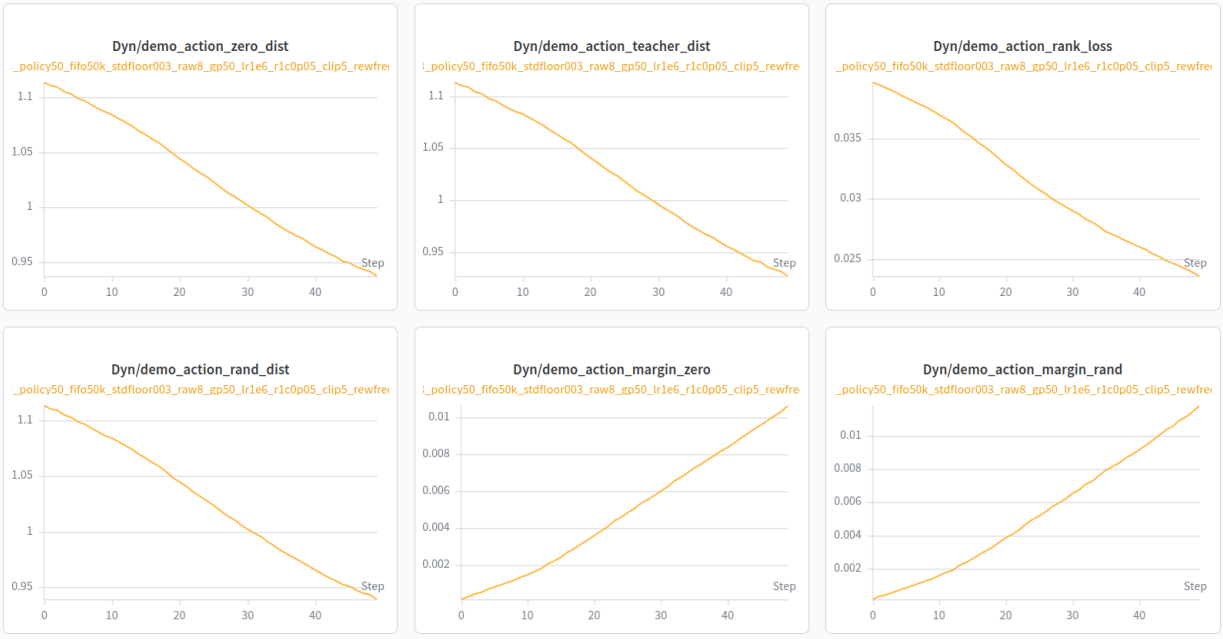

Env 11

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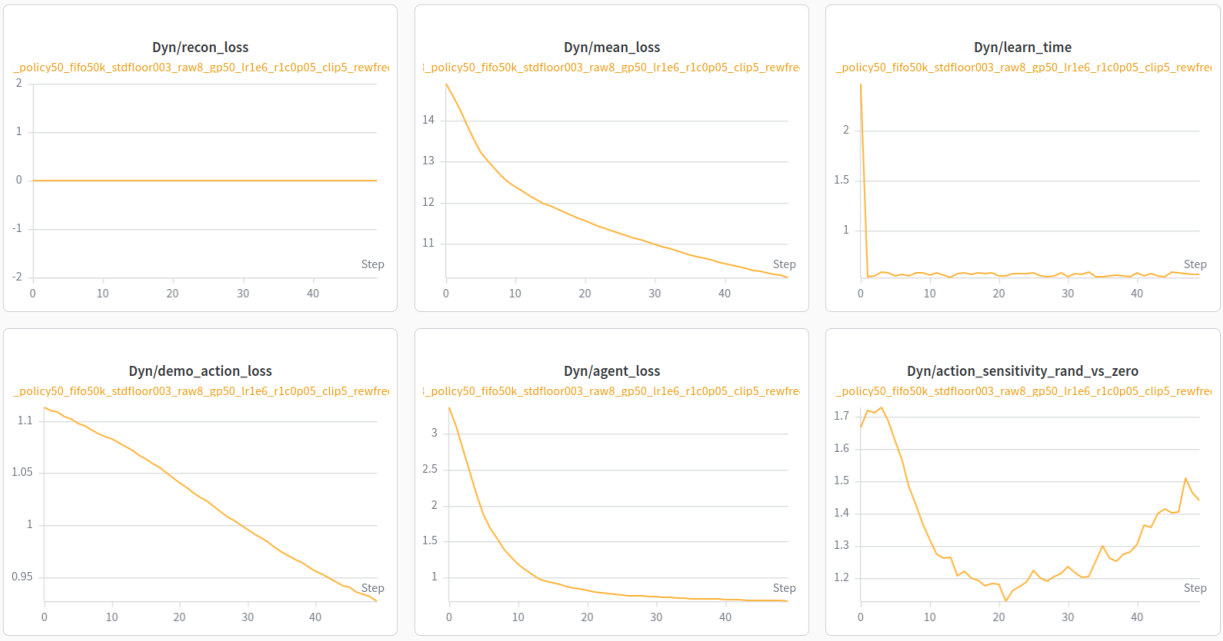

Env 11

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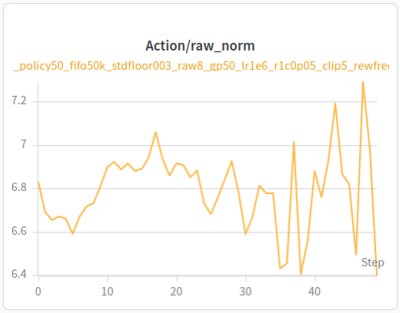

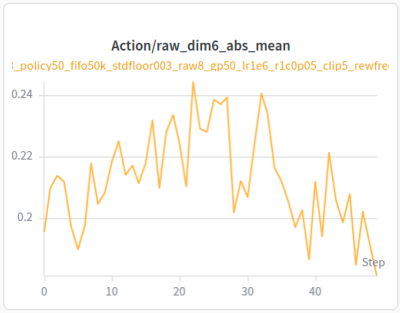

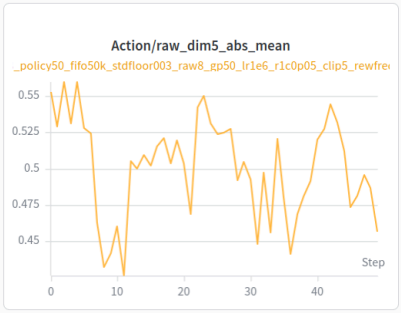

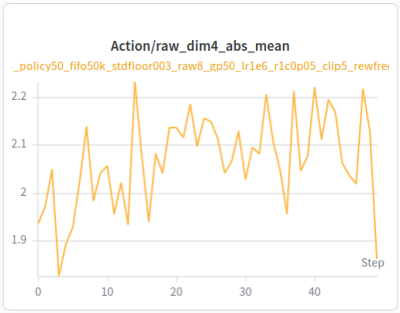

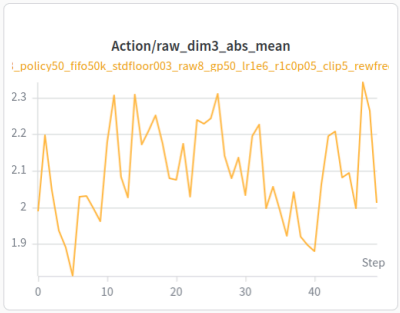

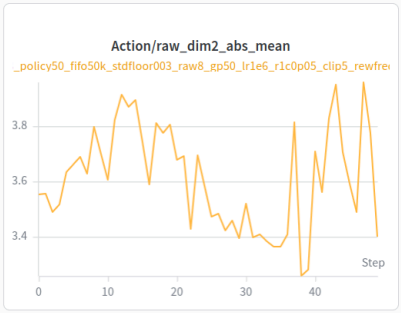

Charts 2

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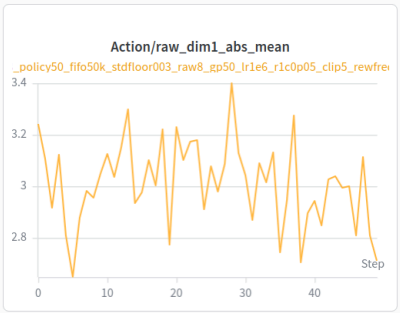

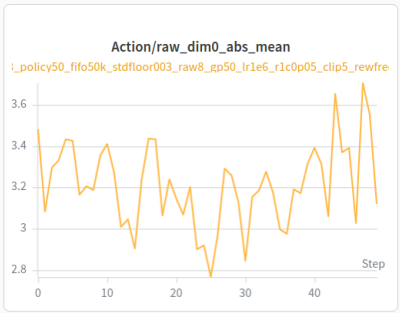

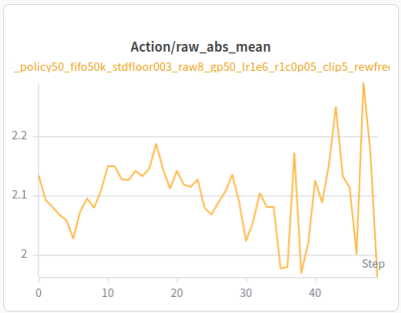

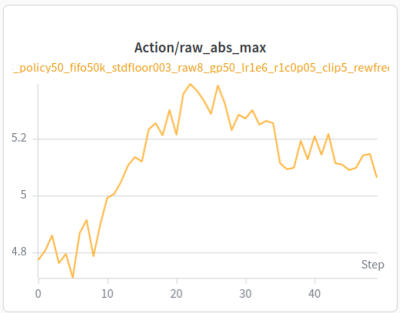

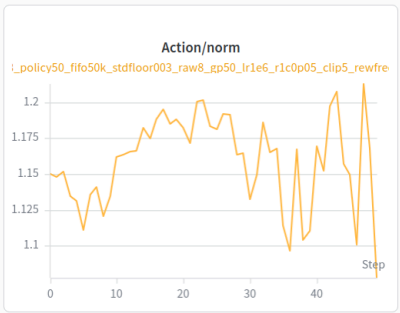

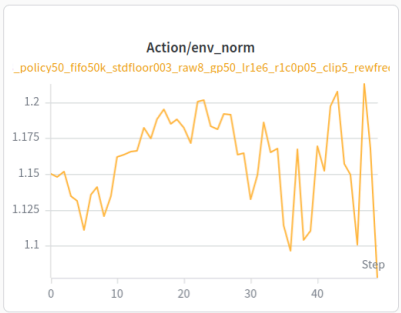

Charts 2

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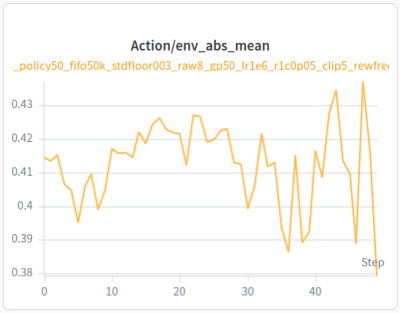

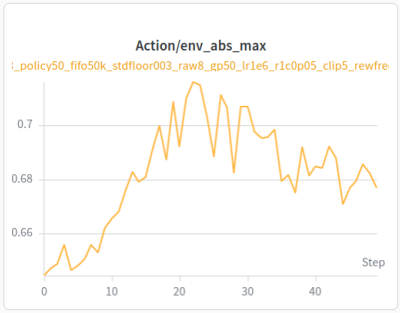

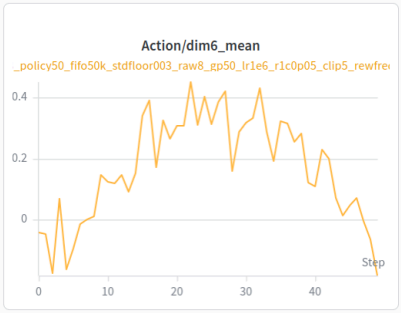

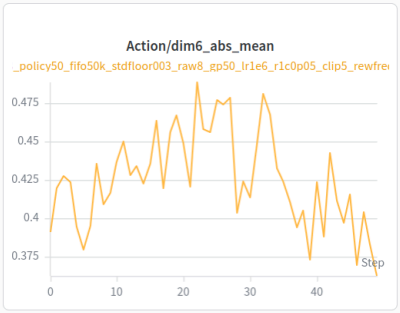

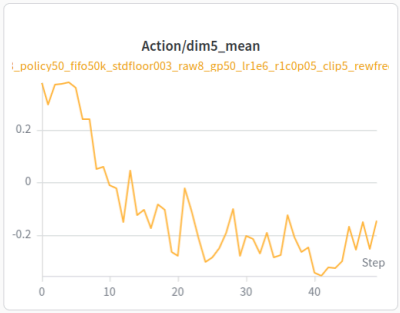

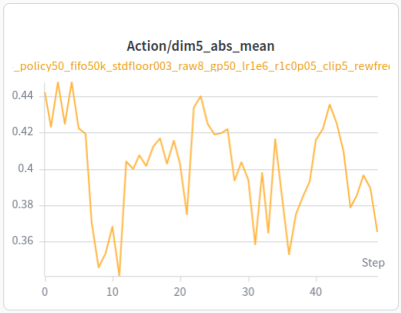

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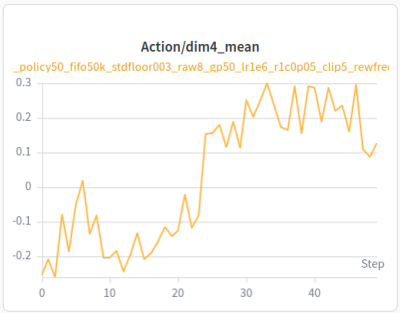

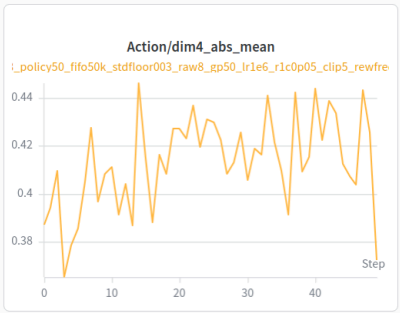

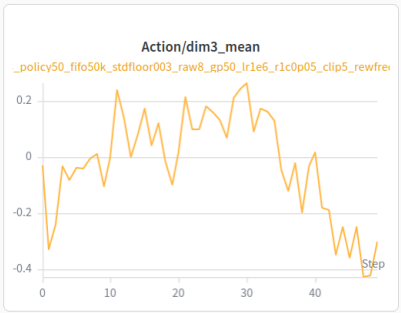

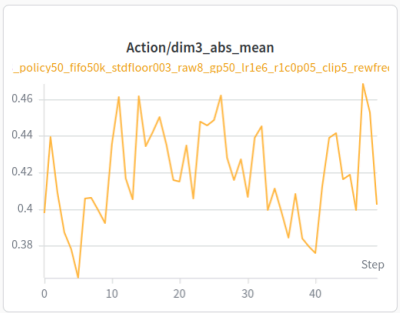

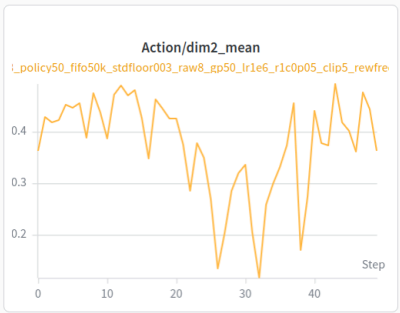

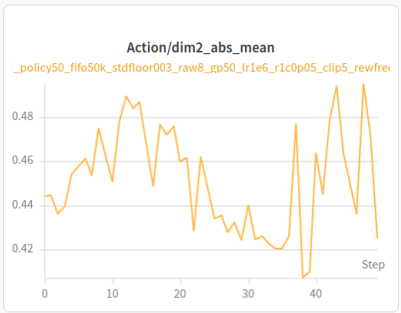

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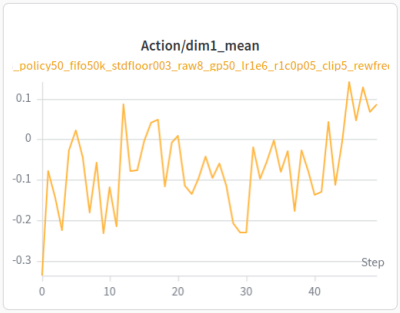
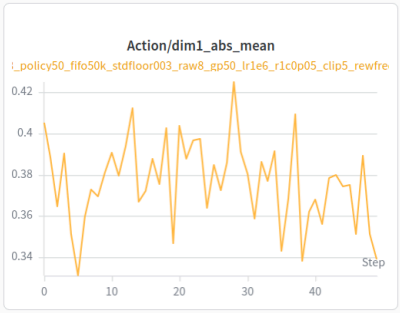
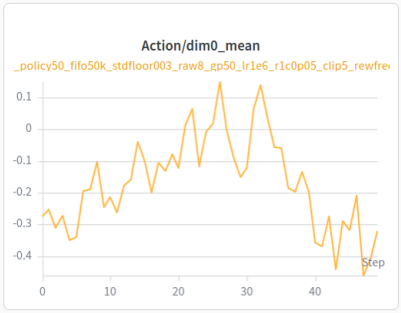
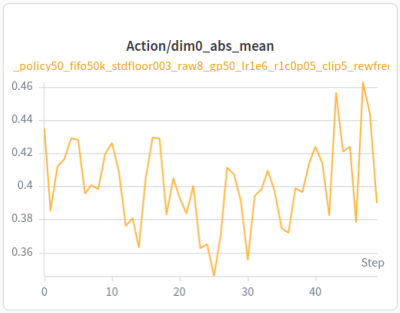
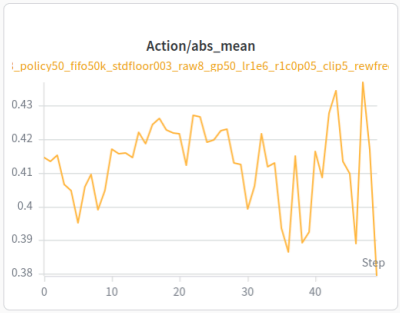
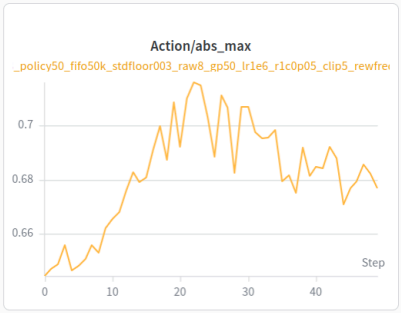






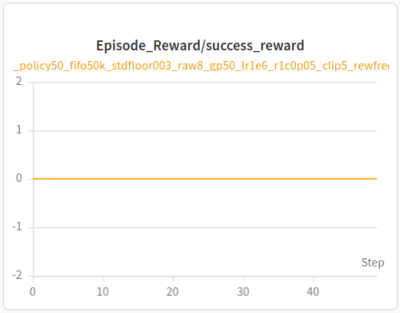


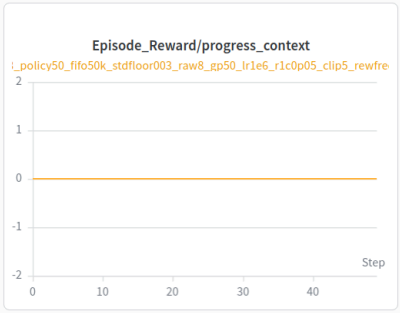


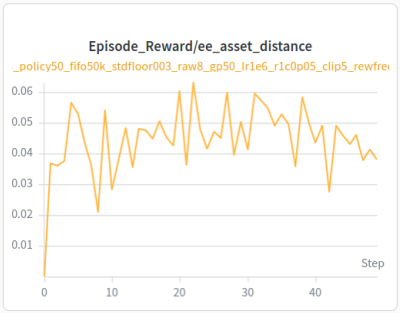


Charts 2

Episode_Reward 8 1-6 of 8







Episode_Termination 2

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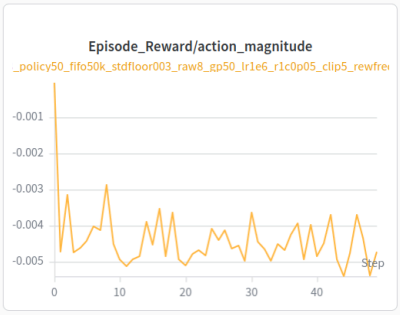
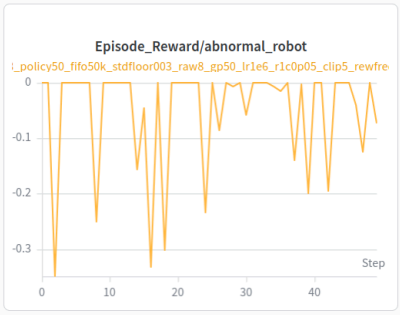
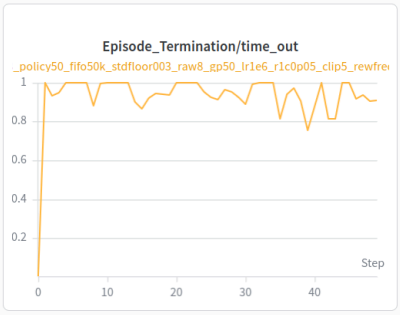
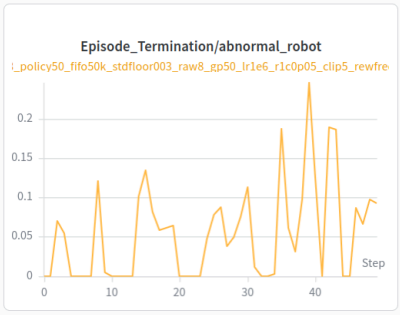
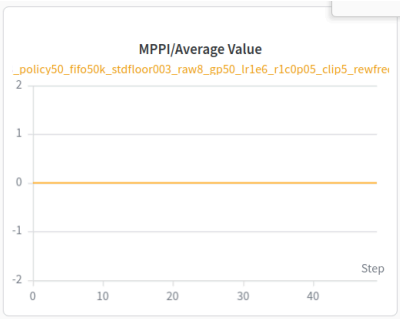


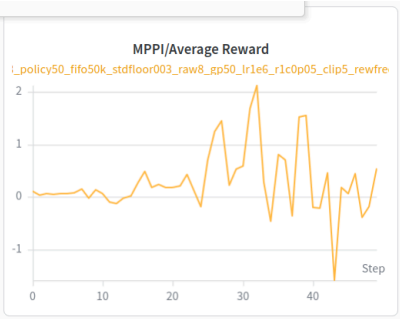


Episode_Termination 2









Perf 3

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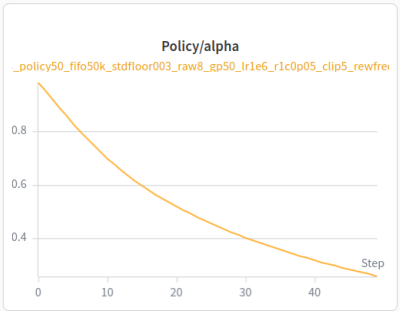

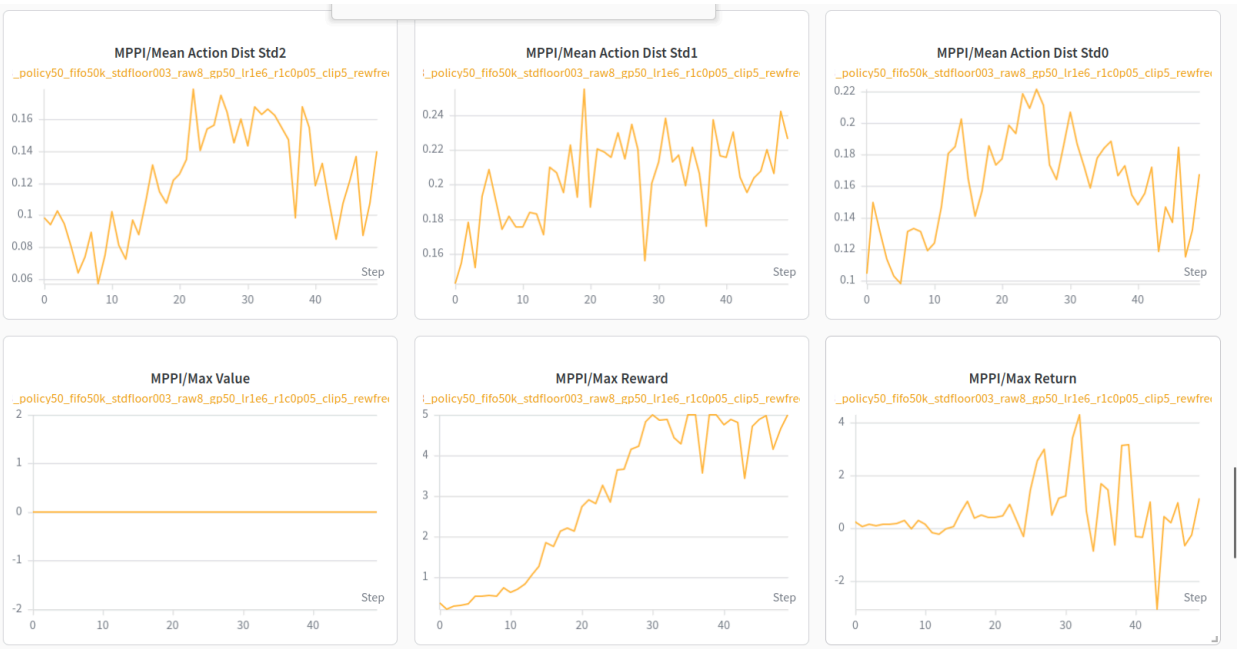

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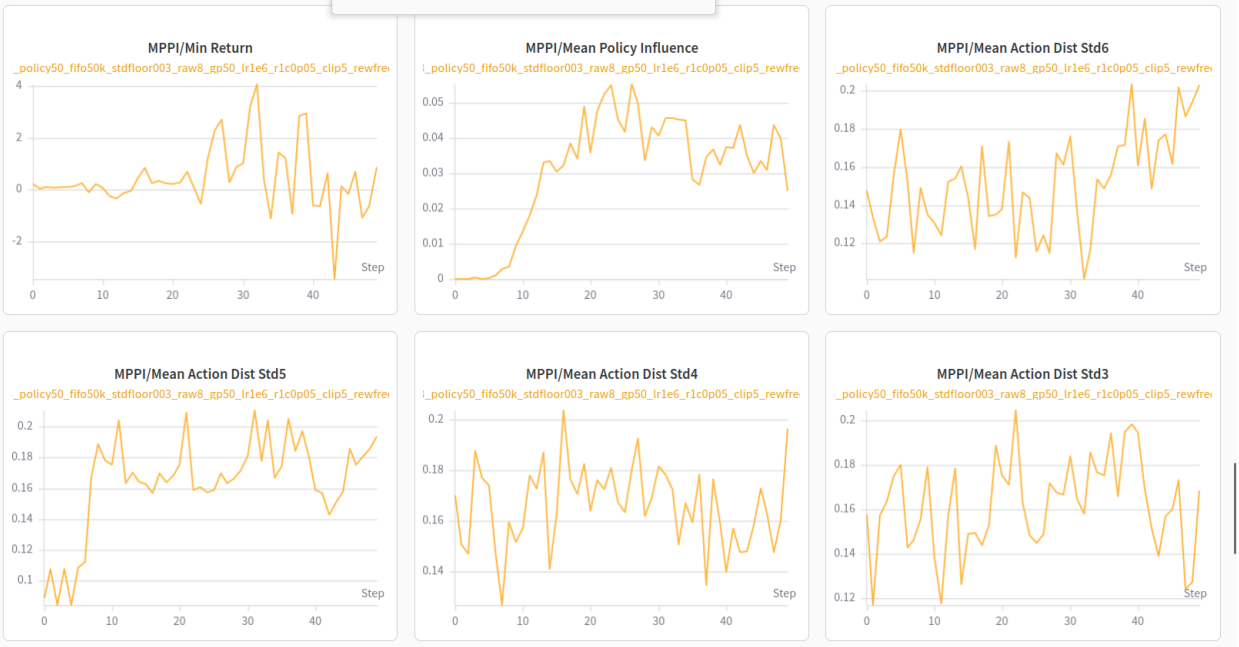

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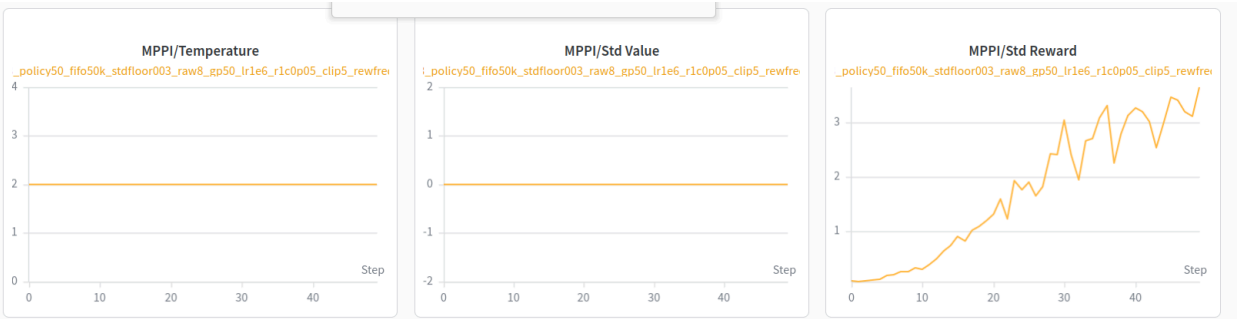
Policy 7



Perf 3

Policy 7

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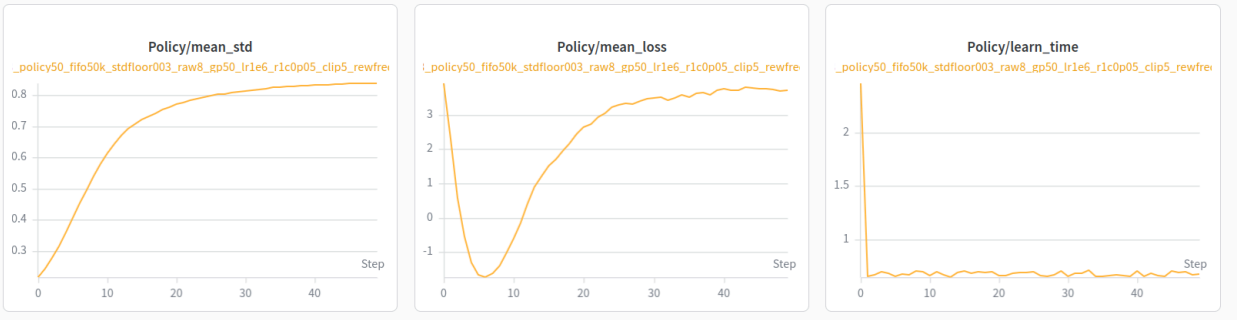

Reward 22

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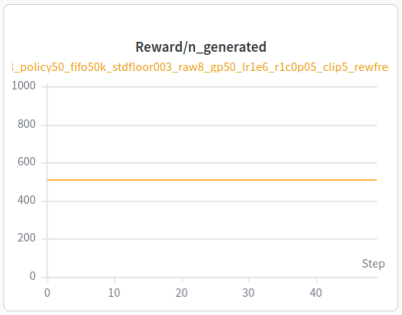

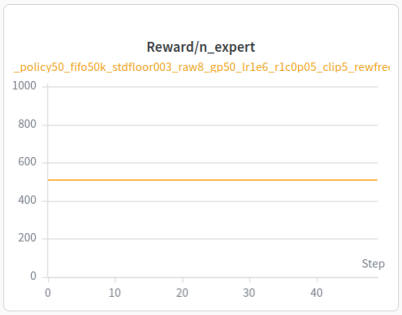

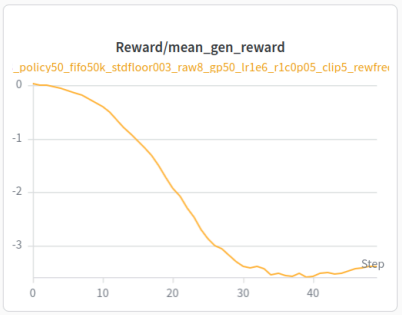

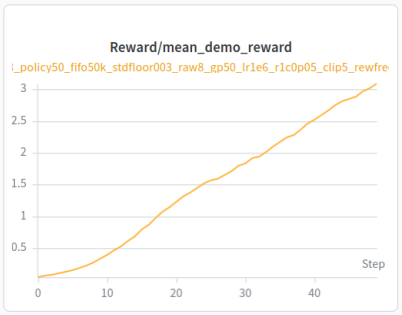

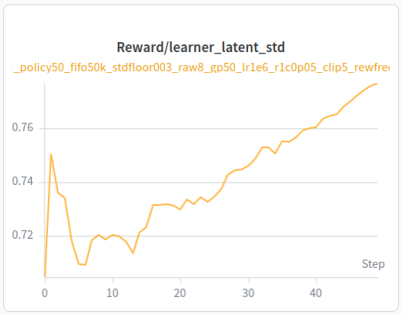

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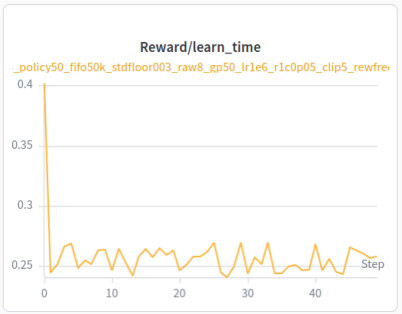

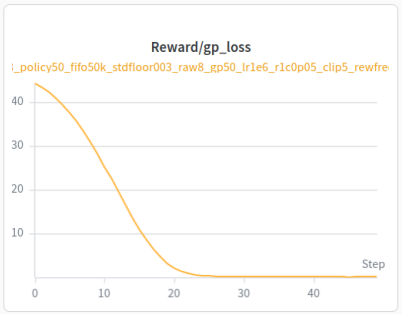

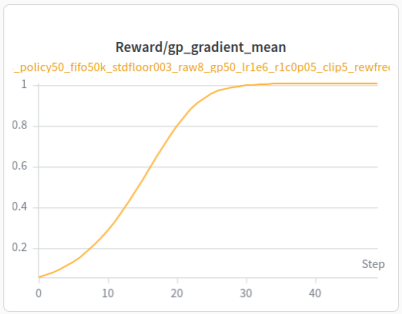

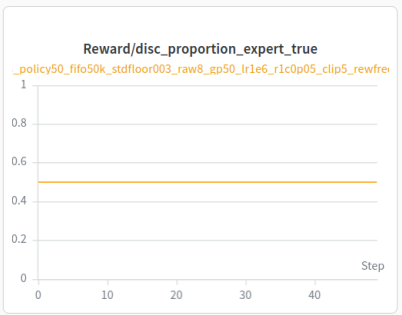

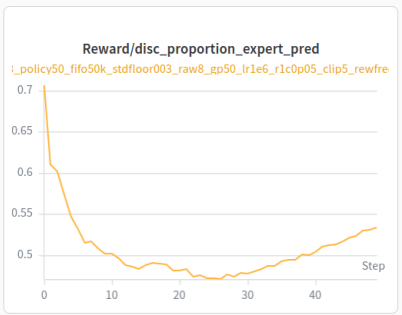

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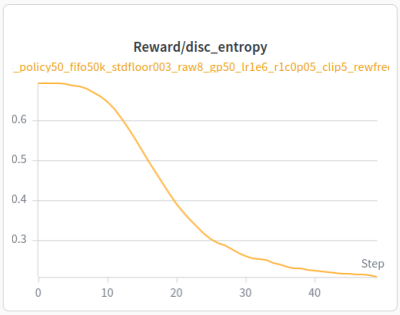
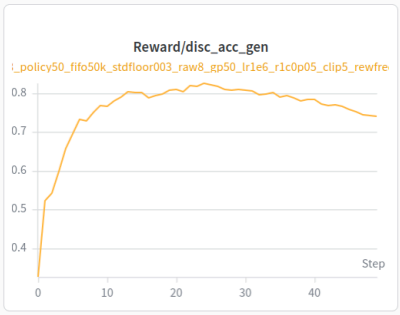
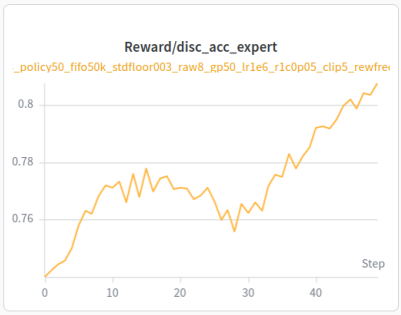
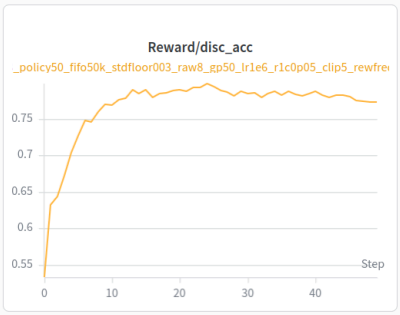
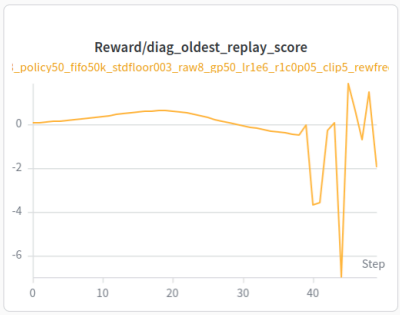
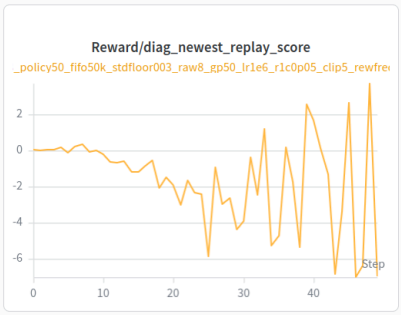




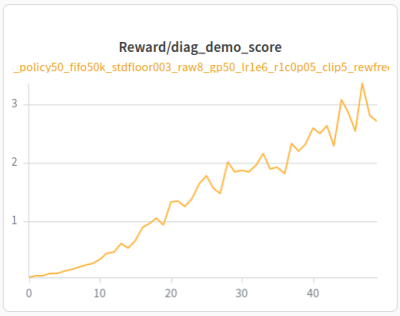


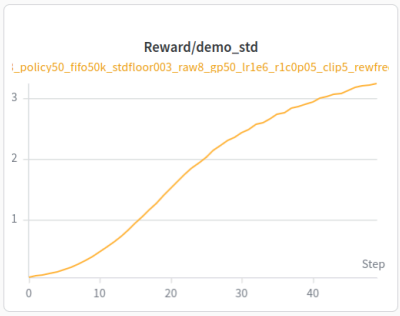


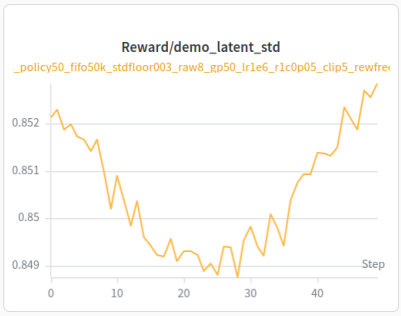


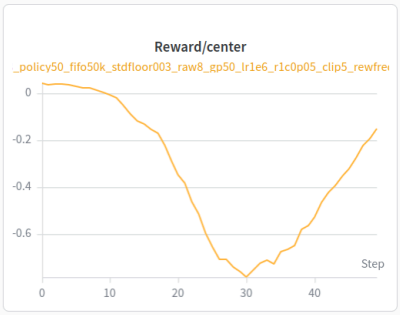


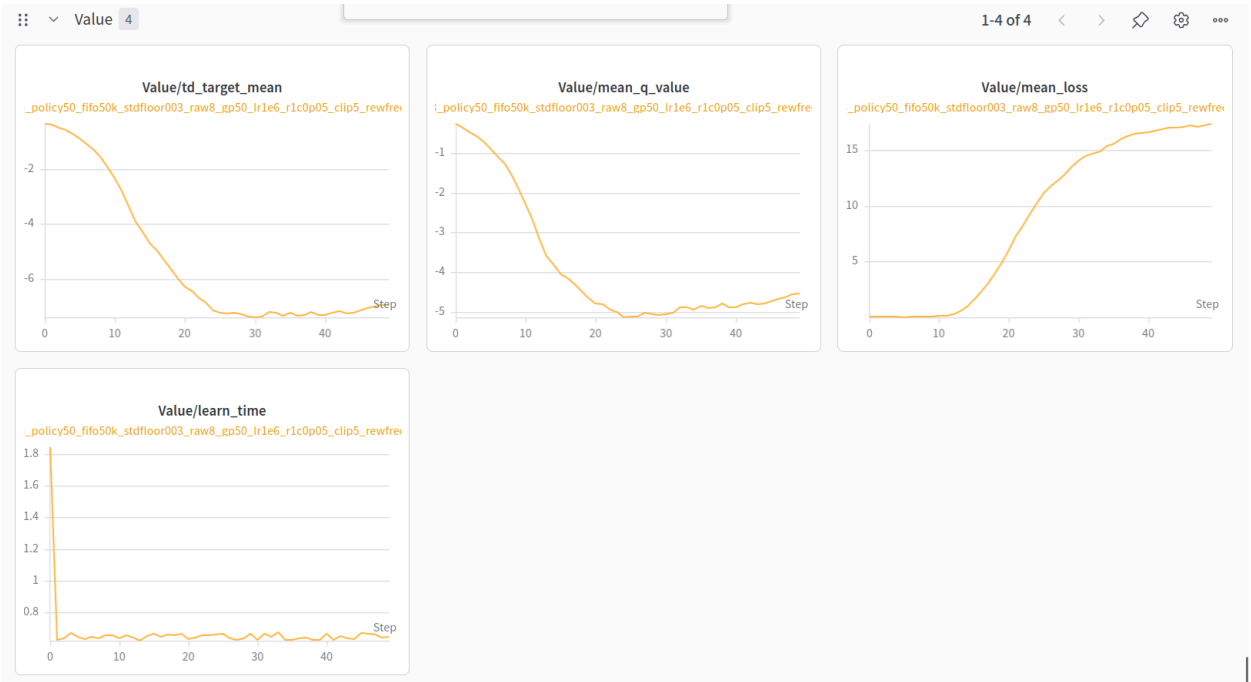
Reward 22 19-22 of 22











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